

Aldo Romano
Giustina Secundo
Editors

Dynamic Learning Networks

Models and Cases in Action

Springer

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Edited by

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ISBN 978-1-4419-0250-4 e-ISBN 978-1-4419-0251-1
DOI 10.1007/978-1-4419-0251-1
Springer Dordrecht Heidelberg London New York

Library of Congress Control Number: 2009926316

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Printed on acid-free paper

Springer is part of Springer Science+Business Media (www.springer.com)

Acknowledgements

We would like gratefully acknowledge some people, without whom this book has not been possible.
First, the Scientific Committee of the Advanced International Summer School, 2008 session, on "*Managing Uncertainty and Competition Through Dynamic Learning Networks*" of eBMS-Scuola Superiore ISUFI, University of Salento (IT) for having organized and led the research tracks and discussions in which the book contributions are developed. A special gratitude goes to Prof. Ernesto Damiani of University of Milan that provided us the right energy and direction to start this project. Moreover, a special thank to the chapters' authors for their enthusiasm and punctuality in providing their contributions and last but not the least, Dr. Claudio Petti for his valuable comments and insights for the organization of the work.

Contents

Acknowledgements ..... **v**

Contents ..... **vii**

Preface ..... **xi**

List of contributing authors ..... **xiii**

Notes on Editors ..... **xvii**

Notes on contributing authors ..... **xix**

Introduction **1**
by Aldo Romano, Giustina Secundo
An integrated perspective on Dynamic Learning Networks 1
What is Learning? 2
What is Network and Learning in network? 4
Why Dynamic? 5
Defining Dynamic Learning Networks 6
Dynamic Learning Networks design variables and typology 7
Structure and contents of the book 9
Key features and guiding principles 16
References 20

Chapter 1. Networks for manage change in international commercial organisations **23**
by Tom Osorio
Introduction 23
Major change and extreme events 24
Organisational Complexity or Complication 25
Research challenges in the hydrocarbon exploration industry: Shell case study 27

Changing a consultancy from partnership to public company: Scott Wilson case study 33
Effective medium term responses to a major event in an International energy company: BP case study 36
Conclusion 40
References 42

Chapter 2. The Relevance of Organizational Learning for High Performing Social Networks 43
by Jeff Clanon
Organizational Learning: Historical Perspective 43
Learning from Research on "Hastily Formed Networks" 47
Introduction: The Hastily Formed Networks (HFN) Project 47
Core Research Questions 48
Findings: How action was mobilized throughout the network 49
What minimal organization structures are required 49
Preconditions that are necessary for Hastily Formed Networks to be effective 50
The leadership capabilities and characteristics required 50
The Relevance of Organizational Learning for High Performing Social Networks 51
Conclusion 54
Acknowledgements 55
References 56

Chapter 3. illycaffè and the Brazilian espresso knowledge network ... 57
by Ingrid Rauscher, Pierpaolo Andriani
Introduction 57
An Award for Quality 58
A network for generating and disseminating knowledge 61
Multi-layer 65
Multi-channels 65
Trust-based and high knowledge density 67
Non-exclusive and open 67
The insertion of coffee in the information and knowledge economy ... 68
Final considerations – Changing the landscape through knowledge 70
References 73

Chapter 4. Learning dynamics within the solutions network 75
by Stefano Miraglia, Andrew Davies
Introduction 75
Integrated solutions: a collaborative business model 76

The shift toward solutions: origins, drivers and recent evolution..... 76
Refocusing the business and reorganizing for customer-centricity .. 78
Capability building and organizational learning..... 79
Learning dynamics within the solutions business model..... 80
The configuration of the necessary capabilities..... 80
The dynamics of the learning process 82
The integrator perspective 83
The supplier-customer interface: emphasis on the front-end..... 84
Learning for repetition..... 85
The customer perspective: feeding the learning process 86
Future trends..... 88
Getting ready for environmental changes: network proactivity 88
Toward complete integration..... 90
Conclusions 90
References 92

Chapter 5. Towards "Stakeholder University" as dynamic learning network: the Finmeccanica case..... 95
by Roberto Maglione, Giuseppina Passiante
Challenging the new Competitive Landscape: the emerging Networked Learning Organizations 95
The central role of Human Capital in Networked Learning Organizations..... 99
The emerging pedagogical approaches for creating Human Capital .. 101
Towards the Stakeholder Universities for creating suitable Human Capital..... 104
Creating human capital in Finmeccanica Group: towards the Stakeholder University 107
The start-up: the Finmeccanica Virtual Corporate University..... 108
Towards the Finmeccanica Stakeholder University: open issues... 112
Conclusions 118
References 118

Chapter 6. Expanding the Value of Corporate Universities: The Stakeholder Approach..... 121
by Mark Allen
Introduction 121
The history and evolution of corporate universities 122
Corporate University Functions..... 124
Traditional University Functions..... 128
Stakeholders of Traditional Universities 128
Stakeholders of Corporate Universities..... 129

The Value of the Stakeholder Approach 132
Summary and Conclusion 134
References 135

Chapter 7. Corporate Universities 2.0: The Future Networked Learning Organization 137
By Jeanne C Meister
Introduction 137
A Worldwide Shift in Demographics 138
Corporate Universities: A Target in Tough Economic Times? 138
The Era of Social Computing Arrives 139
Personal Broadcasting Leads To Personalized Learning 141
Sophistication of Mobile Phones Allows For Mobile Learning 141
The Emergence of Corporate Universities 2.0 142
Corporate Universities 2.0 In Action 144
Corporate Universities 2.0: Blending Learning with Working 144
Corporate Universities 2.0: From Cerner Virtual University To Cerner Knowledge Works 145
Becoming A Corporate University 2.0 147
Implications for Corporate Universities 148
References 149

Chapter 8. Emergent management learning in dynamic learning networks 151
by Walter Baets
Complexity emergence and a quantum interpretation of business 151
On to biology: Sheldrake and "implicit order" 152
Sustainability 153
Sustainability principles 156
A holistic model 157
Applied to management: A systemic management interpretation 159
Leadership: the driving force of sustainability 163
Summary description of the Innovation School and its conceptual model: sustaining learning networks in corporations 164
Some conclusions 166
References 168

Glossary 169

Index 177

Preface

When a firm moves into a new technology or market base developing complex and integrated products and solutions, it needs dynamic process of building capability over time. For this purpose firms have to rely on information and knowledge, creating new managerial roles to deal with uncertainty and competition. Their adaptive dynamic aim is to create sustainable sources of competitive advantage: as a consequence, firm centric model of learning are inadequate and should be more and more supported by network learning and learning networks. The diffusion of networks is generating a new context for organisational learning. These processes are therefore "sine qua non" condition for firm's strategy and become important vehicles for managing emergent strategies. And that is where this book comes in, presenting an integrative view on dynamic learning networks. Just as the focus of computing in recent years has moved from the node to the network, learning networks possess the potential to harness greater efficiencies and greater capacities (both in individuals, organizations and inter-organisation). In other words firms must develop processes to:

- integrate learning that occurs at the core and multiple edges of organizations
- combine learning that takes place at individual, organizational and inter-organizational and network levels
- organize themselves in networked form to manage the uncertainty and the complexity of the competitive environment.

By connecting people in networks, we can leverage learning in organizations, and by networking organizations, we can exponentially expand the reach and value of learning. To compete "on the edge" organizations have to learn "on the edge".

The book represents an attempt to provide a network perspective of organisational learning to drive dynamic competition through extended firm learning processes, exploring enabling and constraining mechanisms associated with effective learning in networks.

This book draws on topics related to discussions from the 10th session of the Advanced International Summer School on *"Managing Uncertainty and Competition Through Dynamic Learning Networks"* held in July 3rd–5th, 2008 in Ostuni and organised by the e-Business Management Section of the Scuola Superiore ISUFI (University of Salento, Italy).

Target Audience

Dynamic Learning Networks: Models and cases in action is a book targeting three categories of readers:

- business strategists providing them with a conceptual framework and a set of guidelines for the development of learning networks as a key process to lead change in complex environments;
- academics through insights and best practices for analyzing learning processes in terms of different literature perspectives including organizational development, strategic management and organizational learning;
- practitioners and consultants, providing them with a holistic view of emerging trends in dynamic learning processes arising in network context, whose nodes are partners, suppliers, users, customers, employees, teams and organization, depicting the emergence of the next-generation's learning organizations.

We are interested in hearing your comments about this book. In times of proliferating management bibles, roadmaps and receipts for success, our objective is to share with the reader our belief that in management, as in life, everything starts with Learning, Interaction and Networks. We hope you enjoy the journey.

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Lecce, February 2009

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- Modelling and development of learning environments enabled by Web 2.0 technologies.

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Introduction

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An integrated perspective on Dynamic Learning Networks

This book is about the extended firm learning processes, formal or informal, enabling mechanisms associated with effective learning in networks to drive dynamic competition. Hence, we argue that learning network perspective seems to describe organizational learning better than its predecessors.

Organizational learning is one of the most important sources of a sustainable competitive advantage that companies have as well as an important driver of corporate performance (Stata 1989). Given the turbulent environments that organizations work within, continuous learning is a key driver of their ability to remain adaptive and flexible – that is to survive and effectively compete (Burke et al. 2006). Organizational learning has emerged as one of the most promising concepts in strategic management literature in late 1980s in relation to the concept of competitive advantage. Nevertheless, the network perspective of organizational learning stretches much further and is embedded also in different schools of thought.

To this purpose we'll take in consideration the concept of "Dynamic Learning Networks" composed by three words; for coming up with a complete definition and a general meaning, we'll start from analyzing the single concepts.

A. Romano and G. Secundo (Eds.), *Dynamic Learning Networks: Models and Cases in Action*, DOI: 10.1007/978-1-4419-0251-1_0,

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What is Learning?

Learning is one of the primary means through which individuals and organizations innovate and change. Modern economies can be characterized as learning economies in which knowledge is the crucial resource and learning is the most important process (Lundvall and Johnson 1994; Lundvall 2002). By learning we mean the acquisition and use of existing knowledge and /or the creation of new knowledge with the purpose of improving economic performance (Boerner et al. 2001), an increase in knowledge that is, an increase in the capacity to produce desired results through reflection and experience (Senge 1990).

Continuously in this century, Western economies have undergone a transformation from bulk-material manufacturing to design and use of technology—from processing of resources to processing of information, from application of raw energy to application of ideas. As this shift has taken place, the underlying mechanisms that determine economic behavior have shifted from ones of diminishing to ones of *increasing returns*. Increasing returns are the tendency for that which is ahead to get farther ahead, for that which loses advantage to lose further advantage. They are mechanisms of positive feedback that operate—within markets, businesses, and industries—to reinforce that which gains success or aggravate that which suffers loss. Increasing returns characterize the knowledge based industries and learning processes can feed the underlying mechanism at different ontological levels: individual, organisational, and at inter-organisational level. The purpose of our book is to go further these levels to explore also the network level.

Learning at individual level: In the literature of innovation, strictly speaking only individuals posses the capacity to learn, to acquire new knowledge and to increase their capacity for effective action (Senge 1990).

Learning may happen through different informal modalities:

- learning by doing: it entails learning as a product of experience that takes place during activity since it usually occurs through the attempt to solve a problem (Arrows1962).
- learning by interacting*:* it refers to the idea that new knowledge accrues through social interaction and cultural experience, and that learning and social negotiation are inseparable practices.
- learning by using: is the process by which the performance and maintenance characteristics of a new product are determined through feedback from customers who have extensive experience with the product (Rosenberg1982).

- learning by searching: it entails searching and discovering the optimal design of a new product or process or service (Rosenberg 1982)
- learning by R&D or scientific learning: it entails acquiring knowledge about the fundamentals laws of science and nature. As the stock of scientific learning increases, the costs of undertaking science-based invention decreases (Rosenberg 1982).

Among the mentioned learning modalities occurring in an informal way, it's necessary to include also the more formal investments in training to increase individual productivity and skills of employees. To this end, a growing number of private organizations have begun to shift from the traditional old-time training events to creating a continuous learning culture where employees learn from each other and share innovations and best practices. The challenge is to activate learning processes at organizational level to transform individual skills and competencies in organizational competencies. These considerations are at the basis for the concept of organisational learning.

Learning at organizational level: If from one side it can be argued that it is only the individuals within an organisation who can learn, from the other side there is some evidence for viewing organisations as capable of learning behaviours which extends beyond that of individuals and which emulates key activities like experimentation, association and remembering (Hedberg 1981). The literature on organizational learning is now voluminous, and topic has been explored from a variety of angles. Organizational learning can be thought as the capacity of a firm to facilitate knowledge creation or acquisition, disseminate it throughout the organization, and embody it in products, services and systems (Nonaka and Takeuchi 1995). Marengo (1992) describes organizational learning as the process for creating new competencies and increasing the existing ones. Organisational learning is the process through which the organizations transform themselves in Learning organization, i.e. organizations with the capacity to support individual knowledge creation in specific directions through adaptive and generative learning processes (Senge 1990). Adaptive learning is focused on making known and accessible knowledge that already exists, and contributes to nurture continuous improvement and problem solving processes. Generative learning is focused on creating new knowledge and commercializing it as a valuable product, service and/or a new business.

With this aim, a lot of companies have traditionally created the "Corporate Universities" (CUs), specific intra-organizational units focused on developing education programs, suitable for the core competences needed and targeting especially internal employees. The term "University" has been used both as a metaphor for outlining the importance of its learning initiatives and for branding the organizations' educational programs,

courseware and processes. The term "Corporate" has given it because of it also offers learning contents to suppliers and customers.

A research study conducted on 100 learning departments and organizations (Meister et al. 2005) found that companies with high-performance learning processes return better results compared to their competitors and industry peers in terms of productivity measured by sales per employee (27% greater), revenue growth (40% higher), and net income growth (50% higher).

However, most contributions of organizational learning have been focused on learning within the organizations, whereas the extension of learning processes outside a company's boundaries, at inter-organisational level has received few attentions.

Learning at inter-organisational level: The few studies that cover external learning either deal with the diffusion of information in highly general way (Levitt and March 1998) or focus only on strategic alliance and learning with R&D activity (Dodgson 1993).Various scholars have recognized that inter-organizational learning is critical to competitive success, noting that organizations learn by collaborating with other firms as well as by observing and importing their practices (March and Simon, 1958: 188; Powell et al. 1996). For example, von Hippel (1988) found that a firm's customers and suppliers were its primary sources of innovative ideas. He argues that a production network with superior knowledge transfer mechanisms among users, suppliers, and manufacturers will be able to 'outinnovate' networks with less effective knowledge-sharing routines. In fact, although there has been considerable theoretical discussion on organizational and network learning, there is little empirical research to date. These considerations are useful for the second word *Network.*

What is Network and Learning in network?

Most contributions of organizational learning have focused on learning within the organizations, whereas learning in networks and networks that learn has received much less consideration. A 'network' is a collection of connected entities, where a connection is something that allows one entity to send a signal to another entity (individual, team, organization, etc). The metaphor of "the network" is seen by some researchers as the fundamental organizational form in today's society (Castells 2004; Hargreaves 2004) Perhaps the most well known author to place networks at the centre of modern societies is Manuel Castells (1996, 2000). Castells has written about the architecture of relationships within and between networks, the

ways that they are enacted by information technologies, which configure the dominant processes and functions in our societies.

When networks are properly designed, they reliably facilitate learning and to acquire competitive advantage for network nodes. This because through the process of interaction and communications (learning by interacting), the entities that constitute the network will form a mesh of connections. Knowledge is embedded in this mesh of connections, and therefore, through interaction with the network, the nodes can acquire the knowledge. It is the organization of the network that supports learning, and if the network is designed appropriately, it will organize itself in order to best support learning. Network Learning is defined as learning by a group of organizations a group. The learning entity is the network collectively, and the learning outcomes are indicated trough changes to properties of the network such as network level or network wide routines, strategies, processes and systems (Knight 2002). In this sense we intend 'Network' as 'Knowledge Network' or "Learning network". In this case network offer the organizational context in which learning occurs or the supporting infrastructure for learning.

Why Dynamic?

Finally, the third word is *Dynamic*. Several network theories suggest explicitly or implicitly that networks evolve through time; this implies the concept of dynamic network. But also the temporal dimension of the learning process in network should be considered with necessary attention.

The second point is strictly connected to the today's competitive environment, characterized by dramatic and radical innovations. These coupled with the high frequency with which they occur have created shorter business cycle that implies competences obsolescence. Thus organizations are asked to continuously and dynamically update the knowledge and experience of people. In analogy with product, that can be seen as a mix of functionalities and technologies used to manufacture it, also the knowledge and experience, at the foundations of competencies, have a life cycle and can become obsolete. When a competence is declining emerge the need to develop a new competence (or enhance/update the old one). This requires continuous learning processes for developing competencies via real experience rather than through purely theoretical knowledge transfer. This is only one of the challenges of developing peoples' competencies in a dynamic, timely and effective manner. Learning Networks can provide a context for this.

Defining Dynamic Learning Networks

After the considerations about the single words composing the concept of "Dynamic Learning Networks" we can define it as:

'a network formally or informally set up for the primary purpose of enabling any kind of learning to take place over the time for building capabilities, managing change, gain competitive advantage in the perspective of learning organization, eventually supported by the information and communication technologies''.

This definition implies a number of features for Dynamic Learning Networks, besides the features already described:

- They can be formally or informally established and defined especially at intra, inter-organisational level.
- They have as primary learning target some specific learning/knowledge/capabilities which the network is going to enable.
- They have a structure for adaptation and change management with boundaries defining participation, in case of network formally set up.
- Learning processes arise especially from the participative and acquisition perspective involving all the network nodes.
- Measurement of learning performances include to which extent a network is supporting the learning processes and how the network structure is addressing the major changes in different situations.

Behind this definition lies a challenge. The key feature which learning networks share is an ability to adapt and learn to deal with their rapidly changing and uncertain environments. Research suggests that there are two important components involved in such learning: the first involves the accumulation and development of a core knowledge base - the 'core competence' - which differentiates the firm from others and offers the potential for competitive advantage. Acquiring this is not simply a matter of purchasing or trading knowledge assets but the systematic and purposive learning and construction of a knowledge base (Pavitt 1990; Prahalad and Hamel 1994; Tidd et al. 1997; Teece 1998).

The second component is the long-term development of a capability for learning and continuous improvement across the whole organisation. Recognition of this need has led to growing emphasis on the concept of 'learning organisations' and on the mechanisms through which this capability can be developed (Senge 1990; Nonaka 1991; Garvin 1993; Leonard-Barton 1995).

Dynamic Learning Networks design variables and typology

Several theories suggest that there are several influencing variables which need to be considered when setting up a learning network, and these include the following (Bessant and Tesekouras 2001):

- *Purpose* - what is the main learning target for which the network is being set up? A clearly defined and shared sense of purpose with a focused and measurable learning target need to be defined.
- *Nodes* - who are the members and nodes of the network going to be? Although learning networks are essentially 'open' in character, there is a value in defining boundaries, particularly in terms of providing focus and coherence.
- *Structure* - is it a 'formal' network with a coordinating hub or a radial network, diffusing information from the centre, or an informal network model? A number of different models exist for learning networks and much appears to depend upon the type of learning involved. Social networking practices also enable the creation of virtual communities, as well as the building of relationships and sharing of common interests and ideas. These social experiences are very often the foundation of learning.
- *Roles* - It is also important to identify and define roles within the network, particularly those of coordinator and facilitator. Networking as a process may not happen without some assistance, both in terms of arranging relevant mechanisms and also in energising and motivating learning and in enabling effective interchange of experience to take place.
- *Learning processes*: the primary feature of learning networks is that they all use the principle of shared learning to enable capacity development. All the learning processes can be interpreted according the acquisition metaphor and the participation metaphor. The former represents a passive-receptive view according to which learning is mainly a process of acquiring information, while the latter perceives learning as a process of participating in various cultural practices and shared learning activities (learning by using, by interacting, by doing etc.)
- *Content of learning* - how will the learning be diffused? Assuming that there is a clear focus to what needs to be learned and exchanged, it should include multiple informal and formal sources that are media rich, interdisciplinary, and global in reach. *Learning patterns* should not be fixed but dynamic, consisting of “bite-sized” modules, interdisciplinary in focus, and blending formal and informal learning.

- *Enabling tools and techniques* – all the networks nodes should be offered multiple opportunities for open, social, peer to peer, and multi-faceted forms of visual, face – to face verbal communication, using multiple media types to achieve immediacy.

Tab i. shows the Learning Networks designing variables.

Table i. Key variables of Dynamic Learning Networks

Aspect	Trend
Purpose	• Enabling any kind of learning take place over the time • Increasing knowledge capacity over the time • Gather complementary assets • Managing rapid change • Sustain competitive advantage • Creating Managerial Competencies • Developing synergies with all the network's nodes
Nodes	• Employee, partners, stakeholder, learner, consumer, supplier
Structure	• Formal or informal network • Virtual or physical network • coordinating hub or a radial network • Knowledge network
Roles	• Facilitator • Coordinator
Processes	• Interactive learning, learning by doing, learning by using, learning by searching, generative learning, adaptive learning, shared learning, situated learning etc.
Content	• Business processes driving curricula design • Integrated rather than functional/specialist knowledge • Hands-on, action and on-demand learning approaches • Dynamic and interdisciplinary learning patterns
Tools	• Collaborative learning tools • ICTs for managing content, competencies and communities • Social computing, open source and web 2.0 applications • Virtual meeting and campus • eLearning, podcasts, wikis, blogs

They have in common some of the variables described and independently from the learning network typology, it's possible to identify the following 'principle' characterizing them (Downes 2007):

First, diversity. Entities in a network should be diverse. In a society, this means involving the widest possible spectrum of points of view. In a human mind, this means being exposed to a wide spectrum of experiences. Diversity allows us to have multiple perspectives, to see things from a different point of view.

Second, and related, autonomy. Each entity operates independently of the others. This does not mean that it operates without input, but rather, it means that it operates according to an individual and internal set of principles and values.

Third, interactivity, or connectedness. The knowledge produced by a network should be the product of an interaction between the nodes. Web 2.0 software supports the interaction that happens between individuals and the other network's nodes. And so, too, the personal learning environment supports not just content consumption but interaction and communication.

Fourth, and again related, openness. Each entity in a network must be able to contribute to the network, and each entity needs to be able to receive from the network. Openness is what makes interactivity possible; barrier that make it difficult or impossible to communicate within the network limit the network's capacity to learn. The Information and Communication Technologies (ICTs) and the emerging Web 2.0 software can provide a support for this.

Different typology of Networks will be described and analyzed in this book. Among these we can we can remember the Hastily Formed Network, Social network, Practitioner Network, Corporate University, Customer – Supplier Network, Knowledge Network, Stakeholder University, Corporate University 2.0 etc.

Structure and contents of the book

Starting from the definition of "dynamic learning networks", from their variables and principles, we can talk about networks in two distinct ways: first, we are talking about the *use of networks for manage change and learning*, and second, we are talking about *networks that learn*. Though these may seem to be very distinct, the central thesis of "learning networks" as a theory is that these two things are strictly interrelated.

An integrated perspective on dynamic learning network consists of the following main components:

(a) Network structures: organizational forms mainly based on network structure for managing change and uncertainty; they offer many opportunities for learning to take place (by interacting, by sharing ideas, etc.) but such learning is just a "by product" of the main purpose of the network.

(b) Learning in Networks: development of formal or informal learning processes and initiatives in the context of networks to gain capabilities and competencies, involving a plurality of nodes, such as employees, manag-

ers, suppliers, partners and organizations belonging to a formal or informal network. The primary purpose of these networks is to enable any kind of learning to take place.

To address the integrated perspective about "Dynamic Learning Networks" the book is organized in the following eight chapters:

- *Chapter 1:* Networks for manage change in international commercial organizations
- *Chapter 2:* The Relevance of Organizational Learning for High Performing Social Networks
- *Chapter 3*: Illycaffè and the Brazilian espresso knowledge network
- *Chapter 4:* Learning dynamics within the Solutions Business Model
- *Chapter 5:* Towards "Stakeholder University" as Dynamic Learning Network: the Finmeccanica case
- *Chapter 6:* Expanding the value of Corporate University: the Stakeholders approach
- *Chapter 7:* Corporate Universities 2.0: the future of networked learning organization
- *Chapter 8*: Emergent management learning in dynamic learning networks.

Fig i shows the organization of the Chapters with the key issues

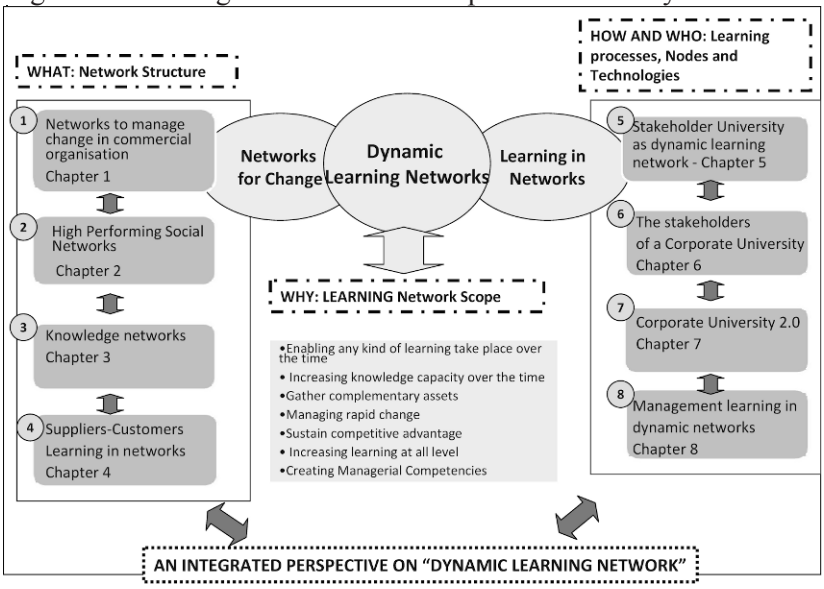


Fig. i. The Book structure

Environmental changes lead organizations to face the challenge of adapting their core practices or risk decline and failure. Network theory provides researchers with one method which to study organization adaptation. Traditional models of adaptation, in which an organization on its own "confronts an environment and decides on responses in social isolation," display certain shortcomings that network theory helps to address. Rather than acting alone during periods of uncertainty in which organizations need to make decisions that will affect its core practices, there is evidence to suggest that an organization will use its network in order to access different sorts of information, "affecting its ability to recognize and respond to environmental threats" (Kraatz 1998). Chapter 1 will move from these points key points showing the cases of commercial organizations.

Chapter 1: Networks for manage change in international commercial organizations *by Tom Osorio, Safety&Operations, BP, UK*

Competitive advantage as well as survival requires commercial organisations to manage major change successfully, as well as performance improvement in stable environments. Particularly during periods of rapid global change, there is great potential in the use of networks to exploit major changes. The chapter uses the concept of networks to assess effective organisational responses to major change. Three case studies examine differing management interventions focusing on the role of internal and external networks influencing organisations in the following examples:

- Response to a fundamental change in the nature of competition and technology control that required leadership to initiate new blue-sky research aligned with long-term strategy whilst redirecting traditional research and technical support activities: Shell Upstream.
- Delivering improved risk management and other requirements of new capital sources by exploiting existing informal organisational networks: Scott Wilson case.
- Change of formal organisations focus and exploiting informal networks to apply best practice to improve management of high impact risks: Bp case.

This chapter will demonstrate the insight provided by networks when studying major change as well as their power to deliver fundamental, rapid, sustainable change within traditional organizations in addition to individual and organisational learning.

When changes in the competitive environment become urgent in "crisis situation", Hastily Social Networks appear to be the primary vehicles for

the way contemporary work gets done in and across the organization. This is the main thesis addressed in Chapter 2.

Chapter 2: The Relevance of Organizational Learning for High Performing Social Networks *by Jeff Clanon, Society for Organizational Learning, Cambridge, USA*

Over the past two decades, Organizational Learning concepts, methods, and tools have been extensively researched and applied and are now taught around the world. However, both the research and application have been conducted largely within the boundaries of an organizational context. The time was ripe for new thinking about organizations as well as thinking about new ways to effectively lead and manage them.

The advent of and ever-accelerating use of the Internet, along with the proliferation of social networking websites such as Facebook, MySpace, and Twitter, demonstrate an increasingly interconnected world. Many organizations find themselves in situations where they need to accomplish work effectively across organizational, geographic, and often cultural boundaries. Based on recent research, social networks appear to be a primary vehicle for accomplishing contemporary work in and across organizations. This research points to the increasing impact and importance of networks that extend beyond traditional organizational boundaries. Given these dramatic changes in life and work contexts, what is the relevance of organizational learning in today's world? More specifically, what organizational learning concepts, methods, and tools are most relevant for enabling effective networks? This chapter will address these main issues.

Change equals learning and survival must be based on a process of constant learning also at an inter-firm level (Bessant and Tsekouras 2001). A real case of Dynamic Network supporting the knowledge creation and exchange processes will be provided in Chapter 3.

Chapter 3: Illycaffè and the Brazilian espresso knowledge network *by Ingrid Rauscher, ADS Assessoria de Comunicações, São Paulo, Brazil and Pierpaolo Andriani, eBMS- Scuola Superiore ISUFI, University of Salento, IT.*

In this chapter, the authors describe key points in the development of a network generating and disseminating knowledge gained from an initiative that aimed to enhance the quality of Brazilian coffee, undertaken by Italian roaster illycaffè. The emergence of a complex network with numerous

nodes, mostly suppliers in Brazil, is described and studied as it relates to the information and knowledge economy, and to the learning process that ultimately led to the creation of wealth for those linked by the network, who gained access to new markets and higher prices as rewards for better quality. In parallel, the text implicitly shows the "de-commoditization" process triggered by initiatives and strategies shaped within a modern network geared to sharing and disseminating information and knowledge.

Even if the work of Nonaka (Nonaka and Nishiguchi 2001) does not leave the supplier networks (customer-suppliers networks) unconsidered, it does not reflect sufficient attention to some aspects of the learning dynamics: what specific capabilities a firm has to acquire to become a solutions provider; how such capabilities can support the repeatability of high-value integrated solutions; and what is the interplay between intra-and inter-organizational learning. All these points will be addressed in Chapter 5.

Chapter 4: Learning Dynamics within the Solutions Business Model *by Stefano Miraglia, eBMS-Scuola Superiore ISUFI, University of Salento, IT and Andrew Davies, Imperial College London, UK.*

This chapter provides an overview of the learning dynamics associated with the provision of integrated solutions in the capital goods sector. In that context, the typical configuration of integrated solutions is that of a project-based business model adopted by a firm able to develop and hold a network of industrial relationships with suppliers and competitors around the specific needs of a customer firm. Within this "solutions network", multiple sources of knowledge trigger learning processes that are intra- and inter-organizational at the same time.

The learning dynamics are interpreted holding, first, the supplier perspective, aiming to understand: what specific capabilities a firm has to acquire to become a solutions provider, how such capabilities can support the repeatability of high-value integrated solutions, and what is the interplay between intra- and inter-organizational learning. Subsequently, the customer perspective is considered, by emphasizing the central role it plays in reorganizing for outsourcing, reporting new categories of needs from the interaction with an increasingly complex environment, and feeding the supplier's learning process. Finally, the reflections on the heightened environmental complexity lay the basis for the introduction of the new concept of "contingent proactivity" as a core capability of the modern networked enterprise.

Another perspective of learning network regards the development of life-long learning processes within networks composed by individuals (employees) to maintain updated the core competencies of the emerging networked organisations. To this end, a growing number of private organizations have begun to shift from the traditional old-time training events to creating the "Corporate Universities" (CUs), intra-organizational units focused on developing specific education programs, suitable for the developing the needed core competences. Following the organizations' evolution trends towards business networks, these "Corporate Universities" are evolving towards networked configurations, the so-called Stakeholder Universities (SUs). This evolution will be addressed in Chapter 5 and in Chapter 6. The analysis will be concluded in Chapter 7, presenting the innovative Web 2.0 tools to create the Corporate University 2.0 and in Chapter 8 with a management model to address workplace learning in Stakeholder University.

Chapter 5: The "Stakeholder University" as Dynamic Learning Network: the Finmeccanica case *by Roberto Maglione, Finmeccanica Corporation, IT and Giuseppina Passiante, eBMS-Scuola Superiore ISUFI, University of Salento, IT*

Since last 90s, the new global and knowledge-based economy has determined radical changes within institution and enterprises. The current complex and dynamic competitive environment has induced organizations to become more flexible, by developing both internal and external networks, for sharing risks and for gathering complementary assets. These new organizational structures call for new competencies: indeed the so called "core competence" has become a strategic immaterial asset, although rapidly obsolete. To maintain them updated, "life-long learning paradigms" have emerged. With this aim, companies have traditionally created "Corporate Universities", specific intra-organizational units focused on developing specific education programs, suitable for the core competences needed. Following the evolution trends towards business networks, these "Corporate Universities" are evolving towards networked configurations, the "Stakeholder Universities". In this chapter the authors present a model of "Stakeholder Universities", undertaken by a big company Finmeccanica, pointing out its main initiatives, leaning processes and results obtained.

Chapter 6: Expanding the value of Corporate University: the Stakeholders approach *by Mark Allen, Pepperdine University, USA*

This chapter discusses the concept of corporate universities as stakeholder universities. It begins with a brief history of corporate universities. This is followed by a definition of corporate universities and a discussion of the elements of the definition. A list of the many different corporate university functions is presented and discussed, including a discussion of next generation corporate university functions. Among these is wisdom management, a systematic process by which an organization manages how its employees use and apply knowledge.

The stakeholders of a traditional university are examined. This leads to a look at the stakeholders of corporate universities. The basic stakeholders of employee, department, and organization are expanded to include business partners (vendors, suppliers, etc.), customers, and society as a whole. These stakeholders not only benefit from corporate university offerings, but provide a way to leverage learning beyond the traditional organizational benefits. By opening systems to expanded groups of stakeholders, organizations foster networked learning and are able to transform their corporate universities into stakeholder universities which can fully leverage learning as a driver of value.

Chapter 7: Corporate Universities 2.0: the future of networked learning organisation *by Jeanne C. Meister New Learning Playbook, USA*

Today's "knowledge workers" need to quickly assimilate new information and knowledge critical to performing their jobs. Because no one can know it all in today's complex world, they need to collaborate with others like never before. To do so, leaders of corporate universities must assess their current operations in today's social media landscape where social computing personalization and mobility dominate the business environment. In this networked age, corporate learning must adapt or die. This means companies must redefine their corporate university to leverage the latest collaboration tools and technologies to enable learning that is both more personalized as well as more unstructured. Corporate Universities 2.0, as the name might suggest, use social media Web 2.0 tools such as blogs, wikis, podcasts, online forums and video sharing sites to seamlessly blend learning and working. These trends will be explored in depth along with the implications for how companies must re-design, re-think and re-configure their corporate universities into corporate universities 2.0.

Chapter 8: Emergent management learning in dynamic learning networks *by Walter Baets, Euromed Marseille, Ecole de Management, FR*

The foundational concepts in the complexity realm emerge from such fields as neurobiology, cognitive sciences, physics, and organizational theory. New developments in knowledge management such as connectionist approaches (complex adaptive systems) for the visualization of emergence give promising results. In fact, instead of causality, it appears that the networked economy is ruled by synchronicity (occurring together in time) in line with findings in quantum research. Once holism, constructivism and emergence are accepted as fundaments of a new paradigm in management, a paradox, comparable to the one in science, emerges. Despite the theory of relativity and quantum mechanics, almost the whole scientific management community is still focused on Newtonian principles, that is to say fixed space and time.

Causality and learning seem to be, indeed, opposed concepts. Learning can only take place in disruptive situations, and those could be provoked in dynamic learning networks. The contribution of this chapter is to discuss and develop a possible way to create and support dynamic learning networks in companies, and between companies with their clients and suppliers. It shares a particular application and its use presenting the model of Innovation School.

Key features and guiding principles

The integration of organizational learning, organizational development, strategic management and network perspectives may be considered as a distinctive feature of the book, as well as the description of real cases and authors' direct observation in focusing topics investigated.

The main need that the manuscript addresses is to provide academics and company managers with an extended view of organisational learning and dynamic networks to manage change including evidence from real cases and different perspectives.

The integrated perspective and the meaningful contributions coming form the above mentioned fields are addressed through the following key research questions and findings.

- What is the relationship between major change and organisational systems? Is it appropriate for management to rely on mechanistic use of co-

re organisational controls and hierarchy during major change? Are networks a practical concept to assist understanding in the business world?

The guiding principle is that rapidly escalating change and chaos in highly dynamic and complex environment produces multiple states of coexisting temporary stability and instability. These conditions require flexibility, speed and innovation and above all the necessity for the firms to reorganize themselves by making rapid and effective changes. The time was ripe for new thinking about organizations as well as new ways to effectively lead and manage them. Particularly during periods of rapid global change, there is great potential for commercial organisations in the use of networks to respond to and exploit major changes. Case studies from the commercial world will examine the concept of networks to assess effective organisational responses to major changes.

- What organizational learning concepts methods and tools are most relevant for enabling effective networks? How was action mobilized throughout the network?
- What preconditions are necessary for these "hastily formed networks" to be effective? Which leadership capabilities are required to manage effectively Social Networks?

The guiding principle is that in "crisis situation", Hastily Social Networks appear to be the primary vehicles for the way contemporary work gets done in and across the organization. Hastily Social Networks address how action is actually activated at all the levels of organizations discovering that the primary function of leaders is to mobilize action across multiple constituencies. There was significant and consistent data that showed that traditional hierarchical leadership was often ineffective in these urgent situations and, at times, was an obstacle for getting effective action accomplished.

- What are the success factors for developing a Knowledge Network in the Espresso Sector? How to develop and reinforce the dynamic learning process arising especially in the espresso network?

The guiding principle is that the strategic actions the illycaffè case to improve quality spontaneously evolved into a complex network, shaped not only by trade relations but also by the pride to belong to the top producers who were glad to share information and knowledge about their activities. The tissue of the illycaffè network in Brazil is created by the direct

trade between company and loyal suppliers of fine coffees and by the strong ties between all links in the coffee producing chain. It is enabled by efficient communication that generates and disseminates information and knowledge, and above all, it is nurtured by the mutual trust of all parties committed to improving the quality of coffee in Brazil. In this ecosystem, the transmission of information and knowledge is faster and more efficient, and is shared in a viral manner, as a "healthy epidemic", reaching all links in the large scale production chain.

- How to develop learning capabilities from suppliers and customers to become a solution provider? Which learning dynamics are developed at intra- and inter organisational level?

The guiding principle is that the analysis of the learning dynamics within integrated solutions has revealed the interplay existing between two levels of organizational learning. First, at the intra-organizational level of the integrator firm, the learning process supports the acquisition and development of a set of four fundamental capabilities: systems integration and operational services, complemented by consulting and financing services. Second, at the inter-organizational level of the relationships between systems integrators and supply and demand actors, far more sophisticated processes of knowledge creation, acquisition and integration happen. Intense and long-term exchanges of information and collaboration with (co)suppliers and competitors generate new knowledge, which is internalized and compared with internal capabilities within a second loop of supply-driven learning. At the intersection between demand and supply-driven learning, a process of knowledge integration occurs, which assigns system integrator the role of mediator of industrial relationships.

- Which are the specific intra-organizational units supporting continuous learning processes to develop the managerial competencies? Which are all the stakeholders of a Corporate University?
- Which are the most promising Web 2.0 technologies supporting a Corporate University? And they can help the transition from corporate university to Stakeholder University?

The guiding principle is that the "Corporate Universities", interpreted as specific intra-organisational learning unit for developing employees competencies, are evolving towards networked configurations, the so-called Stakeholder Universities (SUs) (Bleiklie and Kogan 2007). The SUs are conceived as hubs of learning networks, with learning processes distrib-

uted also outside of the company, including the customers, the suppliers, the partners and the entire society. SUs have then become skilled not only at creating knowledge for the improvement of individuals learning, as well as at speeding up the learning processes of a whole organization, but also at modifying their behaviors to reflect their new learning needs. They have radical changes in learning processes, learning technologies and learning contents. In short, SU have become themselves learning organizations (Dill 2004). By opening systems to expanded groups of stakeholders, organizations foster networked learning and are able to transform their corporate universities into stakeholder universities which can fully leverage learning as a driver of value. For realizing this, Web 2.0 technologies guide some implications how companies must re-design, re-think and re-configure their corporate universities into corporate universities 2.0 and Stakeholder University.

- Which management principles are useful to manage learning in networks? How learning can be improved at network level?

The guiding principle is that if we accept that reality is self-organised on the basis of synchronicity, non-locality and entanglement, the responsibility for learning can only be shifted to the learner. Projects, courses, any real life activity, or any kind of experience could be of help in order to develop management competencies. The focus is no longer the activities, but the development of competencies and to realize this no formal curriculum is necessary. Workplace learning should typically be such a kind of a setting, just as much as corporate universities all together. However today, most of those corporate universities are mainly copies of what business schools do currently, copying precisely all the weaknesses that for business schools are due to the limitations of structure, accreditations, etc. Companies have a much larger freedom to really organize workplace learning, according to the principles of the *Innovation School*. This network concept of learning, this paradigm shift to one of interconnectedness, entanglement and meaning is a sound basis to move from a corporate university towards a stakeholder university.

References

Arrow KJ (1962), The Economic Implications of Learning by Doing, The Review of Economic Studies 29(3):155-173.

Bessant J, Tsekouras G (2001) Developing learning networks. AI & Society 15: 82-98.

Bleiklie I, Kogan M (2007) Organization and Governance of Universities, Higher Education Policy.

Boerner CS, Macher JT, Teece DJ, (2001) A Review and Assessment of Organizational Learning in Economic Theories in Dierkes M, Nonaka I, Child J, Antal AB (Eds.) Handbook of organizational learning and knowledge, Oxford University Press.

Burke SC, Stagl, KC, Klein C, Goodwin GF, Salas E, Halpin SM (2006). What type of leadership behaviors are functional in teams? A meta-analysis. Leadership Quarterly 17:288-307.

Castells M (1996, 2000 2nd.ed.) The Rise of the Network Society. Oxford, UK, Blackwell Publishers.

Castells M (2001) The Internet Galaxy: Reflections on the Internet, business, and Society. Oxford: Oxford University Press.

Castells M (2004) Afterword: Why networks matter. In: McCarthy H, Miller P, Skidmore P (eds) Network logic: Who governs in an interconnected world? London, England: Demos.

Dill DD (1994) Rethinking the Planning Process, Planning for Higher Education:8-13.

Dodgson M (1993) Organizational learning – A review of some literatures. Organization Studies 14(3):375-94.

Downes S (2007) Learning Networks in Practices in Ley D. (ed.) Emerging Technologies for Learning, National Research Council of Canada.

Garvin D (1993) Building a Learning Organisation, Harvard Business Review. July/August:78-91.

Hargreaves DH (2004) Networks, knowledge and innovation. In: McCarthy H, Miller P, Skidmore P (eds) Network logic: Who governs in an interconnected world? London, England, Demos:79-88.

Hedberg B (1981) How organizations learn and unlearn. In: Nystrom PC, Starbuck WH (eds) Handbook of organizational design. London: Cambridge University Press.

Knight L (2002) Network learning: Exploring Learning from interorganisational Networks. Human relations 55(4):427-454.

Kraatz A, Matthew S (1998) Learning by Association? Interorganizational Networks and Adaptation to Environmental Change. Academy of Management Journal 41(6):621-643.

Leonard-Barton D (1995). Wellsprings of Knowledge: Building and Sustaining the Sources of Innovation. Harvard Business School Press, Boston, MA.

Levitt B, March JG (1988) Organizational learning, Annual Review of Sociology 14:319–340.

Lundvall BA, Johnson B (1994) The learning economy, Journal of Industry Studies 1(2):23-42.
Lundvall BA (2002) Innovation, growth and social cohesion: The Danish model, Cheltenham, Elgar Publishers.
March JC, Simon HA (1958) Organizations. NewYork:Wiley.
Marengo L (1992) 'Coordination and organizational learning in the firm' Journal of Evolutionary Economics 2(4).
Meister JC, Andrews J, Kraack T (2005) Increasing the Business Impact of Learning: Lessons from High-Performance Learning Organisations. In: Paton R, Peters G, Storey J, Taylor S (eds), Handbook of Corporate University Development, Gower Publishing Limited.
Nonaka I (1991) The Knowledge Creating Company, Harvard Business Review. November-December: 96-104.
Nonaka I, Takeuchi H (1995) The knowledge creating company: How Japanese companies create the dynamics of innovation. New York: Oxford University Press.
Nonaka I. and Nishiguchi T. (2001) Social technical and evolutionary dimensions of knowledge creation, in Nonaka I. and Nishiguchi T. (eds.) Knowledge Emergence. Oxford University Press, London.
Pavitt K (1990) What we Know about the Strategic Management of Technology, California Management Review 32:17-26.
Powell WW (1996) Interorganizational collaboration in the biotechnology industry, Journal of Institutional and Theoretical Economics 152:197–225.
Prahalad C, Hamel G (1994) Competing for the Future. Harvard University Press, Boston, MA.
Rosenberg N (1982) Inside the Black Box: Technology and Economics. Cambridge University Press, Cambridge.
Senge PM (1990) The fifth discipline: The art and practice of the learning. Doubleday, New York.
Stata R (1989) Organizational learning: The key to management innovation. Sloan Management Review 30(Spring):63-74.
Teece D (1998) Capturing Value from Knowledge Assets: The New Economy, Markets for Know-How and Intangible Assets, California Management Review 40(3):55-79.
Tidd J, Bessant J (1997) Managing Innovation: Integrating Technological, Organizational and Market Change. Wiley, Chichester.
Von Hippel E (1988) The Sources of Innovation. Oxford University Press, New York.

Chapter 1. Networks for manage change in international commercial organisations

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Introduction

Competitive advantage and long-term survival requires commercial organisations to manage major change successfully. During periods of rapid global change, there is great potential in the use of networks to actively exploit as well as simply respond to major changes.

This chapter uses case studies from the commercial world to examine four questions:

1. For commercial organisations what defines major change or an extreme event?
2. Is it appropriate for management to rely on traditional mechanistic organisational controls and hierarchy during major change?
3. Are networks a practical concept for understanding managing change in the business world?
4. Can leadership use networks to deliver successful organisational responses to major change?

The chapter will demonstrate the insight provided by networks when studying major change. Furthermore the potential of networks is illustrated as a practical concept to help management identify options that will deliver rapid and sustainable change in traditional organisations

A. Romano and G. Secundo (Eds.), *Dynamic Learning Networks: Models and Cases in Action*, 23
DOI: 10.1007/978-1-4419-0251-1_1,
© Springer Science + Business Media, LLC 2009

Major change and extreme events

Much of organisational theory focuses on managing organisations during steady state conditions. Management develop systems to optimise and compete in stable or gradually evolving circumstances. Mechanistic models aid understanding of such systems. Application is often in the form of integrated data management, well defined metrics, tools such as balanced scorecards, clear responsibilities and key performance indicators based on a simple mechanistic premise.

Major organisational change is a large field of study, principally focused on the 'project management' of the change itself. This ignores the challenge of managing ongoing operations and improvement after major changes. Typical major organisational change theoretical models assume that the conventional management structures above the unit subject to re-organisation are effective to implement the necessary changes at the required speed.

Management initiates a small proportion of major change internally; however, the most difficult organisational change is a response to external extreme events. Organisations evolve and managers optimise them to cope with the steady state conditions, and ensure they return to stability after minor perturbations from the mean. Concern with the average, and minimising variability is embodied by the 'six sigma' methodologies widely expounded by General Electric and other US corporations in the 1990s. Six Sigma methodology not only includes focus on reducing variability assuming binomial type distributions, but it also focuses on structured management control using value based measurement of improvements and hierarchical reporting through a pyramid type organisational structure. This traditional paradigm relies on the premise that the world is satisfactorily understood using Gaussian binomial distributions.

In contrast, a study, of over 80 natural and social phenomena (Andriani and McKelvey 2006) ranging from earthquakes to sales of books on Amazon, clearly demonstrated that rarer events are better-modelled using power law distributions. Such systems are better-modelled using complexity and networks. The consequence is that extreme events are more significant, being larger and more frequent than predicted by Gaussian based statistics. The impact of the subtle but important difference between Gaussian distributions and power law distributions, particularly when considering rare events has been well explained and popularised in 'The Black Swan', (Taleb 2000) page 281. Taleb demonstrates the impact of ignoring the importance of extreme events and describes how this lead

to the collapse of the firm 'Long Term Capital Management' (LTCM) in 1998. The response of banks and governments, clearly accepted the potential for the collapse of LTCM to have an impact well beyond that which simple analysis of the company's direct contracts would predict.

More recently, the knock-on effects on the world's financial system of the US mortgage markets and lack of interbank lending has again dramatically demonstrated the importance of extreme events. Conventional regulatory and commercial mechanisms proved inadequate to respond in these circumstances. In mitigating the potential consequences of the collapse of LTCM, financial organisations and governments quickly determined measures were required which had an influenced across many extensive networks, operating with complex relationships. Individual actions, had consequences which could not be predicted using traditional economic analytical tools or concepts.

These financial collapses have many similarities to the circumstances experienced by organisations in managing major change. Following major internal or external changes traditional management controls, theories and metrics are ineffective. Understanding and management theory is generally lacking in guidance of actions and interventions, which are effective across complex networks as required in these circumstances.

Theory does not recognise that the existing traditional formal organisational structures and systems are no longer effective. A simple and important characteristic of an extreme event is that it is outside the control capabilities of existing systems. Many organisational major changes take place in a situation, where it is vital to explicitly challenge the assumption that existing systems provide adequate control.

Organisational Complexity or Complication

Managers aim routinely state a desire to simplify their organisations, reduce bureaucracy, deliver clear reporting lines and delayer. Essentially this is eliminating complication of unnecessary interactions, controls and components. In contrast, most organisations also invest heavily in encouraging healthy 'complexity', supporting non-linear interaction of multiple systems. Examples include knowledge management, peer reviews, data-mining tools, staff-rotation and cross-postings. In the commercial world, authors often described such systems as networks, but the academic concepts associated with complexity are also powerful.

Elizabeth McMillan (McMillan 2002) compares classical science and complexity science models as:

Classical		**Complexity**
Linear	Vs	Non-linear
Hierarchical	Vs	Non-hierarchical
Reductionist	Vs	Holistic
Controlling	Vs	Self organising
Inflexible	Vs	Flexible
Uniform	Vs	Diverse
Centralised	Vs	Networked

Conventional management science paradigms and suggested approaches concentrate on optimisation and exploitation of the classical aspects. In practice, any real large organisation displays characteristics of both models, and practitioners should consider both paradigms. A recent study (Hirotaka et al. 2008*)* identified the degree to which many successful large organisations demonstrate the constructive reconciliation of apparently contradictory characteristics. A case study of Toyota showed that it:

- Moves slowly, yet takes big leaps
- Long term steady growth, yet paranoid
- Highly efficient, yet uses employees time in seemingly wasteful ways
- Frugal, yet splurges on key areas
- Insists on simple communications yet builds complex social networks
- Strict hierarchy yet encourages all employees to 'push back'

In a mechanistic organisational model, these characteristics appear conflicting and contradictory. A network organisational model helps understand, develop or exploit the apparent conflicting characteristics.

Further examination of these apparently contradictory characteristics exposes contrasting situations for application.

1. The former of each pair above is entirely appropriate for control in steady state conditions; the world of Six Sigma, limiting variability and controlled optimisation. Such an approach will be characteristic of a strict hierarchy.
2. The latter of each pair is required for helping the organisation to initially detect and then responding to extreme events, those which outside the control of the routine management systems. For example paranoid consideration of potential new technology or long term development, encouraging all staff to push back and devoting large amounts of time to ensuring rich communications across all levels.

It is also interesting to compare these characteristics with those of 'high reliability organisations', such as air-craft carriers, nuclear power stations and air traffic control. A study by Weick et al identified organisational culture and behavioural characteristics that increase propensity to disasters.

These included, ‘accuracy rather than plausibility, advocacy rather than active listening, constant rather than periodic exchange of information, complete consensus rather than partial working consensus’. (Weick et al. 1999)

The latter desirable characteristics identified by Weick aid the effective operation of networks as a vital complement to formal explicit control mechanisms. It is illuminating to consider these characteristics in the context of NASA’s organisational culture, which is widely recognised to have contributed to both Space Shuttle disasters. On January 28, 1986 O-ring damage, exacerbated by low temperatures on the launch pad, destroyed the Challenger. This followed a decision to launch which was against the clear advice of the relevant expert from Morton Thiokol who produced the booster rockets. On February 1, 2003, foam damage to a wing leading edge destroyed Columbia. Experts in ceramic tile damage assessed the foam damage, although the damaged area of the wing was not tiled.

Both disasters demonstrated poor active listening by leadership, an inability to act on the knowledge of a small group of experts. Decisions by senior managers were supported by a majority of staff that lacked relevant expertise. All these factors meant that NASA did not detect that circumstances had moved beyond the norms of prior experience and existing conceptual models. Leadership ignored widespread concerns within the support team, which engineers could not quantify. Tragically, poor management of networks can cause extreme events, as well as hinder organisational response to them.

Case studies now describe look at practical actions which leadership used to exploit networks and organisational ‘complexity’ to successfully major change.

Research challenges in the hydrocarbon exploration industry: Shell case study

This case study examines the response to a fundamental change in the nature of competition and technology control. To survive the research function leadership had to initiate new blue-sky research aligned with long-term strategy and redirect traditional research whilst continuing routine technical support activities.

In the mid 1990’s, Shell’s upstream business was facing some major challenges. The 4,000 central staff supported a profitable but highly cyclical production of oil and gas. In addition, Shell was making a lower rate of return on capital when compared to its competitors. The research

function paradigm was that smaller companies and countries with major hydrocarbon resources could not do what the company and its direct multinational competitors could manage.

Fundamental changes had occurred which meant that countries and smaller competitors had direct access to sources of long-term capital and advanced technology. Furthermore, some aspects of hydrocarbon exploration and development technology had become so extreme that international specialist technical companies were better suited to deliver the technology in the field rather than small departments within traditional oil companies. Improvements in capital and technology markets had transformed the competitive market traditional oil companies faced.

The research function, supporting the two parts of Shell's, upstream and downstream took very different approaches to managing this change within their research functions. The downstream business, supporting refining and large-scale process plants, reduced costs and numbers in conjunction with a focus on providing chargeable services to internal and external customers. This created a cost-neutral base for the staff, which retained and developed the company's technical capability. In the short-term, it did not offer a clear competitive advantage to the core business.

The technical function of the upstream business took a radically different approach. Leadership recognised that changes in the commercial environment and speed of technical evolution in the industry meant they could not produce long-term success with top down directives and hierarchical controls. Instead, it responded to the evolving new industry structure by creating an environment that stimulated networks right across the company and exploited links with external networks to direct and deliver successful research. Interestingly Shell recognizes its long history of managing change in a sustainable and creative way, based on a sophisticated and durable culture as a basis for competitive advantage (de Guess 1997). Shell's technical experts recognised that they did not have the understanding of emerging technology that would allow them to select research projects in isolation. Internally senior managers had no experience or the required approaches to research management, whilst available external experience was in very different industries. Leadership recognised that organizational learning in many areas would be required to develop solutions to new and evolving challenges.

Changes eliminated the formal hierarchy of the research function, and staff allocated to small teams of typically 15 staff. A simplified three-element structure was developed, which covered allocation or resources as well as individual and team inventive programmes. All technical teams within Shell's upstream research function, had to deliver in three areas of service, research and 'game-changer' work. End-users in worldwide

operating units paid for service work at commercial rates, removing the incentive to use central group staff.

As important as the positive creation of networks, were leadership actions, designed to weaken traditional control mechanisms in the organisational culture. For example, disincentives for staff to provide simple services capped provision, unless aligned to long-term strategic priorities including supporting major projects, new regions or clear technology themes. This offset the natural tendency of the organisation to focus resources and attention on the needs of the traditional major operating companies, which dominated the current business but not future growth and profit opportunities.

Shell Upstream took an innovative approach to developing detailed research programmes. Directors communicated strategic imperatives based on quantified long-term business opportunities for each area of the business. Technical teams put together research proposals to obtain funding allocated using a simulated market place, proposals purchased by parts of the business, which 'owned' the opportunity. These energetic market places took place over intensive three-day sessions involving staff from Shell's operations around the world. Research teams sold projects passionately re-defined scope and agreed new deliverables, to attract required funding and promises of trial locations. Management evaluated subsequent projects based on successful delivery of benefits to trial operating units. This required the combination of research teams delivering the technology and the business sponsors providing successful pilots, trials and demonstrations that did actually deliver the claimed potential business impact. The approach exploited global communities of technical experts within the company from research and operating units. Successful teams focused projects on strategic value creating areas and business benefits with defined timescales.

Whilst the bulk of research could be managed using artificial markets and global technical networks, leadership recognised that this approach would not support radical high risk, high gain research, vital to give Shell the edge over specialist technical service companies. Leadership delegated control of this part of the portfolio to an approach known as Gamechanger (Hamel 1999). Every technical team had metrics to ensure full involvement in radical research. Targets included submitting at least one proposal per team member and at least 10% of total team time on approved 'Gamechanger' projects. Whilst not all staff were comfortable with this program, it created a strong incentive for teams to recruit innovative and risk taking individuals, who would not have been welcome in a service or more controlled research environment. The 'Gamechanger' process was

set up to encourage seeking technology from outside the industry. Potential projects often came from connections with new fields or sources of ideas including mathematics, physics or aerospace academic research. These have since delivered successful and commercialised technology.

The change program adopted by Shell Research leadership can be tested against the key elements proposed by Jennings and Dooley for 'the re-conceptualization of organisation for the 21st information era and knowledge economy' which compared a mechanistic bureaucratic system with a complex adaptive system. (Jennings and Dooley 2007). The actions taken by Shell leadership in this case study can be assessed against the structure proposed by Jennings and Dooley which is shown in italics below:

Mechanistic Bureaucratic System

Machine Metaphor*: Static states, Rigid structure, inert mechanisms*. Shell removed the formal organisation, and eliminated barriers that allocated individuals into particular teams.

Control Dynamics: *Specified ends; causal certainty, exogenous forces, Hierarchical relations, Top-down flow. Alternating change and stability.* The legacy traditional research programs were cancelled. Leadership and central management control of technical content or allocation of budgets was cancelled.

Equilibrium seeking*: Stability and coherence. Dampen feedback loops. Converge on norms.* Traditional methods of matching research programs to available resources were removed. Senior managers who had built up large teams or researchers lost their line and budget responsibility to provide long term funding and technical program continuity.

Complex Adaptive System

Living organism metaphor: *organized complexity, dissipative structure, co-evolving system.* Management was in the form of simple processes, such as clear market places for research projects to identify and elevate effective programs aligned to business objectives. Leadership focused on balancing multiple processes to ensure that diverse networks could develop for service, research, innovative and external research scouting.

Emergent Dynamics: *Causal uncertainty, unspecified ends, endogenous agents, heterogeneous agents, bottom-up flow, self organizing emergent structuring.* Objectives were limited to business objectives and filed types, without any defined technology approaches. Strong incentives ensured that resources clustered to exciting and credible ideas developed by anyone, at

any level of the company, including those outside research. Initial funding for ideas from outside the company from academics, specialist contractors or other industries was subject to very rapid approval and simplified controls.

Adaptive Striving: *Dynamism and complexity, amplify feedback loops, deviate from norms.* Incentives for all levels from individual upwards including teams and senior management ensured a minimum allocation of resources to innovative and new research areas. Where a high risk, high reward program was identified, this was funded and participation championed by leadership.

At a practical level, transforming an organisation in this way clearly provided severe challenges for individual leaders, financial systems and a organisational culture generally. Despite these challenges, it would have been impossible to energise such connections using traditional directive management mechanisms. The research tradition of providing a tolerated subsidy for a low percentage of 'behind the fume cupboards' personal research time had not worked for Shell's research staff, with the bulk being used simply to cross-subsidise existing service work or hide over-spends on traditional research budgets. The approach led to a series of very productive networks developing, which rapidly provided real benefits for the company.

For example: In the critical area of technical services to support ongoing operations two networks developed. Firstly catalysed by the need to demonstrate high value for work, rather than simple time based support on routine problems research based staff moved to spending more of their time supporting high gain innovative work in the field, with particular focus on problems that could potentially provide further research programmes. This helped identify technically advanced and very staff based in operating businesses and engaged them in the advanced research team networks. Secondly to provide assistance to routine queries and share best practice across remote global operations a small investment was made in community support IT infrastructure, followed by launch events designed to ensure effective social ties between participants. This led to the development of many specialist global communities, which rapidly responded to hundreds of queries every month. Although many central research staff were active participants in many of these specialist communities, it quickly became apparent that global communities had been created with staff based within local operating companies providing the majority of solutions. In addition, analysis of these operating support networks demonstrated that once they reached a level of maturity where participants were prepared to use advice as well as provide knowledge, the

number of queries posted to the networks accelerated leading to significant improvements in operational performance.

Traditional research programmes change was initially dominated by the disruption caused by radical changes in funding levels after the market place allocation of funding. The first effect of this was that a significant minority of staff had to move out of their traditional specialism's to work on new programmes, where the funding had been allocated by the market. Management continuing to support the professional development of staff irrespective of their 'home team' facilitated a beneficial effect of the distribution of specialists. The non-line functional networks quickly found new and innovative applications of their traditional technology in new team areas. In addition, because the senior staff lost their budgetary and programme responsibility they were freed to devote time to identifying new and profitable applications for their specialist skills which in some cases resulted in new valuable research programmes. Perhaps most valuable were the networks created around each research programme of the operating units, which were nominal funders but more importantly were committed to demonstrating business benefit if the research was technically successful. These early adopter networks provided real benefits to the research directly such as identifying improved trial sites or advice and assistance during trials. In addition they led to very strong communities around particular business opportunities which improved sharing of other advanced technology available in the market but also accelerated adoption of new techniques after initial trials were successful.

The Gamechanger program in its first phase operated by funding pre-existing ideas, accelerating their development or enabling research which would not have been funded in the old model. As the process matured it became necessary to provide stimulus to faciliate identification of new ideas. The visible mechanism for this were workshops focused on particular problems, science or apparently unmatched teams. Although initial results from these workshops were encouraging, sustained long term value was created by the secondary networks that resulted. For example new post-graduate recruits provided links to academic research of value to other teams, which lead to productive sources of innovative ideas. The workshops and resultant initial projects created more durable multi-discipline communities, which generated new solutions to traditional problems. These networks operated outside research and traditional academic links, with some of the most productive relationships developing with specialist service contractors, who provided knowledge and approaches to complement Shell's fundamental understanding of the businesses problems and opportunities.

Changing a consultancy from partnership to public company: Scott Wilson case study

This case study describes the rapid delivery of improved risk control and financial reports that new capital sources required. The company faced rapidly change in many parts of its environment, which rendered existing control and management systems ineffective. Success had created a scale of operation beyond managerial experience and capability of formal systems. Diversity of services meant that senior managers could not remain familiar with operational details required for success. Client expectations for a complex service adapted to each business sector went far beyond simple the provision of individual expertise. Capital funding sources required formal reporting and risk management that had previously been exercised by the working owner's intimate involvement with operations and strong personal relationships. Required degree and pace of change was outside the capabilities of existing formal systems. Leadership chose to exploit informal organisational networks and simple tools rather than traditional investment in computing systems and formal management structures.

Scott Wilson is an international engineering consultancy that had grown rapidly changing into a private company from a traditional partnership, after having double in size to over 2000 staff in less than three years. A diverse range of services, projects and international locations delivering rapid growth; the company expected to reach 4000 staff by the end of 2006. To support this growth leadership decided to take the company public, as it provided the benefit of access to further funds, a clear value for shares and increased profile. In addition, public listing required a number of changes to the business in particular better risk management with improved financial reporting and controls. Public listing required consistent business processes across all the activities.

The challenges for Scott Wilson included a diverse range of weak financial management systems in use, management expertise in engineering and project management rather than business controls. Strengths included ongoing rapid growth with underlying profitability, a highly professional and autonomous workforce and support from a small community of commercial experts.

Traditional options were for management to exploit mechanistic controls, change targets for existing metrics, implement new financial systems, formalise procedures and recruit new managers with experience of such controls. These options would have been costly, disruptive and take at least two years to implement. Scott Wilson instead focused on

using networks to strengthening current systems, delivering the same results. For success staff at all levels would need new business skills and solutions quickly. Although management science and service providers offered a range of model answers, the application of such solutions in Scott Wilson was impracticable without serious disruption to ongoing business operations. The organisation needed to exploit external and internal expertise, learning to deliver required functionality with existing business systems.

Actions to increase the effectiveness of existing networks were simple, and had very rapid impact. A team with representatives from main developed a very simple projects process, with management control focused on only four gateway decision points, avoiding the need to formalise controls and use a conventional computerised financial management system. In developing the model, discussions created a community of champions who had ownership of the shared model but more importantly understood the strengths and weaknesses of their Division's approach combined with potential learning from other Divisions. The team did not have to change the wide range of project management models, which had evolved to match specific client sector requirements. Analysis identified gateways to deliver maximum business value, with minimum impact on client service or working project teams. The simple process focused limited commercial and senior management resources where they would have maximum impact, without destroying the effectiveness of existing managers. Early in the application of the simple project management approach, there was rapid sharing and adoption of very specific local changes. The combination of a locally based team-members and a simple vehicle to support the adoption of new practices within local operations quickly led to the application of simple and powerful new practices. Example benefits included more consistent financial reporting, improved profitability by better selection of potential bid opportunities and focusing marketing resource on high return work of increased revenue from existing clients and sectors, rather than long shot development with potential new clients. In addition, by sharing experience internationally, on a shared project model the existing commercial team produced simple tests to quantify commercial and project risks, applying these rapidly across all projects across the entire company.

Traditional improvement in this area would have been delivered in the form of new financial tools, project management software and strong directives to ensure consistent utilisation. Inevitably such a change would have had high direct costs and caused a large amount of disruption. Often such formal systems in a diverse operating environment result in much duplication and even reduced quality of service to clients. In this approach,

a very small team and a simple model created powerful networking. The initial collaboration was the formal part of the change project itself. In the second phase of work this core network moved on to drive adoption of a small number of changes across a wide range of projects and divisions. The team members viewed the role as improving divisional projects rather than implementing central directives and approaches. In the mature phase, the project model delivered a shared business model and language across the company. This shared business model enabled functional experts in a single Division to assist and apply tools across the entire company. Support function experts that had previously been isolated in a single division with no exploitation of their skills more widely included finance, risk management, marketing and project management specialists.

Clearly, in addition to the formal controls of each piece of work or project, there existing a large volume of formal documentation and less controlled guidance. , Instead of issuing new equivalents, Scott Wilson concentrated on ensuring better understanding and application of existing policies, procedures and technical standards. The traditional approach would have been to identify corporate leads, to write a new top-level document. Often this leads to unhelpful debate of precise wording then followed by slow or reluctant adoption. At its worst the new documents is not drafted by the person with most knowledge and are less helpful than existing material. The approach breaks provides the appearance of control, but can destroy working relationships and trust between distributed experts at an operating level. Scott Wilson did not have the central resource to deliver a top-down approach, and there was a tradition of respecting working consultants across the company. Consequently leadership was determined to maximise the use of existing material and distributed expertise. Before any discussion of new material was considered an extended debate was created on proposed knowledge management solutions, this helped to develop a clear understanding of the existing huge range of material. The discussion focused on staff at all levels of the company, but most importantly those with many years experience in their positions. The outcome of this process was a clear model, which recognised information status and applicability. Leadership then formalised the results of this debate by launching a simple document management system on the company intranet system. The intranet clearly positioned all information within three levels of company level policy, functional procedures and project specific material. The structure put a huge range of material into a structure that meant specialists could locate and utilise material from peers in other divisions. Historically Scott Wilson had invested time and effort in creating communities of particular

technical disciplines with little success. By providing visibility to working material, initially there was increased sharing of the information, but this quickly and more powerfully enabled increase sharing of the experts themselves. Rapid improvement in understanding and information application occurred, without a major investment in communication or re-drafting manuals. Most importantly working teams retained ownership their own work methods adapted for specific business needs. The created visibility of material across all levels of the company greatly reduced the number of company wide policies that management had to issue. In addition, typically a company policy was limited to a one page statement of intent, as existing divisional and project procedures already delivered the intent in a form adapted to local conditions. The high level policies catalysed sharing or best practice and between experts themselves.

The approach taken by Scott Wilson focused on simple interventions to increase the capability of existing systems and maximise effect of specialists across a wide range of business projects. By exploiting the organisation's traditions, leadership delivered a short-term improvement without disrupting ongoing business or client relationships.

In traditional terms, the confluence of business environmental changes and rapid growth meant senior management lacked the resources to deliver adaptation of the organisation with directive controls. Using a complex organisations paradigm, integrative influence by leadership delivered change. 'When the discrepancy between the contextual requirements in the external competitive environment and the actual resources of a team to perform the tasks reaches a certain threshold of adaptive tensions the team may produce creative responses that have the potential to increase the order within the system as a whole.' (Panzar et al 2007).

Effective medium term responses to a major event in an International energy company: BP case study

This case study looks at how changes of formal organisation focus and exploiting informal networks was use to apply best practice to improve management of high impact risks.

Based on high commercial performance and successful deal-making BP had grown rapidly into a world-class entrepreneurial energy company. A series of major mergers had underpinned growth, resulting in a wide range of diverse businesses and cultures within the company. On 23rd March 2005 a major explosion occurred at BP's Texas City refinery near Houston in the USA leading to the tragic loss of 15 lives and many serious injuries. The company quickly recognised the need for major change in the

company and the ability to demonstrate to all stakeholders commitment to manage its operations to the highest standards of safety and environmental performance. BP had a history of successfully using networks to deliver business transformation, having transformed its exploration function with the use of 'amplifying devices to evoke disequilibrium and move an organisation to edge of chaos'. (Pascale et al. 2000).

A detailed independent investigation by the previous US secretary of state James Baker identified serious organisational failures including a lack of technical skills, adequate formal standards, and cultural factors (Baker et al. 2007). These organisational weaknesses precluded an effective short-term traditional mechanistic response relying on just existing robust command and control systems focused on the safety of operations. Satisfactory response to the findings of the Baker report required new skills and understanding of highly complex facilities and business operations at all levels of the organisation. The extent to which expertise can be imported is very limited. Leadership teams needed to increase understanding of managing operations and risk management, whilst continuing to run and develop the company. Operations teams needed to develop new skills in risk applying these in conjunction with their expertise in running the existing large and complex process units. For survival, individuals and the organisation as a whole required rapid learning, which conventional training models alone would not deliver.

After the tragedy, management immediately took all practicable actions to improve formal control systems. This case study focuses on the additional actions were taken to improve the effectiveness of existing networks and less direct control systems, which contributed to rapid improvement of standards across the company's diverse range of operations. In addition to focusing on building new systems from scratch, leadership took simple measures that energized existing networks and capabilities. Leadership made these changes to improve management of operational risks at the same time as a high profile initiative to simplify the company, which effected all functions and staff, not just those within high-risk operations. Harter's paper 'Leadership and the promise of simplification' describes the common challenge that leadership faces to maintain a balance between order and complexity as organisations evolve cycling between degrees of the two (Harter 2007). The Baker Report identified that BP's growth had created organisational complexity, which had led to breakdown in effective communications and application of the company's resources at the Texas City refinery. In this situation BP's leadership faced the challenge of improved control of major risks whilst delivering the desired organisational simplification.

To improve capability of operations staff at all levels, leadership started an ‘operations academy’. This introduced parallel training programmes for company leadership, senior management and front line supervision. The programme used existing relationships with external academics that BP had already used to build skills in the area of project management. Part of the training was communication of formal concepts and techniques. The programme also provided structured sharing between participants, in particular improving understanding and application of existing business tools and the skills already within BP. A large part of the academy’s benefits derives from the way in which it energised existing networks. Directly the training provided core skills for all participants, but in addition it created a common language for non-technical issues, a forum for sharing best practice and clear indication for staff at all levels that their supervision were committed to achieving excellence in this area. Consequently, the academy generated energy and motivation across many staff within to share and apply best practice although the number of staff who were directly involved in early phases was small. Best practice sharing is straightforward, when the techniques are straightforward and not context specific. The operations academy extended the focus of operations networks to more conceptually challenging areas including leadership styles, organisational culture, risk management and leading indicators.

‘Quantitative Risk Assessment’ (QRA) is a conventional and widely used tool in high-risk industries for managing technical risks on complex process plants. Although commonly used on some facilities and decisions QRA is expensive with many different methods in use. BP experts had already developed standardised approaches, which businesses could apply to assess risks more quickly but management had not mandated application of these techniques. After the Texas City tragedy, BP leadership took the simple measure to mandate all significant hazard sites to assess operational risks using a simple consistent approach, reporting any major issues. The objective was to ensure that all high risks were visible to leadership and introduce an incentive for local business management to implement practicable measures to mitigate hazards where practicable. Although the tools used were not ideal for every facility or operation, the consistent simple methodology, company expectation was that clear timescale for completion, coverage and direct reporting to company leadership would create powerful improvement momentum. Simple assessment tools and linked to senior management reports should improve understanding and accelerate practical action across the diverse business portfolio.

In practice, the action achieved the expected benefits in the challenging area of low probability, high consequence events that are notoriously hard to manage using conventional metrics. The mechanisms for delivery of the

benefits are of interest. To complete initial application of the standard techniques, each operation had to seek the assistance and approval from the central team of experts. To prevent duplication of effort, the central team could share approaches and results for similar units, this started with ensuring a shared understanding of risks, quickly moving on to groups of experts that developed common solutions. In addition to the predictable internal company communities, the focus helped link the entire company into networks operating in the region or specialist industry of a particular site. Because of the diverse focus and background of internal and external networks associated with high-risk activities, the focus of improving risk management systems encouraged was similarly diverse. Activities that the networks focused on included simple operations procedures, competency, knowledge management, inspection techniques, plant design and fundamental risk assessment techniques.

At the time of the Texas City tragedy, BP had a series of management standards designed to control operational hazards. Standards had evolved over time and increased as BP grew through mergers. Traditionally audits within operating businesses monitored application against the standards using teams with local knowledge and experience of the particular business environment. This had led to a wide range of interpretations and approaches. Rather than waiting for the development of more explicit management standards, BP created a central audit group including external experience and clear protocols for managing the audits in a consistent manner. The function audited businesses with higher risks. Audit reporting was directly to a sub-committee of the main board and produced clear dated actions for local management.

As expected, this approach led to rapid improvement in a number simple ways. To avoid poor reports to senior leaders, local management took all possible measures to improve risk controls before audits. Where required the need for additional resources audits highlighted this enabling allocation of specific expertise. By identifying clear actions, which were then tracked to closure it catalysed a high level of discipline across the company on with risk mitigation actions which had no short term impact on business results. These benefits are interesting as they resulted from no explicit change in requirements or management activities. One model for describing these gains is simply that the approach to audit ensured that complete and effective management processes were in place, including identification of actions to rectify gaps and ensuring the actions are completed.

The approach to auditing had a more subtle and unexpected impact, which enabled the evident improvements of simply making management

systems work as intended. By judging operating units against a simple standard and enforcing resolution of gaps, leadership gave a large incentive for operational units to ask for and apply best practice from across the company. Inevitably many weaknesses were shared across many operations, in some cases it was straightforward to provide a site with an action, with a number of different solutions to the problem, which initiated dialogues between the various locations often leading to improvements and changes at all of the locations going well beyond the original narrow focus of the action. When solutions were less evident all the locations, which needed a solution, could work together, using internal and external expertise to develop a single and robust standard. The audit process also identified excellent sites, which assisted management to use these sites as a source for integrated and inclusive management practices.

Under extreme pressure to deliver rapid improvements leadership, in addition to direct mechanistic improvements, they applied three clear and simple approaches described here across the entire company. Shared operations training, consistent risk assessment and consistent auditing of existing management standards all complemented the traditional direct interventions by leadership. In addition to conventional improved controls, leadership focused the capabilities of the existing networks and energised existing capabilities to deliver a rapid improvement in risk management across a diverse international business without disrupting existing formal organisations and operations.

Conclusion

Managers, researchers and students can consider major organisational change as an extreme event, which are by definition outside the capabilities or normal focus of business control systems. In this context, an organisation's leadership has to exploit internal and external opportunities to improve performance. Actions that will energise or re-focus capabilities of the company culture and range of existing networks, which inevitably exist in any company, will maximise the value delivered. As large interconnected systems, organisations can behave in unexpected ways with small changes triggering fundamental shifts. Considering the principals of complex systems, offers leadership a range of powerful actions to use in response to extreme events. The use of complexity science to understand the recent behaviour of the global financial markets is increasingly recognised (MacKenzie 2008). This chapter demonstrates that leadership

can exploit concepts from complexity science to manage change in large commercial organisations.

It has been demonstrated using complexity theory that economic systems stabilise in different sub-optimal conditions, depending on the sequence of events and route at which they have arrived at their state (Ormerod 1998). In the field of international development, 'the challenge of implementation' is that the sequence and relative strength of reforms can have a profound impact on the success and stability of countries and entire regions (Stiglitz 2006). Likewise in managing organisations the timing, sequence and conditions of simple management interventions has a major influence on their impact, success and durability. Application of complex organisations theory provides vital guidance to management, especially during periods of major which are outside the capabilities of conventional control systems. Theory provides models that management science can be use to describe techniques that excellent business leadership is already consciously applying as part of integrated change management programmes. Better guidance from complexity management theory for traditional organisations managing extreme events would provide great benefits.

The case studies in this chapter describe situations that provided completely new challenges to individuals in the organisation. Rapid and poorly understood environmental changes threatened company business models. Staff at all levels of the companies needed to develop new skills and understanding, including senior management and technical experts with international reputations. Adoption of advanced technology from other industries and new academic fields had to accelerate. To support the required rate of change, traditional models of organisational and individual learning were inadequate. In some cases expertise simply did not exist, in others the time and cost of acquisition would have been impracticable. Leadership created an environment that exploited and refocused networks on urgent business challenges. In the absence of simple sources of expertise and advice, networks provided unique mechanisms for both individual and organisational learning in the challenging environment of rapid change.

Commercial case studies in this chapter have described the potential of networks in conventional organisations to contribute to rapid, successful and sustainable change during major organisational change.

References

Andriani P, McKelvey W (2006) Beyond Gaussian Averages: Redirecting management research toward extreme events and power laws, Durham Business School, research paper, June 19, 2006.

De Geus AP (1997) The Living Company: Habits for survival in a turbulent business environment, Longview publishing, Washington.

Erwin G, Priest S, Tebo PV, Baker JA III, Rosenthal I, Bowman FL, Hendershot D, Leveson N, Wilson DL, Gorton S, Wiegmann DA (2007) The report of the BP US Refineries independent safety review panel, Houston, Jan 2007.

Hamel G (1999) Bringing Silicon Valley Inside, Havard Business Review Sept-Oct 1999.

Harter N (2007) Leadership and the promise of simplification, Page 333-348, Complex Systems Leadership Theory, Ed Hazy, Jeffrey & Lichtenstein, ISCE Publishing, Mansfield USA.

Hirotaka T, Osono E, Shimizu N (2008) The contradictions that drive Toyota's success, Harvard Business Review, June 2008.

Jennings PL, Dooley KJ (2007) An emerging complexity paradigm in leadership research, page 17-32, Complex Systems Leadership Theory, Ed Hazy, Jeffrey & Lichtenstein, ISCE Publishing, Mansfield USA.

MacKenzie D (2008) Facts of life that banks can't ignore, New Scientist, 25 October 2008.

McMillan E (2002), Tackling industrial complexity: the ideas that make a difference, Papers from the 2nd International Conference of the Manufacturing Complexity Network, University of Cambridge, UK 9-10 April 2002 Edited by Gerry Frizelle, Huw Richards Published by the Institute for Manufacturing, University of Cambridge on behalf of the Manufacturing Complexity Network.

Ormerod P (1998) Butterfly Economics: a new general theory of social and economic behaviour, Faber & Faber, London.

Panzar P, Hazy JK, McKelvey B, Schwandt DR (2007) The paradox of complex organizations as integrative influence, page 305-332, Complex Systems Leadership Theory, Ed Hazy, Jeffrey & Lichtenstein, ISCE Publishing, Mansfield USA.

Pascale R T, Millemann M, Gioja L (2000) Surfing on the edge of chaos, page 94, Three Rivers Press, New York.

Stiglitz J (2006) Making globalization work, W W Norton, New York.

Taleb NN (2007) The Black Swan, Allen Lane, New York.

Weick KE, Sutcliffe KM, Obstfeld D (1999) Organizing for high reliability: Processes of collective mindfulness. Research in Organizational Behavior, Volume 21, pages 81-123, JAI Press Inc.

Chapter 2. The Relevance of Organizational Learning for High Performing Social Networks

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Organizational Learning: Historical Perspective

In 1990, Peter Senge published his landmark book *The Fifth Discipline: The Art and Practice of the Learning Organization*, which generated worldwide interest in the concept of a "learning organization." Senge defines a learning organization as an "organization where people continually expand their capacity to create the results they truly desire, where new and expansive patterns of thinking are nurtured, where collective aspiration is set free and where people are continually learning how to learn together." (Senge 1990). At the core of his hypothesis are five practices or disciplines, which constitute three major areas of competence. If they truly are to become learning organizations these five disciplines are foundational for organizations to learn and apply. The three core areas of learning competency and the five disciplines include:

- Aspiration, and the disciplines of Personal Mastery and Shared Vision
- Reflective Conversation, and the disciplines of Team Learning and Mental Models
- Understanding Complexity, and the discipline of Systems Thinking

This theory, the foundation of organizational learning, is that on-going learning and adaptation are fundamental to an organization's performance improvement and ultimately for organizational sustainability. Furthermore, the five learning disciplines are instrumental in enabling both individual and collective learning to occur.

A. Romano and G. Secundo (Eds.), *Dynamic Learning Networks: Models and Cases in Action*, 43
DOI: 10.1007/978-1-4419-0251-1_2,
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In 1991, Peter Senge and a group of colleagues founded the Center for Organizational Learning as a sponsored research center at the Massachusetts Institute of Technology, Sloan School of Management. The Center was organized as a collaborative of corporations working in partnership with MIT researchers to study what would happen when the five disciplines were applied in practical business settings. The MIT Center for Organizational Learning grew rapidly and by 1995 the consortium included nineteen corporate partners (Clanon 1999). The companies took on a number of significant organizational experiments, which all involved the introduction and application of the five disciplines. The results were often transformative for organizations in cases where the disciplines were rigorously applied. In a project at Ford Motor Company, Vice President Bob Womack reported "taking dialogue, systems thinking and personal vision from the management team to the front line workers transformed the way the organization worked" (Clanon 1999). Similarly, at Harley Davidson Motor Company, an organizational learning project that focused on transforming a new product development process had significant impact. According to Rick Teerlink, who was CEO at the time, "the company was saved from the brink of bankruptcy in the mid 1980's by the heroic effort of more than 2000 people, but it also created a legacy of reactive crisis management that threatened the future. Our organizational learning work established a new language of basic concepts and is gradually building a base of new skills leading to a second 'turn around' – from crisis managers to continual learners"(Clanon 1999).

As a result of the continued sales of The Fifth Discipline, a second book, The Fifth Discipline Fieldbook was published in 1994. These two books, along with the publication of a number of research papers describing the results of the various projects, continued to accelerate interest in organizational learning. At the same time, this worldwide interest overwhelmed the capacities of the Center for Organizational Learning at MIT. Thus the major stakeholders of the Center, (including representatives from the research and administrative staff, several faculty from MIT, senior managers from companies that had been central to the Center's development, and several consultants who had been involved in project initiatives) decided to embark on an extended process of reflection and inquiry focused on the Center's purpose, design, structure, and operation. In 1997, the Society for Organizational (SoL) was established as a non-profit member organization comprised of organization, consultant, and research members. Today, there are thirty-two communities around the world interested in building learning organizations, which comprise SoL; it has become a globally networked organization.

Part of the discovery from the early research work at MIT in the area of organizational learning was that the three core areas of learning competence – Aspiration, Reflective Conversation and Systems Thinking-- were vastly underdeveloped within many organizations, and appeared to be missing entirely in a few organizations. In some instances, there was no clarity about what an organization was trying to accomplish; if there was clarity, there was certainly no alignment through the organization about what the future should look like and what needed to be accomplished. A few people at the top of the organization, such as the CEO and the senior management team, may have generated a "vision statement" that was meaningful to them but not others. When researchers asked employees in the middle or lower levels of the hierarchy what the vision for the organization was, the employees could not remember, or they fumbled for a laminated card or something on a bulletin board where the vision statement was written down. Clearly they had not been involved in the process and the vision statement had little meaning for them. It was rare to find an organization that supported individuals' personal visioning work. Research showed work in the areas of both shared and personal vision was critical, as the connection between the two was the factor generating engagement and energy for doing an organization's work.

Research also demonstrated that skills in the area of reflection and quality conversation were often missing, or at least not applied. There was so much emphasis on completing tasks and "doing" in most organizations that reflection wasn't even considered a legitimate way of spending time except for those at the top of the organization; finding the time for it was difficult even for them. More often than not the quality of conversation in staff and planning meetings was quite low. The real open, honest conversations would happen outside of the meeting in the hallways or men's or women's rooms. Frequently, political positioning was a problem. Employees had a tendency to hold back their actual thoughts if they felt they differed from their bosses. People reported that they didn't like this dishonesty, but at the same time they accepted it as the normal "political" reality of their organization.

It was even more rare to find organizations that dealt with complexity in a sophisticated way. Rarely was there both familiarity and application of systems thinking concepts and tools within an organization. Thinking about the whole system, including its interdependencies and possible unintended consequences of management decisions, was simply not a part of what most management teams did. This was not surprising; before the early 1990s, this area was simply not a part of management education.

An important overarching finding of the research work at MIT and SoL is that the three core areas of competence are very interdependent. Focus and application in just one of the areas of learning competence did not produce meaningful results. As mentioned above, competence in the area of individual and shared vision generates energy. Open, honest, direct, reflective conversation enables a team to share their collective view of the system and to deal with it. To teach the five disciplines, we have come to use the metaphor of the three-legged stool: If one leg is missing, the stool falls over (Fig. 2.1). So we emphasize that teams and organizations learn and apply the disciplines in concert.

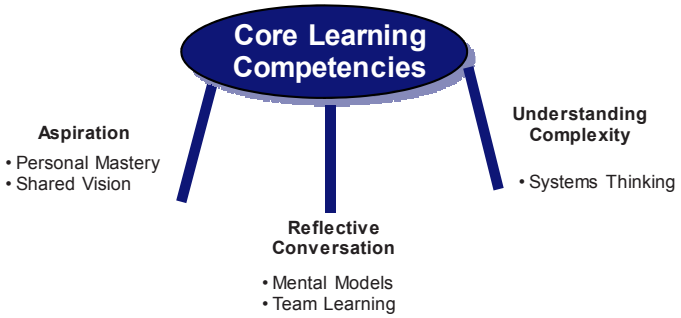


Fig. 2.1. Core Learning Competencies

Organizational Learning has come a long way since 1990. It is now taught as part of management curricula literally all over the world.

In a recent Society for Organizational Global Forum held in Muscat, Oman in April 2008, over 450 people gathered from forty-seven countries to share their work. It now appears that more organizational learning application work is underway outside the United States than inside it. However, organizations that actually fit Peter Senge's original definition of a leaning organization are still very much the exception, not the rule. At the same time, the context in which people live and work has shifted; the world has become more interdependent and connected.

Learning from Research on "Hastily Formed Networks"

Introduction: The Hastily Formed Networks (HFN) Project

Over the past two years, a group of organizations that share an interest in organizational learning (including the Ford Motor Company, Boeing, Detroit Energy Company, the US Department of Defense, Tufts University Feinstein International Center, and the Society for Organizational Learning) have been researching what enables extraordinary action in social networks focused on urgent situations or crises. It is interesting to contemplate what interest such a diverse group of cross-sector organizations could have in common? What research interests do the largest organization in the world, the US department of Defense, share with the Feinstein International Center, a small university-based humanitarian relief research organization? The answer is a common interest in social networks.

The US Department of Defense at the Naval Post Graduate School in Monterrey, California initiated this particular research on social networks. Increasingly, the US Military has become involved in complex, large-scale disaster relief efforts at home and abroad, such as the Tsunami in Southeast Asia and Hurricane Katrina in New Orleans, Louisiana. This work is beyond the traditional scope of military activity, yet they were aware that, as a global force in the world, they would be called upon more and more frequently to assist in these kinds of situations in the future. At the same time, the military realized that some of the capacities for effective action in these situations were limited. . In particular, the Department of Defense needed to develop military leaders who could quickly and successfully organize the networks required to deliver effective action in such urgent circumstances.

In February 2006, The Naval Post Graduate School hosted a Society for Organizational Learning Liaison meeting. As SoL organizational members discussed the context for this work, a number realized that as global corporate entities they often faced similar "urgent" challenges in getting work done beyond the boundaries of their organizations. Accomplishing such networking work effectively, in fact, was essential for their success. There was a collective deepening understanding that today's world is increasingly interdependent as it is comprised of global organizations, nation states, government agencies, NGOs, and formal and social networks. Today, these networks are working across cultural boundaries often driven by off shoring, across organizational boundaries driven by Original Equip-

ment Manufacturers (OEMs) and sourcing arrangements, and across functional boundaries often driven by the need for shared co-design. The need was clear: Organizations must learn to mobilize action in new and non-traditional ways.

Core Research Questions

As a result of the discussion in Monterrey, a formal cross-sector, cross-organizational research project was organized to learn about enabling extraordinary action in "urgent" situations and in a variety of organizational conditions. The research work involved defining the concepts, engaging "learning practitioners" to initiate projects in their particular contexts, monitoring the projects and activities, generating and collecting data, reflecting and synthesizing, and proposing new questions which would further this learning.

Though binding the research to something that was relevant and of interest to all the participating organizations proved challenging, it was achieved. The study centered on four main questions relating to the concept of Hastily Formed Networks. The first question focused on how action was actually mobilized at all the levels in the eight examples of hastily formed networks that were studied. This we felt was important, as a primary function of a network is to mobilize action across multiple constituencies. The second question focused on what were the minimal organizational structures, which were required for effective network work. We sought to uncover what structures were needed to enable clarity of intent and distributed coordinated action, both of which we found crucial for the success of the network. This is an area where accelerated learning is critical. The third question, which we all were interested in, was the issue of preconditions, which were helpful or required for the effective action of the network. Finally, there was across the board interest in what kind of leaders and leadership capabilities are needed for these hastily formed networks to be effective. There was significant and consistent data that showed that traditional hierarchical leadership was often ineffective in these urgent situations and, at times, was an obstacle for getting effective action accomplished.

As mentioned above, the study focused on eight instances of hastily formed networks, which occurred within or across the organizational boundaries of the participating organizations. Several were very successful in achieving their objectives and desired results, several achieved some but not all their objectives and two essentially failed to get off the ground

and/or achieve any results. The following is a summary of the key findings from the study.

Findings: How action was mobilized throughout the network

A core function of effective networks is to mobilize action across constituencies. In each instance, mobilization of these hastily formed networks required upfront communications as well as connecting and convening people to clarify a situation and its conditions as quickly as possible. Establishing a safe, open "communication space" was essential for mobilizing action. The space should be face-to-face when possible; technology should be used when face-to-face is not possible. Clear factors identified that enabled mobilization included: a genuine belief in others and the capacity to depend on them, and the ability of the leader or convener to be vulnerable, in the sense of relinquishing control. Making and maintaining the connections among people also facilitated action. Quickly recognizing who had expertise and/or establishing quick ways of gaining that understanding among the network was critical. It was also important for the network to collectively clarify the situation so individuals could gain a sense of accountability, which ultimately lead to predictability and independent, coherent, intelligent action. A network's capacity to recognize and quickly deal with naysayers was also a factor for effective action.

Too much central authority, when distant from what was actually happening in the network, was in some cases an obstacle for effective action.

What minimal organization structures are required

The study's findings showed that less structure is best, particularly in the early stages of organizing the network. As mentioned above, perhaps a network initially needs only some "communication space" to create and model minimal but important norms and decision-making mechanisms. This conversation space can enable clarity of the intent of the network, assessment of the current reality, feedback on the conditions of the situation, and critical feedback on the performance of the network. However, the level of structure needed changed over time and evolved based on the need for the network to: (1) cohere and sense the situation, (2) coordinate and act, (3) complete the work and (4) disband. In some instances, organizations jettisoned hierarchical and procedural structures in order to move most quickly and efficiently. Over time, however, organizations needed

some additional structure to enable the process of action tracking and follow-up.

In each of the instances of the hastily formed networks studied, there was an individual who served as a convener for the network who brought people together. In some instances, though not all, the convener also had content expertise and positional authority. When working across organizational, functional or cultural boundaries, it was essential to have all constituencies of the network represented if at all possible.

Preconditions that are necessary for Hastily Formed Networks to be effective

Some of the preconditions necessary for these networks to work effectively are intuitively obvious, such as a group of people who are willing to come together and who share a common sense of urgency or crisis. Others preconditions are more subtle. People who hold the belief that they can contribute, who are willing to subscribe to an overarching goal, and who are ready to utilize, and contribute whatever knowledge or skills they have to help the situation comprise effective hastily formed networks. This goes well beyond their specific organizational role and job description. Another characteristic of effective networks is that people are willing to share what they do not know as well as what they do know. This enables more rapid decision-making and action on the part of the network as well as accelerates the collective learning of the network.

The ideal precondition for an effective hastily formed network is a previously recognized formal or informal network already in place comprised of people who have a shared value system, who may know one another, and who perhaps have worked together previously. In situations where people participating in a hastily formed network either know or know of one another, and have already established some level of trust and respect, those involved in the network work described it as satisfying and meaningful as well as more fun than work. Some participants described their participation as a highlight in their life.

The leadership capabilities and characteristics required

Establishing a collective system of distributed leadership within the network was the fundamental leadership work involved in a hastily formed network. Effective leaders were able to remain open and sensitive to the direction required. They could communicate a situation's reality accurately

and effectively, including the extent of the dire conditions of the situation. Finally, they could clarify decision-making processes and relinquish control of them when it was advantageous. In addition to enabling the actions of others, leaders could act quickly and decisively when necessary. These are "agile" leaders who understand and can carry out the new kind of leadership nimbleness. They can adapt: step up, or back and let others lead depending on the situation.

These leaders also possessed some interesting common characteristics. They seemed to have a clear sense of personal identity. In crisis situations, they did not panic and others perceived them as a calming influence on the network. They appeared to have a high tolerance for ambiguity and participants in the network saw them as genuine, empathetic, caring, curious, and willing to listen to others' "answers." Others perceive an effective network leader as having integrity and acting with integrity. A network leader's character really matters (Clanon 2008)!

The Relevance of Organizational Learning for High Performing Social Networks

How do these recent findings in the context of social networks and the world we now live in relate to the five organizational learning disciplines Peter Senge articulated eighteen years ago? Interviews were conducted by the author with the research team for the Hastily Formed Networks Project, each of whom has worked extensively in social network settings and is also well grounded in the application of Senge's five learning disciplines (HFN Research Team 2008). A seven member SoL faculty team currently involved in the design and delivery of a leadership development curriculum that integrates organizational learning disciplines and social system technology for a large US Government agency was also interviewed. (Acknowledgements)

An overarching realization of the Research Team that participated in the Hastily Formed Networks project was that to large extent social networks are the primary vehicles for actually getting work done. Although formal planning and management still largely occurs vertically in organizations, work typically gets accomplished horizontally across and beyond organization borders.

The relevance and importance of the learning disciplines of shared vision and personal mastery relate to the HFN Research Team's shift in mind regarding how work gets done. A key component for the success of hastily formed networks is a common understanding of the network's cur-

rent situational reality, the network's overall objective, and who could do what for the network. Achieving a shared level of clarity as rapidly as possible is essential for the success of the network. This clarity enables a shared sense of individual accountability, which leads to predictability and independent, coherent, intelligent action. In the US Military, they refer to a "common operating picture" among individuals as essential for coordinated, effective action. This often is achieved virtually through the use of on-line social networking tools and through face-to-face exchanges.

A key difference in the discipline of shared vision in an organizational context, versus a social network context is that in an organizational context the vision most often comes from the top down. It might be enhanced or modified based on input from other levels of the organization, but in most instances, it originates at the top levels of management. In social networks, on the other hand, the vision more often originates from the bottom up, at a grassroots level, or horizontally across the network. The intent of alignment across an organization, team or network is the same but the context and origin can be quite different. It can be collectively created via on-line networking tools, face-to-face exchanges, or some combination of both.

Another finding of the Hastily Formed Networks project was that conveners of the networks who were successful in achieving their objectives were individuals who others perceived as having a clear sense of their own identity, as being calm in urgent situations or a crisis, and as having and acting with integrity. These characteristics may not be quite as critical in non-crisis network situations but still may be important in social networks where there is no formal leadership authority structure. Peter Senge defines personal mastery as "the discipline of continually clarifying and deepening our personal vision, of focusing our energies, of developing patience, and of seeing reality objectively" (Senge 1990). We see the discipline of personal mastery as enabling and sustaining these characteristic and practices.

Another characteristic of effective conveners that the discipline of personal mastery may help enable is the capacity to be "agile," that is to be able to move beyond one's own ego needs, to be directive when situations call for that, and also to step back and allow others to assume leadership roles when that will enable the overall success of the network.

Although the five disciplines can be taught individually, those interviewed maintain that the five disciplines precondition quality relationships that enable learning and effective performance in social networks. What matters most is applying and integrating the disciplines in doing work together. Deborah Reidy is a consultant member of SoL and faculty team member for a large US Government agency's leadership development cur-

riculum that is integrating organizational learning methods and tools in social networks contexts. She sees the primary leadership activities in a social network as "fostering and strengthening enriching exchanges and relationships and helping people to keep focused on their collective performance and results by 1) creating containers (minimal structures) for such exchanges to happen, 2) appreciating and acknowledging and 3) facilitating the movement of learning and resources throughout the system."

This is consistent with the results of the Hastily Formed Network project, which cited the need for a "communication space" as a minimal structure. The particular form of the communication space is less important than the quality of relationships that develop within the communication space. In a recent article, Dennis Sandow, who has conducted research in the area of social networks and social capital for over a decade says, "effective social networks that are based on mutual acceptance, respect, and co-inspiration realize large gains in performance. They also generate well-being in their workforce and that "an understanding of social networks will become a core competency for organizations that find themselves in an integrated global economy" (Sandow 2007).

The quality of relationships and communication in teams within organizations, as well as the quality of relationships in networks that cut across and extend beyond organizational boundaries, correlate with a high level of performance. The findings of the Hastily Formed Networks project pointed out that in effective networks, people felt they could and would share not only what they knew but also what they did not know. In urgent situations there is no room for posturing. Valuable qualities included authenticity, openness, empathy, caring, curiosity, and a willingness to listen carefully to others. These qualities are quite consistent with those Senge articulates for practicing the disciplines of team learning and reflective conversation to elevate mental models. They are also very consistent with the practices Dennis Sandow lists as required for building effective social networks, including listening, understanding, trusting, collaborating and reflecting (Sandow 2007).

Skills that are clearly useful in a structured organizational team context become essential in the context of a social network where there are no clear lines of authority or designated roles. The study on Hastily Formed Networks elevated the importance of leadership agility as an important capability – the ability to adjust one's leadership role from directive if the situation requires to empowering others to assume leadership roles when that would be serve the objectives of the network. But what competencies underlie a leader's capacity to be "agile"? No doubt there are a number of them, but one that clearly ties back to the Five Disciplines in the area of re-

flective conversation is the ability to balance advocacy and inquiry. In other words, one must communicate effectively and advocate one's views while also retaining the capacity to inquire, to hear, and to be open to other points of view.

There are also new skills emerging that are necessary in effectively operating within social networks. The most obvious are the technical skills required to navigate the new online social networking tools, which seems to be largely a generational issue. My 18-year-old son and 21-year-old daughter will enter the workforce with these skills well intact, whereas my generation may struggle a bit as they learn to use these new tools.

Other important new skills for effective participation in social networks include the capacity to distill large amounts of data in order to get to what is most important for the success of the network, along with setting personal boundaries for your participation. Generally, no one outside of oneself will set these boundaries for participating in a social network. For example, there appears to be a growing dark side problem of addiction to using some of these websites. Here, again, the discipline of Personal Mastery seems quite relevant to this issue.

Of the Five Disciplines, the most important and relevant for today's global context, according to those interviewed, is the discipline of Systems Thinking: perceiving systems from a holistic perspective, understanding interdependencies and time delays in the system, and predicting unintended consequences. Our very survival as a species may depend on it. Ecosystems that provide clean water, breathable air, fertile soil, and pollination are in decline worldwide. The gap between the rich and the poor continues to widen. According to Oxfam, 2007 "Climate change represents the biggest threat to poor people in the 21st century."

To successfully address the profound issues we face as cohabitants on our shared planet, a shift to a larger social systems network context needs to occur on a global level. It is obvious now that no one country, government, or organization can solve these issues on its own. New levels of cross-sector, cooperative, collaborative-networked approaches are required.

Conclusion

Perhaps the most striking example of the integration of the practice of the Five Disciplines and work implementation via high-performing social networks is the recent US presidential campaign of Barack Obama, which resulted in his election as the forty-fourth President of the United States.

Some describe it as the most successful campaign in history from virtually every perspective: most money raised, most people engaged, best use of online technologies, all of which enabled the election of the first black president of the United States, a startling result.

A recent television program interview with the core team who managed the Obama campaign focused on understanding why the campaign was so successful. The team described elements of the campaign that relate directly to the practice of the Five Disciplines. They spoke of their affection for one another, the quality of their communication, the fact that although they certainly encountered issues and frustrations there was no drama or screaming at one another. They cited the clarity of Obama's vision for the country, which they all shared as a motivating and uniting factor. They also cited Obama's character and calm in the midst of intense issues that helped the team stay on track. Last but not least, they credited their community organization based systems approach and the use of technology, which enabled the engagement of an unprecedented number of people in the campaign, as well as an unprecedented level of funding.

Humans are by nature social beings; we are neurologically wired to be social and at our best are loving to one another. Social networks are a primary way contemporary work gets accomplished. In fact, it may already be the primary way. Humans are also biologically constructed as learning beings. The practice of the Five Learning Disciplines that enable learning is as relevant in our contemporary, virtually connected world context as it was almost two decades ago; perhaps it is even more relevant. It is now more apparent than ever that our success, well-being and sustainability as a species depends on our capacity to learn, to communicate effectively with one another and, ultimately, our capacity to sustain quality relationships with one another.

Acknowledgements

My sincere gratitude to The Research Team for the Hastily Formed Network project who provided input for this Chapter including: Greg Clark, Shelia Covert-Weiss and Eric Wingfield from the Ford Motor Company, Carol Gorelick - SoL Research Member, Sue Higgins – US Department of Defense, Naval Post Graduate School, George Roth – MIT, SoL Research Member, Jason Schulist – Detroit Energy Company, Jean Tully – SoL Consultant member.

Also my sincere gratitude to The SoL faculty for the Social Systems Thinking Leadership Development Curriculum who also provided input

for this Chapter including: Rick Karash, Deborah Reidy, Fred Simon, Cynthia Way and Nick Zeniuk all consultant members of SoL and Dennis Sandow who is a research member of SoL.

References

Clanon J (2008) Hastily Formed Networks: Organizing for Extraordinary Performance. Research Project Synthesis

Clanon J (1999) Organizational Learning From the Inside Out: Reinventing the MIT Center for Organizational Learning. The Learning Organization, Vol. 6, Issue 4

Sandow D (2007) Extending Systems Thinking to Social Systems. The Systems Thinker Vol. 18, No 5, Pegasus Communications

Senge PM (1990) The Fifth Discipline. Doubleday, New York

Senge PM (2008) Keynote Address, Annual International Coaching Federation Confederation Conference Montreal, November 14, 2008, Quebec, Canada

Chapter 3. illycaffè and the Brazilian espresso knowledge network

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Introduction

Italian roaster illycaffè celebrated its 75th anniversary in 2008. It is known worldwide for the quality of its *espresso*, serving 6 million cups in more than 50 thousand shops spread over 144 countries. The illy *espresso* is a refined blend of nine types of coffee beans purchased in 13 producing countries. Out of those, the Arabica variety produced in Brazil accounts for 50% of the coffee used in the blend.

In order to acquire quality coffee beans in the volume and scale needed to serve its customers while meeting the stringent quality requirements of the company, illycaffè developed an extensive and complex supplier network. Aiming to ensure quality along all processes, the company brought into the network all the links of the coffee production chain, from the plantation to the processing of the coffee beans in the company facilities in Trieste (Italy). To show its appreciation for suppliers and ensure a reliable supply of fine Brazilian coffee, the company developed a series of strategic actions that went beyond business relations to include an educational effort, generating and disseminating information and knowledge on how to produce quality coffee (also known as specialty or gourmet coffee), while also forging strong ties with all the elements in the net.

We should bear in mind that in the early 90's, when illycaffè started building up its relationship with Brazilian coffee producers, the international coffee market was very different from today. First, flat global

A. Romano and G. Secundo (Eds.), *Dynamic Learning Networks: Models and Cases in Action*, 57
DOI: 10.1007/978-1-4419-0251-1_3,
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prices did not differentiate products with different characteristics. Therefore, in coffee producing countries, the widespread practice was to mix good quality and poor quality beans in the same lot. This caused losses to illycaffè: the company had to discard up to 70% of the volume purchased because the beans did not meet the standards set by the company. Second, the coffee sector was ripe for radical transformation. Until 1990, the coffee market had effectively been regulated by a cartel (ICO: International Coffee Agreement) lead by Brazil via the Instituto Brasileiro do Café (IBC) (Luttinger and Dicum 2006). The cartel controlled price and price volatility by agreing quota among the producer countries and controlling the amount of stock (Daviron 2005). The end of ICO in 1989 and IBC in 1990 opened the gate for the emergence of new business models.

If on the one hand there were flat prices for the producers, on the other hand demand for premium coffee experienced a boom in importing countries. Consumers wanted a fine drink made with quality beans, something that would add to pleasant relaxing moments. At the same time, by re-inventing the century-old tradition of the coffeehouse (Allen 2001), the so-called 'Latte Revolution' (Daviron and Daponte 2005) changed the habit of coffee drinking and transformed the perception of coffee in the US first and then in most other industrialized countries.

Challenged in its efforts to provide this growing market with a great *espresso*, illycaffè undertook initiatives that would enable the company to reach its strategic goal of bringing about structural changes in the cultural values of Brazilian coffee producers to promote premium quality. The company devised mechanisms and developed tools to map producing regions and to identify those producers who were willing to adhere to its concepts of superior quality. Another objective was to ensure their loyalty, so that the company could rely on them to provide coffee beans in the volume and scale needed to meet not only current demand but also its market expansion plans.

An Award for Quality

When it first came to the Brazilian market, the first strategic action of illycaffè to change the prevailing culture among coffee producers and promote quality was to launch the "Brazil Award for Quality in Coffee for *Espresso*", together with ADS Assessoria de Comunicações, the partner agency chosen for company's public relations in Brazil. After 17 annual editions, it was renamed "The Ernesto Illy Foundation Award for Quality

in Coffee for *Espresso*" to honor the late president of the company who died at the beginning of 2008.

With the award, the company kicked off its strategy of identifying producers of fine coffees who were also interested in adhering to premium quality concepts, and mapping the more favorable regions for quality crops. This effort led to the formation of a base of loyal suppliers who could support the growing demand.

Today, illycaffè distributes more than US$ 100,000 in prizes every year to the 50 top contestants, besides committing to purchase all lots submitted for evaluation that pass a quality test. The prizes are awarded in a gala event promoted by the company to celebrate the quality of the coffee traded and the successful acquisition of products meeting their high standards.

Besides ensuring the purchase of coffee from finalists, the company rewards quality with prices that are approximately 30% higher than the market average. For producers, it makes sense to adhere to the concepts of premium quality, as it represents an alternative to the flat prices they would otherwise get. It is also good business for the company, since it minimizes losses and provides a reliable supply of the most important type of coffee beans in the blend served to discriminating customers wherever the Italian brand is present.

Besides the contest and prizes, strategic actions by illycaffè included the forging of direct relationships with suppliers, eliminating the so-called intermediaries. The company made arrangements with the export organization Porto de Santos and started purchasing directly from Brazilian producers and taking charge of the export procedures.

Having created the contest and set a direct channel to producers, the company helped establish the Assicafé, a laboratory led by one of the most respected specialists in coffee quality in Brazil, Dr. Aldir Alves Teixeira. Assicafé analyzes the samples submitted to the contest and the samples being offered for purchase by the company. The lab also assists coffee producers on technical matters and trains the classification specialists who work with the coffee producers.

The actions undertaken by illycaffè around the award drew the attention of the agricultural market in Brazil. Communication actions involving cooperatives and associations in major producing regions and the press helped disseminate company concepts, as well as its vision and values, and this was a driving force for transformation in the local culture, promoting greater commitment to superior quality. In time, the award came to involve other agents, such as classification professionals, technicians working in cooperatives and associations who help coffee producers identify the best beans. Starting to reap the positive results of its actions, the company went

on to create new strategic actions to further stimulate superior quality. Some examples are the creation of a special network group called the "illy Coffee Club", and the establishment of the illy Coffee University, Unilly, currently called Universidade do Café Brasil (Brazil Coffee University). Both initiatives aim to disseminate knowledge among the links in the production chain. The university was created with the cooperation of professors and technicians of the Program of Studies of the Agro-Industrial System Business (Pensa), of the Foundation of the Institute of Administration (FIA), associated with the School of Economics and Administration of the University of São Paulo (FEA/USP).

The results of the actions can be seen not only by illycaffè but also by its Brazilian suppliers. It was on the one hand beneficial for the company, for it changed the prevailing culture among coffee producers and prodded them to embrace quality concepts through actions that went beyond creating commercial ties to weaving a complex network geared to generating and disseminating knowledge. Some figures for context: to date, illycaffè has distributed close to US$ 2 million in prizes, received more than 10,000 samples, and built a base of more than 1.6 thousand suppliers who are proud to belong to a differentiated class of producers of fine coffees, the "elite" among their peers, and whose products are welcome in the most demanding international markets.

On the other hand, Brazilian coffee producers and the local market also perceive significant results. In the wake of the pioneering contest, a number of other awards for coffee producers have emerged sponsored both by public and private institutions. Another positive point was the differentiation of premium coffees introduced initially by the company, which contributed to the launch of a number of other fine coffee brands in the country. This was essential for shaping a culture of coffee appreciation, and led to the establishment of a number of new businesses, besides enabling restaurants to offer their patrons a special way to close their meals, and generate extra income.

In the case of the special premium brands that followed the illycaffè award, an example worthy of mention is "Café do Cerrado". In the first years of the contest, the positive results obtained by producers in the region called Cerrado, in the Midwest State of Minas Gerais, led to a local strategy of using the high ranking positions obtained in the contest as a marketing tool, positioning the region as a denomination for premium coffee. Regional associations and cooperatives got together and created the Council of the Associations of Cerrado Coffee Producers (CACCER), in the municipality of Patrocínio (MG), launching the denomination as a means to claim more favorable trade conditions. The results were highly positive: they did in fact climb to a privileged position within the market,

and the image of a region that became known for quality coffee even improved the selling price of land. At present, premium quality concepts are so widespread and ingrained that CACCER features certification programs both for products and for properties to meet the requirements of domestic and export customers. The history of the council featured at their webpage (www.cafedocerrado.org.br) recognizes the actions started by illycaffè to stimulate superior quality in coffee. This is a very clear example of how results in the contest were used as a reference in marketing actions:

The award proved beyond doubt the superiority of the Cerrado coffee. Exploiting this fact, the Caccer started to make use of the award as a marketing tool to promote the coffee from the region. As the illycaffè award was widely known and impacted heavily on the coffee market, we built our strategy around the award. Given the magic presence of Cerrado growers among the award winners, in a short time the image of the Cerrado coffee and of the award spread all over and the quality of the region started to be recognised on the global scale.

A network for generating and disseminating knowledge

The strategic actions of illycaffè to improve quality spontaneously evolved into a complex network, shaped not only by trade relations but also by the pride to belong to the top producers who were glad to share information and knowledge about their activities. The tissue of the illycaffè network in Brazil is created by the direct trade between company and loyal suppliers of fine coffees and by the strong ties between all links in the coffee producing chain. It is enabled by efficient communication that generates and disseminates information and knowledge, and above all, it is nurtured by the mutual trust of all parties committed to improving the quality of coffee in Brazil.

The establishment of the Coffee University of Brazil, the meetings of producers in the illy Coffee Club and the award for quality gradually shaped a true ecosystem generating gains for the Brazilian coffee producing chain and for those regions in the country that embraced the quality concepts in the production of premium coffee. In this ecosystem, the transmission of information and knowledge is faster and more efficient, and is shared in a viral manner, as a “healthy epidemic”, reaching all links in the large scale production chain.

The illy Coffee Club, created in June 2000 by illycaffè to bring together its Brazilian providers of premium coffee, was the first initiative of this

kind in Brazil aiming to generate loyalty among suppliers. Through the club, the company organizes direct communication actions with members, and rallies them for actions geared to conveying, exchanging and sharing information and knowledge. At the club, it is quite common for members to meet on the so-called "field days", where they get together on a property to watch up close what practices are being adopted for the production of superior quality coffee, and the methodologies used for improvement based on information or techniques. The importance of these get-togethers, always featuring a specialist associated with the Brazil Coffee University, can be easily seen: some members will travel close to 1000 kilometers to take part in a field day. The illy Coffee Club is an important knot in this mesh of information and knowledge woven by illycaffè in Brazil; it is an example of collective intelligence whereby coffee producers learn from each other and from the group, leading to new ideas and new knowledge for the improvement of the quality of Brazilian coffee.

Last year, the illy Coffee Club boasted 583 members, 95% more than it had in 2002, shortly after it was created. Besides the possibility to sell their produce to illycaffè at prices that are higher than the market average, members also get better prices for the portion of the crop not sold to illycaffè. The membership card ensures elite status among peers, and illycaffè suppliers are recognized as producers of premium quality coffee. The local market and peers understand they work with the information and techniques aligned with the strict requirements of a brand known for superior quality – press reports and studies by research institutes place illycaffè's brand among the 15 most known in Italy.

At the illy Coffee Club, members are distinguished for the uninterrupted time they have been supplying the company. First year members carry a red card; after two years, members have a silver card. Members who have sold their product to illycaffè for three or more consecutive years receive a gold card. Gold and Silver members enjoy more than the benefits of the club: they have the chance to enter the "Supplier of the Year" contest, in which a number of aspects regarding regularity and quality of lots are evaluated. The award for the winner is a trip for two to Trieste and Venice to visit the illycaffè's facilities and see firsthand the technology adopted for the production of the illy *espresso,* and to enjoy one of the major tourist destinations in Italy. The winner is announced at the award ceremony. Besides the trip, the "Supplier of the Year" receives market recognition given the publicity in the press and the story featured in L´Espresso magazine, an illycaffè publication distributed in the areas in which the company operates.

Another important node in the illycaffè network in Brazil is the Brazil Coffee University, with courses and seminars aiming to train all links in

the coffee production chain, from production systems to management. The professors are specialists from Pensa/USP, and from illycaffè's Università del Caffè in Trieste, Italy. Over the past 8 years, almost 6,000 producers participated in the programs, exchanging information and knowledge among themselves and with specialists.

To tie the knots in this mesh aimed to generate, disseminate and share information and knowledge, illycaffè develops a series of communication actions in cooperation with the agency responsible for its public relations in Brazil, ADS. Some examples of communication pieces are the video "How to make coffee a work of art", with tips and cases selected from within the coffee production chain; the instruction manuals "The roads to excellence", guiding coffee producers on good management practices they can adopt in their properties for better quality in all aspects of coffee farming; and printed technical reports on routine problems that can be avoided or corrected in the field. Other examples of tools that facilitate communication between the company and other nodes in the network, and also serve as a platform for disseminating information and knowledge are the magazine "L´Espresso" and the newsletter "Focus on illy Club", both of which feature reports on initiatives by the company, its partners and suppliers – and ultimately, on all the areas and nodes of the national network.

Other tools also help strengthen the ties between the nodes and build up relationships, like the webpage of the illy Coffee Club – www.clubeilly.com.br. In this digital space, the company shows the actions undertaken by the club and also interacts with members, communicating not only technical information but also "field days", courses, seminars and lectures promoted by the Brazil Coffee University and information on the contest and award ceremony. Other initiatives promoted by illycaffè for customer relationship include a toll free number for customer service, exclusive illy gifts, special sales on illy products (coffee machines, cups, coffee blends among others) as well as traditional greeting cards (birthdays, Christmas, etc).

The initiatives meant to disseminate information and knowledge are aligned with the interests of all parties involved in the mesh woven by illycaffè in Brazil. A recent survey among 96 members of the illy Coffee Club showed supplier satisfaction in a number of strategic aspects related to the proper functioning of the network. Among the suppliers that responded to the questionnaire: 96% said they are very happy with their relationship with the company; 84% participate in the contest; 54% take part in the courses and seminars of the Brazil Coffee University; 92% perceive the benefits offered by the illy Coffee Club as competitive advantage; 69% participate in the "field days" promoted by the club; 92%

enjoy reading the L´Espresso magazine; 88% welcome the information in the newsletter "Focus on the illy Club"; 70% like the club webpage; and 79% stated they are very satisfied with their sales to illycaffè. The findings are very encouraging, confirming that the company has overcome the hurdles it found when it first came to Brazil. The survey also proves that the strategic actions go beyond mere business relations; they have shaped a network for the dissemination of quality concepts, tying together all the parties involved in the trade of coffee beans used in the preparation of a drink enjoyed by customers requiring the highest standards in more than 140 countries all over the world.

The findings show that results exceeded the highest expectations for the initial strategy devised to organize a base of suppliers of fine coffees. The company initiatives in Brazil, especially the creation of the award, the club and the university, became so relevant in the Brazilian scenario that they actually lead to other unpredicted positive outcomes. There are plenty of such examples. One of those is the case of the "Zona da Mata" region, in the State of Minas Gerais. The region was once known for the poor quality of its coffee, and with the support of the nodes in the illycaffè network in Brazil, it successfully turned the situation around. Using the relations in the illy Coffee Club and the suggestions of illycaffè's specialists, producers in the region adopted best agricultural practices and improved the quality of their product, climbing to the top ranks in illycaffè contest, and improving their trading conditions. By adhering to quality concepts, producers united to change the identification of the region, using a new denomination, "Matas de Minas", to distance themselves from the negative image associated with the old denomination "Zona da Mata".

The case of "Matas de Minas" serves as evidence of the viral, epidemic-like effect of the information and expertise exchanged and shared through the network among all the nodes connecting illycaffè relations in Brazil. The efforts geared to the adoption of new agricultural concepts and practices in the region drew the attention of the market not only locally but also internationally. Besides the production-related aspects, the efforts impacted even the personal lives of the producers. Case in point is the example of brothers Ednilson and Walter Dutra, farmers in the municipality of Manhuaçu (MG). By adhering to illycaffè quality concepts, they gained access to new markets and better trading conditions. This brought in more income and improved quality of life for their families – they can now afford to travel to the US and Europe on vacations. Also, the benefits of adhering to quality in agriculture motivated them to adopt the same concepts in their other businesses, with similar positive results.

The structure of the knowledge flows within the illycaffè suppliers network show some interesting features:

Multi-layer

The network shows different layers of connectenedness and control. At the centre sits illycaffè in Trieste, Italy. The centre sets the strategic direction and practises a fairly hands-off policy in terms of implementation of strategy. But illycaffè also keeps a close involvement with the Brazilian Espresso Knowledge Network by participating to the club field tours, administering technical courses, visiting suppliers, etc. These actions create and maintain a network of personal relationships between the centre and the other layers of the network that infuse the network with a human touch, generate long-term trust and reduce the knowledge path length (Watts 2003).

illycaffè's inner network. This is composed by a limited number of companies, in tight contact with illycaffè in Trieste, specialising in the management of all operations related to the Brazilian environment, such as, quality, PR, marketing, export, logistic, training, management of the illy award, etc.

Core suppliers' network: this is the set of suppliers that are part of the Club. We call it core as it shares and promotes by their actions and professional success the quality philosophy of illycaffè. Club members are active participants to the Brazil Coffee University training initiatives and are widely considered as the best quality producers of coffee for espresso in Brazil. illycaffè invests heavily in the relationship with these suppliers, some of whom do collaborate on Research and Development programs with illycaffè and other Brazilian institutions.

Suppliers - not part of the Club. This is a broader set of suppliers from which the core of suppliers is selected. Though this layer is more distant than the Club suppliers, nevertheless, it provides a crucial source of weak links (Granovetter 1992) through which a wide variety of information can flow. Also, it constitutes a basin for further expansion of the suppliers' base.

Multi-channels

Knowledge flows through a variety of different channels:

Top-down

From illycaffè to growers via Brazil Coffee University: this is the channel controlled by illycaffè and illycaffè's inner network in Brazil. This is the preferential channel through which the Brazil Coffee University training

takes place. Two different strands of knowledge flow through this channel: first, technical and scientific knowledge mainly related to coffee growing and acquired in central labs (learning-by-researching) is spread in a top-down fashion to suppliers; second, the channel helps diffuse 'best practice' acquired in a bottom-up fashion (see below).

Bottom-up-down

From growers to illycaffè: visits to farms and conversations with growers in Brazil and beyond enable illycaffè people to observe and select innovations and local practices (largely based on learning-by-doing). These innovations are then diffused within the network through the top-down mechanism described above.

Bottom-out

From growers to growers via Club visits: besides the vertical channels discussed above, we have also observed the spreading of innovations through a horizontal channel. Growers that participate to farm visits organized within the Club observe local innovations and improvements, upon which they can experiment. This process results in a transformation of the initial product/process innovation with a consequent improvement and adaptation to a local context. Occasionally some of the organizational innovations turn into best practices. Further visits will spread the improved product/process innovation throughout the network. The stunning feature of this channel is that it is an unintended consequence of the loyalty program introduced with the Club. It is, in other word, a self-organizing and distributed innovation mechanism. Innovation and knowledge diffusion within the horizontal channel occur outside the planning and intentions of illycaffè's inner network. As it happens with many self-organizing features in network, this one too is exposed to virtuous network effects (or externalities). The larger the network and the more frequent and intense the conversations occurring within, the higher the chance that some innovation will be picked, improved and diffused for the benefit of the whole network. Though in principle this may look like a net loss of competitive advantage, in practice there are powerful reputational and peer-pressure mechanisms at play from which the innovators draw some kinds of returns.

Outside-in

From growers abroad to Brazilian growers: the replication of the Brazilian 'experiment' outside of Brazil has created a further channel for knowledge exchange. For instance, we can document the case of the 'raised beds': this is a technique for drying of coffee cherries, used in the African coffee producing countries, the use of which has been stimulated in various regions of Brazil by an illycaffè executive.

Trust-based and high knowledge density

Our research has revealed that Illycaffè has devised a business model based on four features: a) higher price for coffee supplies, about 30%, or more, than the market price (set at the New York Coffee Exchange); b) investment in improving suppliers' produce quality by freely diffusing technical knowledge; c) direct relationships with suppliers based on personal interactions; d) long-term relationship based on expectation that illycaffè never fails to honour the promise to buy quality coffee.

The synergistic combination of the four business model aspects and the long-term reliability of illycaffè, have generated a high-trust environment, in which at most two degrees of separation exists between any two members of the network. Trust, short path length and wide geographic dimension covered by the network, enable diversity to coexist with a smooth and fast diffusion of information.

Non-exclusive and open

Membership in Club is non exclusive and non-binding. The most entrepreneurial suppliers are usually part of multiple overlapping associations, clubs, cooperatives, and other organizational forms. Participation in multiple fora increases the variety of context to which suppliers are exposed, amplifies the quality and diversity of the circulating information and stories and allows the cross-fertilisation of knowledge among different sectors. In fact, many suppliers have diversified from coffee into other crops, such as soy, sugar cane, wood, etc. and applied the knowledge developed in one sector to the others. illycaffè encourages all initiatives that lead to a quality increase even in the case, as indeed it is often the case, illycaffè's competitors may profit.

This business model together with illycaffè's reputation of the best espresso roaster in the world has generated powerful incentives to join the

network and engendered a high level of trust between illycaffè and suppliers and within the suppliers' community.

The insertion of coffee in the information and knowledge economy

The many connections of the illycaffè network in Brazil show that the strategic actions initiated by the company brought great innovations to the local coffee market. The company always pioneered the use of technology and innovation in its operations, introducing a significant number of advances in the industry. The history of illycaffè is based on the tripod "science, technology and art": these are the factors behind the development of programs for generating, disseminating, exchanging and sharing knowledge and information and guiding the design of tools for communication and relationship management.

Shaping a complex information and knowledge network with suppliers in Brazil is consistent with Spanish sociologist Manuel Castells' description of a "network society". In the late 1990's, Castells published a study on the technological evolution of the many layers of society and its local and global impact. In his book *The network society*, 2007, which first edition was published in 1999 as the first volume of a trilogy called *The information age: economy, society and culture*, the author analyzes the transformations caused mostly by the advance of what he calls the "information and knowledge economy", leading ultimately to a network society.

Although Castells focuses more closely the development of communication and information technologies, the movement towards a network society may help describe the impacts of the network formed by illycaffè and suppliers in Brazil, with connections expanded by interwoven nodes or knots tied by the generation and dissemination of knowledge. Like the movements observed by Castells, this is a local network in terms of operation, yet global in its reach. The knowledge transmitted and generated in Brazil stimulates the production of quality coffee beans exported to Italy, where they are processed together with beans from other countries for the blend used in the *espresso* enjoyed by customers in more than 140 countries. The creation of a network featuring complex connections and embedded in the economy of information and knowledge permeating any intended market (in the case of illycaffè, the premium coffee market) impacts the different economic and social environments described by Castells.

Aware of the emergence of new social and economic scenarios facing global companies, the strategic actions by illycaffè are related to Castells' observation concerning the restructuring process of capitalism itself, characterized by factors associated with the economy of information and knowledge, such as: flexible management; decentralization of companies and organization in internal and external networks (involving other companies and institutions; individualization and diversification of labor relations; incorporation of women in the remunerated workforce; state intervention for the deregulation of markets and dissolution of the welfare state; and fiercer economic competition). (Castells 2007: 39)

Some key points shaping the new scenario described by Castells are also present in the structure developed by illycaffè in Brazil to promote quality. Among them, he mentions economic and social changes leading to an informational, global, networked environment.

It is *informational* because productivity and competitiveness among units or agents in the economy (whether companies, regions or nations) depend basically on their ability to generate, process, and efficiently apply the knowledge-based information. It is *global* because the key productive activities, consumption and circulation, as well as its components (capital, labor, raw materials, administration, information, technology and markets) are organized on a global scale, either directly or through a network connecting economic players. It is a *network* because, under the new historical conditions, productivity and competition arise within a global network of interactions among corporate networks. (Castells 2007: 119)

The characteristics of the new scenario described by Castells show that the unfolding of the information economy is relevant to every economic activity, and very especially to industrial operations. In this context, the network structured by illycaffè to disseminate quality concepts in coffee is aligned with the restructuring of capitalism defined by the author as "informationalism". As for the effects of economic transformations, Castells believes factors such as productivity, competitiveness and profitability are now very much influenced by the accumulation and application of knowledge and the access to information, as well as by the information flow between the different nodes connecting corporate networks in different economic activities. Once again, the concepts described by the author can be used to explain the impact of the knowledge network structured by illycaffè in Brazil, as well as the efficiency of the strategic actions developed by the company to create a base of loyal suppliers using knowledge applied to crops for the production of premium coffee.

Another author who writes on the creation of knowledge networks is Yochai Benkler (2006), professor at the Yale Law School in the US. As

illycaffè's initiatives brought about a cultural change in Brazilian coffee production aiming superior quality through the use of knowledge, Benkler analyzes the function and role of networks within the new scenario where the format in which information and knowledge are disseminated has been altered. The author shows that the traditional communication pattern "one to many" has evolved to "many to many" within networks. This is a more malleable, interactive pattern, allowing for strategies that do not have to conform to the traditional information market, since knowledge is now exchanged and shared.

In his study, Benkler defines the current scenario as the "network information economy", in which the exchange of information through open platforms is increasingly important; where knowledge is built on the decentralized, cooperative, shared traffic of information among peers or individuals sharing the same interests. Albeit driven by clear business objectives, the illycaffè case fits the description, given the fact that the actions went beyond the initial strategies devised to support the purchase of quality coffee beans by the company. In the case of the network structured by the company, the decentralized information flow among coffee producers and the other nodes in the network has fostered the generation of new knowledge on coffee agriculture, as well as the dissemination of proven concepts and techniques as best practices in growing and producing fine coffees.

Although it is a network structured around a company, the nodes in the illycaffè network in Brazil feature a new logic, one that Benkler claims must not be aligned to a market system, or that in other situations may run parallel to such a system. In either system, the logic identified by Benkler and deployed in the knowledge network devised by illycaffè for quality in coffee features clear cooperation between peers in networks, or among the many nodes within them, with information being exchanged and shared and thus generating and disseminating knowledge that can be adopted and applied, and consequently cause ongoing transformation and improvement in reality.

Final considerations – Changing the landscape through knowledge

This article illustrates an example of a knowledge network in the agrifood sector with some notable characteristics both in terms of its evolution (how it came to be) and of the structure of its knowledge flows.

In 1991 when illycaffè changed their purchasing policy they unwittingly triggered a radical change in the coffee value chain (CVC). The global CVC was and still is characterised by two peculiar features: a) geographic isolation between production and consumption - production takes place in the Southern hemisphere, whereas consumption occurs predominantly in the Northern -; b) dramatic imbalances in the CVC (Daviron and Daponte 2005) with a 'coffee boom' in consuming countries and a 'coffee crisis' in producing countries' together with a high supply of 'low quality' coffee and a dire shortage of 'high quality' coffee.

Against this background illycaffè has decided to disintermediate the coffee supply chain and to interact directly with coffee producers in Brazil. This decision had a radical effect: it created a direct link between consumption markets and production markets, through which knowledge and information could flow. Before then, the multiple links between production and consumption (roasters), cooperatives, traders, buyers, financial markets, etc. – effectively blocked knowledge and information flows between users and producers, that is, between consumption in the Northern hemisphere and production in the Southern. One of the great intuitions of illycaffè and specifically of the late Ernesto Illy, the chairman of the company, was to understand that the structure of the global CVC froze the coffee sector and the lives of thousands of growers into a business model geared toward increasing product standardization, ideal for a commodity market (Andriani and Detoni 2007). The way to break the trap of commodity markets was twofold: by designing of knowledge management approach that transferred knowledge of consumption markets directly to the provider of the 'raw' material and by providing a business channel that bypassed the commodity markets and its bottleneck centred around the New York Coffee Exchange.

The network for the generation and dissemination of knowledge structured by illycaffè produced a lasting change in the culture of coffee production activities in Brazil, ultimately leading to the insertion of the country in the worldwide market for premium coffees. Considered as a whole, the strategic actions undertaken by the company to stimulate Brazilian producers to commit to producing superior quality in coffee helped some of the players in the production chain to focus their efforts not only on the commodity market, where quantity alone is what matters, but also on new value-added markets that reward superior quality in coffee beans with higher prices. We can thus say that the illycaffè network contributed to the emergence of a new paradigm for Brazilian coffee.

Other macro-economic reflexes of the strategic actions undertaken by the company in Brazil were the recovery of the image of Brazil as a country producing specialty coffee, conquering international reputation,

and access to new markets both domestically and internationally. In the case of the new market conditions for fine Brazilian coffees, two facts deserve mention. First, the concern illycaffè showed for the quality of Brazilian coffee created a new culture of *espresso* appreciation and consumption. It contributed to creating new habits among Brazilian consumers, and also opened the market to competitors who came to Brazil to offer their products to the newfound consumers of special coffees. The growing number of coffee bars spread out over large cities all over Brazil is evidence of the emergence of new coffee consuming patterns. Also, in the specific case of illy *espresso*, for example, sales went from practically zero in the early 90's to the current 54 tons a year. Second, today there are dozens of special domestic and international coffee brands available in supermarkets as well as coffee bars and many other places for consumers who want to enjoy a good *espresso* and find out more about the different types of coffee. Regarding the larger volume of *espressos* sold today, illycaffè always defended the idea that larger sales are correlated with better quality products available for coffee lovers – that is why the company believes in ongoing investment in new technologies and initiatives to enhance quality in all phases of the production chain, from plant to cup.

For rural producers, after the arrival of the Italian company and the creation of the award for quality, a number of other contests were launched both by public and private institutions, all aiming to identify quality coffees. This is true for all producing regions, both traditional and new areas mapped by illycaffè's pioneering contest. In fact, the contest was decisive for the emergence of new areas associated with quality in coffee, such as Cerrado Mineiro and Matas de Minas (formerly known as Zona da Mata, in the State of Minas Gerais); the Mountains of the State of Espírito Santo; the Western regions in the State of Bahia; the surroundings of the municipality of Piraju, in the State of São Paulo, and the States of Paraná and Goiás.

Among the suppliers that are part of the illycaffè network in Brazil, one can see market recognition for the quality concepts they chose to adhere to, and a new status for being associated with a company whose name is synonymous with premium quality.

The success of the strategic actions leading to the unfolding of the illycaffè network in Brazil inspired the company to export the model to other countries where it also purchases fine coffees for the espresso blend it offers consumers around the world. Today, India and Colombia have their own version of an award that recognizes superior quality in coffee and identifies suppliers and regions producing fine coffees, with a view to

develop a loyal base of producers who can meet the high standards set by illycaffè and its consumers.

References

Allen SL (2001) The devil's cup: coffee, the driving force in history. Canongate Books, Edinburgh

Andriani P, Detoni AF (2008) An interview with Ernesto Illy on complexity, coffee and Management. Emergence: Complexity and Organization Vol. **10**(1): 84–88

Benkler Y (2006) The wealth of networks: how social production transforms markets and freedom. Yale University Press, New Haven and London. Available at: http://www.benkler.org/Benkler_Wealth_Of_Networks.pdf Accessed on Feb. 20, 2008

CACCER – Conselho das Associações dos Cafeicultores do Cerrado (The Council of Coffee Producer Associations of the Cerrado). História do conselho. (History of the council). Available at: http://www.cafedocerrado.org/br/doc/caccer_hist.pdf

Castells M (2007) A sociedade em rede. A era da informação: economia, sociedade e cultura. (The networked society. The information era: economy, society and culture) V. 1, 10th edition. Paz e Terra, São Paulo

Daviron B, Ponte S (2005) The coffee paradox: global markets, commodity trade and the elusive promise of development. Zed Books, London

Granovetter M (1992) Problems of explanations in economic sociology. In Nohria N, Eccles RG (eds) Networks and organizations: structure, form, and action. Harvard Business School Press, Cambridge, MA

Luttinger N and Dicum G (2006) The coffee book: anatomy of an industry from crop to the last drop. The New Press, New York

Watts, D. 2003. Six Degrees: The Science of a Connected Age. Norton, New York

Chapter 4. Learning dynamics within the solutions network

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Introduction

The shift towards the provision of integrated solutions, rather than stand-alone products or services, is increasingly important to the competitive success of leading companies worldwide. This model of industrial organization involves a customer-centric approach to business. Getting closer to the customer, deeply understanding its needs, and fulfilling them with "all-inclusive" offers are essential capabilities of a solutions provider. Such a strategic trend entails a number of organizational changes affecting all the stakeholders involved, both internally and externally with suppliers and customers. As firms develop a new value proposition around, often beyond, their own traditional products or services, new capabilities are generated which reveal fundamental for establishing enduring and far more profitable relationships with customers. These changes offer opportunities of organizational learning, which firms can iteratively explore and exploit in order to gain sustainable competitive advantages.

This chapter provides an overview of the learning dynamics associated with the provision of integrated solutions in the capital goods sector. In that context, the typical configuration of integrated solutions involves a project-based network of industrial relationships with suppliers and competitors oriented towards the specific needs of a customer firm. Within this "solutions network", multiple sources of knowledge trigger simultaneous processes of intra- and inter-organizational learning.

A. Romano and G. Secundo (Eds.), *Dynamic Learning Networks: Models and Cases in Action*, 75
DOI: 10.1007/978-1-4419-0251-1_4,
© Springer Science + Business Media, LLC 2009

The learning dynamics examined in the chapter relate to the specific capabilities a firm has to acquire to become a solutions provider; how such capabilities can support the repeatability of high-value integrated solutions; and what is the interplay between intra- and inter-organizational learning. Subsequently, the customer perspective is considered, by emphasizing the central role it plays in reorganizing for outsourcing, reporting new categories of needs from the interaction with an increasingly complex environment, and feeding the supplier's learning process. Finally, the reflections on increasing environmental complexity lay the basis for the introduction of the new concept of "network proactivity" as a core capability of the networked solutions enterprise.

Integrated solutions: a collaborative business model

The shift toward solutions: origins, drivers and recent evolution

Over the last two decades, leading companies worldwide have striven to provide customers with increasingly comprehensive combinations of products, services and technologies, specifically designed and implemented to work together as tailored integrated solutions. The shift toward solutions started in capital goods in the early 1990s, when pioneers like IBM, General Electric, and ASEA Brown Boveri (ABB) moved from a base in manufacturing into a range of consulting, financing, and maintenance services conceived to complement physical products and systems over their life cycle. A similar trend has also affected consumer goods, whose offerings have been progressively upgraded by consumer credit, maintenance contracts, and short-term warranties (Davies et al. 2007). The emergence of solutions is a response to major economic and market changes, recognized by an emergent body of literature as key drivers for the phenomenon, such as: the intense competition in manufacturing, which has forced firms to move downstream and differentiate their offer (Wise and Baumgartner 1999); the growth in scope and value of services in contrast with the commoditization of products (Vargo and Lusch 2004); and the outsourcing of non-core activities by large business and government-owned companies (Davies et al. 2006). Today, both customers and suppliers benefit from the higher value of solutions. On the one hand, consumers enjoy sophisticated purchase experiences when buying products which embody multiple features or enable the fruition of services, while business customers exploit the typical advantages of outsourcing and keep focused on their core ac-

tivities. On the other hand, suppliers seek to add value to commoditized products, expand their life-cycle, differentiate from competitors, obtain longer revenue streams and larger margins from high-value services, and improve customer satisfaction and retention.

Integrated solutions have so far reached their most mature configuration in capital goods and services (e.g., flight simulators, power generation, telecommunication systems, infrastructures and built environment, global financial services, etc.), where they appear as the ongoing result of networks of exchanges and collaborations among a number of actors involving:

- a typically large *customer* with a pressing business or operational problem;
- a *systems integrator*, i.e., a company that bears responsibility for solving that problem through the integration of products, technologies and services into seamless packages; and
- a number of *(co)suppliers* respectively linked to the systems integrator for the provision of product and service components (Fig. 4.1).

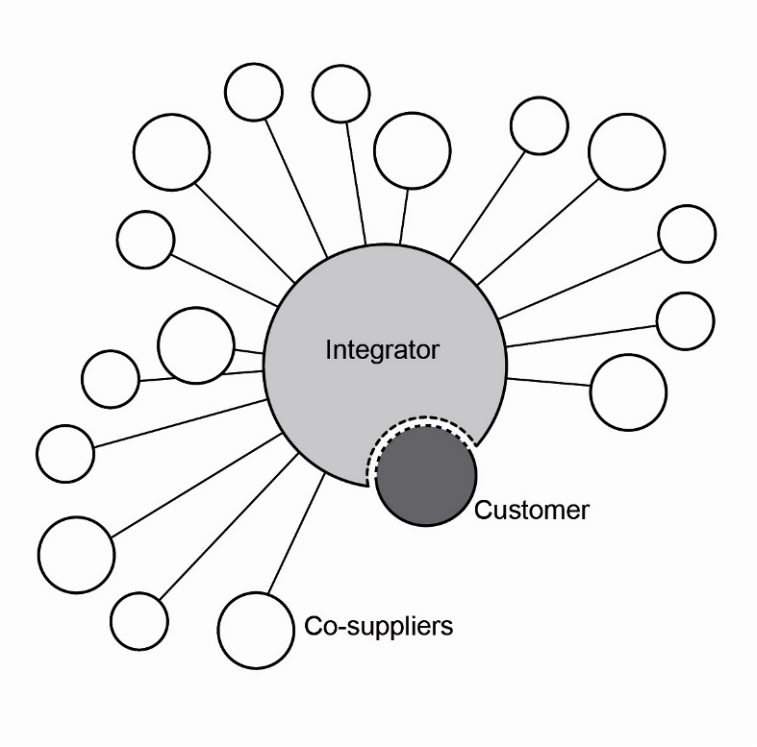


Fig. 4.1. The solutions network.

The customer is at the center of the solutions network, which is built around it by a systems integrator able to interpret its needs and to organize resources, activities and processes for their continuous fulfillment. But what does customer-centricity really mean? How is systems integration pursued, and what does it imply?

Refocusing the business and reorganizing for customer-centricity

The transition from stand-alone products or services to high-value integrated solutions entails a number of radical shifts in the role a firm plays within the industry. Indeed, it can be considered as a change of 'paradigm' in firm's strategy and organization which affects all aspects of a business model, from the relationships with customers, (co)suppliers and competitors to the very concept of value creation and delivery (Bessant and Tidd 2007).

The goal of a solutions provider is the fulfillment of customer's outsourcing needs on a dynamic basis. Initially, firms' attempt to deliver more value to customers generally involves a form of product bundling: products and services are marketed together in more or less inclusive packages in order to address more articulate requirements. Although they represent the first step towards the provision of integrated solutions, bundles of products and services generally do not require relevant processes of organizational learning and do not necessarily entail a tight relationship with customers.

Solutions can be differentiated from mere bundles not only for the higher degree of complementarity and integration among their components, but also because they "involve the transfer of responsibility for various of the customer's in-house functions and – in some cases – the transfer of the customer's physical assets and staff back toward the supplier" (Davies and Brady 2000). Moving from its traditional base in aircraft engines, Rolls-Royce, for example, relies on increasing revenues generated by its 'Power-By-The-Hour' services, which guarantee the availability of perfectly working engines, whose fixed maintenance costs are spread over long term contracts. Similarly, Alstom has become a global leader in equipment and services for train transport by expanding in systems integration and providing its clients with complete packages: fixed infrastructure, rolling stock, signaling systems and services for *operating and maintaining* trains (Davies et al. 2007). However, a high level of commitment to each client cannot be sustained by the traditional product-centric view of

the business, according to which a firm should provide as many buyers as possible with undifferentiated products or services. On the contrary, a solutions supplier focuses on customer's business processes, relentlessly reorganizes resources and activities into new combinations, and looks to assure long term *joint* performances.

Turning away from a product-centric logic yields profound transformations in the relationships between the solutions provider and other two categories of industry actors: (co)suppliers and competitors. A systems integrator is not immune to the same trends of outsourcing and reorganizing experienced by its customers. As it becomes less dependent on its original product and focuses on systems integration and service activities with the sole purpose of creating value for and with the customer, this firm reorganizes internal functions, outsources non-strategic activities to subcontractors, and redefines the whole network of upstream collaborations with suppliers of product and service components, even including competitors. IBM, for example, has been among the first to recognize that the ability to specify and integrate within its own solutions a competitor's technology – if a customer demands it or when it provides superior performances – is what really characterizes a solutions provider focused on customer needs.

From a network perspective, the reorganization process triggered by changing customer requirements can reshape the whole industry through the mediating role of the systems integrator. This firm works as the hub of the solutions network, agenting customer's demands while dealing with outside suppliers (Auramo and Ala-Risku 2004). The solutions provider and its customer start and nurture complex processes of value co-creation by leveraging tangible and intangible resources, specialized skills and knowledge for attaining competitive advantages within a long term symbiotic relationship (Vargo and Lusch 2004).

Capability building and organizational learning

As we have just outlined, organizing for the provision of high-value integrated solutions entails a challenging paradigm shift (Tab. 4.1): from product- to customer-centricity, from output to performance, from transaction to relationship, from value delivery to value co-creation, from competition to co-supply, from value chain to value network (Gummesson 2006; Vargo and Lusch 2004; Ballantyne and Varey 2006; Cova and Salle 2000; Rust 2004, 2006). The scope and depth of change are far-reaching. Firms that fail to fully gauge the scale of such implications, not only miss the chance to gain higher levels of competitiveness, but also run the risk to get relegated to the sideline of the industry. In order to successfully take up the

challenges of the solutions paradigm, firms need to build new sets of capabilities by promoting, monitoring and exploiting an iterative process of intra- and inter-organizational learning. System complexity, service intensiveness, and intricate plots of codified and tacit knowledge, make the learning process far more difficult. However, system integrators that first succeed in acquiring and exploiting the necessary capabilities build a significant edge on competitors (Chesbrough 2004).

Table 4.1. The solutions paradigm shift.

From	To
Product-centricity	Customer-centricity
Output	Performances
Transaction	Relationship
Value delivery	Value co-creation
Competition	Co-supply
Value chain	Value network

Most of the capability building efforts concern the fundamental role of specialized services. In fact, integrated solutions in capital goods are complex combinations of products and technologies designed, implemented and put to work together by means of high-value services. In this context, services are crucial activities which link products, technologies and systems to specific applications and those applications to final performances. Given the high level of interdependence among supplier and customer's contributions to performances, organizational learning has to be promoted and pursued beyond and independently of firms' boundaries. The organization mirrors the solutions (Galbraith 2002), but also – and more subtly – solutions mirror the organization. The success of a solutions provider measured in terms of performance and market acceptance depends on the thoroughness of the combined provider-client organizational learning.

The next section presents an overview of the learning dynamics within the solutions business model, holding both supplier and customer's perspectives.

Learning dynamics within the solutions business model

The configuration of the necessary capabilities

Firms that aim to become solutions providers need to acquire – and develop on an ongoing basis – a wide range of strategic, organizational, technical, marketing, and mediating capabilities. A taxonomy based on the business function they are developed for comprehends four sets of key capabilities: systems integration, operational services, business consulting, and vendor financing (Davies 2004; Davies et al. 2006).

In complex products and systems, a provider of integrated solutions is chiefly a systems integrator, i.e., a company able to design and integrate internally or externally developed components – products, services, and software – into unified systems and to coordinate the activities of component, subsystem or product suppliers (Prencipe et al. 2003). Hence, *systems integration* is a core capability for the solutions provision. Far beyond the scope of the traditional engineering-based task, it is a strategic business activity aiming at the co-creation of value with customers and at the transformation of their own businesses (Davies et al. 2006). Empirical studies in several sectors have confirmed that all companies that adopted a solutions business model also developed systems integration capabilities (Hobday et al. 2005; Ceci and Prencipe 2008).

Soon after integrating and implementing complex systems for their customers, solutions providers are often requested to operate, maintain, upgrade or renovate them along the entire life cycle. Thus, *operational services* are another fundamental set of capabilities associated with the provision of integrated solutions and their specific configuration substantiates the breadth of the solutions provision and the depth of the supplier-customer collaboration. Customers that outsource operational activities recognize that they must reshape the boundaries of their own organizations, open to suppliers and engage in long term partnerships with them.

Additionally, solutions providers often need to develop *business consulting* capabilities, in order to help customers deal with a range of different domains of knowledge, respectively related to conceiving, planning, designing, financing, building, operating, maintaining and, in some case, decommissioning complex systems. The precious knowledge acquired from consulting services is employed in ensuring the compatibility of (co)suppliers' contributions and achieving compliance to customer's needs. From this perspective, even before the integration of products and services, solutions providers perform a fundamental process of knowledge

integration which lays down the foundations for a network of mediated inter-organizational exchanges.

Moreover, the realization of complex systems is usually capital-intensive and entails considerable long-term investments. Some solutions providers have developed specific capabilities in *vendor financing* and offer financial support to customers with limited funds. Finally, in particular industries (e.g., transportation systems, built environment, etc.), a capability of *asset management* is offered to customers seeking to keep efficient and extend the operating life of a significant installed base of products (Davies 2004).

Systems integrators iteratively acquire and develop these four sets of capabilities through a complex process of organizational learning whose dynamics go well beyond the boundaries of one single firm and involve support and contributions from both customer and (co)supplier organizations.

The dynamics of the learning process

The journey toward the provision of high value integrated solutions entails a long and difficult process of organizational learning, which develops along three complementary phases.

In the first phase, the firm has to correctly interpret customer's requirements by fully understanding its business, industry, processes, and relationships with the end users. At the same time, due to the long-term nature of a solutions provision, from design to implementation, also future needs have to be anticipated and taken into account at an early stage (Kalwani and Narayandas 1995). The demand-based knowledge acquired at the interface with the customer organization has to be filtered, internalized and compared with the firm's existing capabilities, in order to start a first loop of *demand-driven learning*.

In the second phase, a comparison between demand-based knowledge and internal capabilities allows the firm to define the possible requirements for additional capabilities and resources to be sought externally. In turn, the interaction with (co)suppliers generates a supply-based wealth of knowledge which has to be internalized and dynamically compared with internal capabilities within a second loop of *supply-driven learning*. The intersections between the two loops of demand- and supply-driven learning give rise to the fundamental process of *knowledge integration* on the basis of which systems integration capabilities can be developed and strengthen on an ongoing basis. Due to the pivotal role of the integrator company,

demand- and supply-driven knowledge is combined through simultaneous processes of intra- and inter-organizational learning.

In the third phase, organizational learning takes place when the solutions provider has accumulated sufficient experience and develops the ability to reuse integrated knowledge for future projects across different industries.

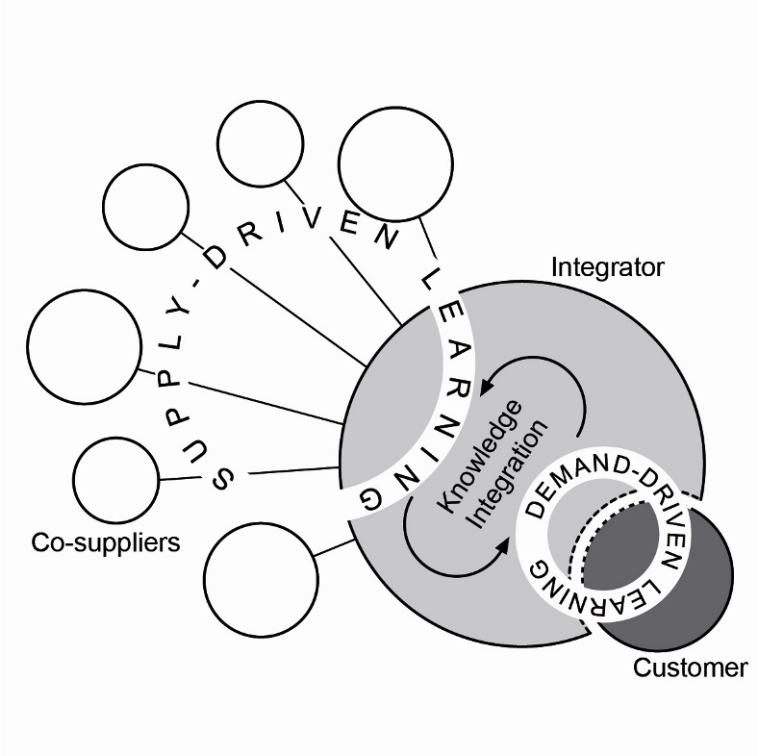


Fig. 4.2. Learning dynamics within the solutions network.

The integrator perspective

From the systems integrator's point of view, the process of organizational learning outlined in the previous section is hindered by a series of challenges peculiar to the solutions business model. Some of these challenges are related to the service-intensive and collaborative nature of solutions. Indeed, integrator and customer equally participate in simultaneous activities of delivery and utilization of the product-system, by defining, reshaping and integrating its service components. In this ever changing picture, the onus is on the integrator to guarantee effective knowledge transfer and

utilization, while coping with systemic complexity (Chesbrough H. 2004). Another challenge facing the integrator company is represented by the continuous tensions between the capabilities requested by customers and those which are functional to the firm's strategy (Miller et al. 2002). These tensions easily affect the relationships among the integrator firm units, especially the front-end unit (FEU) at the interface with the client, and the back-end unit (BEU) responsible for sustaining firm's capability building processes. Finally, a company that moves from a base in products downstream toward industrial services often has to deal with a demand for services whose variability cannot be smoothed very effectively. The solutions provider has to learn how creating adjustable service capacity in order to match the demand, by leveraging the salary structure, cross-training employees, and incentivizing customer participation (Auramo and Ala-Risku 2004).

In order to turn difficulties and challenges into opportunities for organizational learning, the integrator firm has to capitalize on knowledge acquired from clients and integrated internally, by emphasizing the mediating role of its FEUs.

The supplier-customer interface: emphasis on the front-end

The analysis of the learning dynamics associated with the solutions provision has clarified the extent to which this business model is customer-centric and the importance of the collaborative relationship between customer and supplier. The journey toward the solutions provision starts with a deep understanding of customer needs and ends with their fulfillment on a long term basis. The customer is constantly at the center of a dynamic network of relationships within the industry. Within the supply network, the supplier must concentrate on the strategic, organizational, technical, marketing, and mediating capabilities, while its FEU plays the indispensible role of interfacing the client organization.

In many aspects, the FEU resembles a corporate consulting unit (Sandberg and Werr 2003). It works like a knowledge-creating organization (Nonaka et al. 1995), but at the same time possesses the characteristics of a project management unit, as it is definable in terms of specificity and uniqueness of goals, complexity of task interdependence, and criticality to the company's global results (Middleton 1967). Due to constant interactions with customers, the FEU rapidly accumulates explorative knowledge on whose exploitation the overall performances of the supplier depend. Far from being a simple repository of knowledge, the FEU operates as a broker of capabilities and resources, for the punctual fulfillment of customers'

needs. For example Citi Group's FEU is made of account managers each dedicated to serve a single client worldwide. "These are elite individuals with considerable power and connections who negotiate hard to get the best products and services for their customers. Citi's superb functional, product, and regional capability managers at the back-end of the organization serve those account managers and their clients responsively, and on demand" (Miller et al. 2002).

Nevertheless, knowledge transfer from the FEU to the BEU is by no means a trouble-free process. In many cases, only part of the knowledge acquired is actually internalized and used by the firm. The difficulties involved in knowledge absorption are often due to the differences in the business logics between the FEU (customer-centric) and the BEU (by nature, more product-centric), and the fact that the new knowledge introduced by the FEU tends to meet resistance from the back-end as it undermines established routines and power bases (Sandberg and Werr 2003). Firms can overcome these obstacles by actively promoting an intense dialogue between the FEU and the BEU since the very early stages of each project, paying attention to not reducing the autonomy of the FEU in its task of dragging the whole company toward customers and market-triggered innovation.

Learning for repetition

Organizational learning accompanies and underpins the evolution of the solutions business model throughout the three phases: offering of simple bundles of products and services, customization of integrated solutions, and provision of repeatable solutions. Solutions providers need to enter the third phase of this transition in order to assure profitability and long term competitiveness. To do so, they need to learn how to leverage internal resources as well as external collaborations within solutions network in order to balance customization and standardization.

Repeatability – or replicability – is the firm's winning post in the long journey toward the provision of integrated solutions. After undergoing a series of business and organizational transformations and effectively carrying through the process of organizational learning, a systems integrator accumulates enough knowledge to replicate solutions across multiple projects and customers, so exploiting "economies of repetition" (Davies and Brady 2000). Rather than take up unique projects, the company can pursue higher efficiency and effectiveness by delivering 'repeatable solutions' and leveraging the capabilities acquired and developed within the supply- and demand-driven learning loops. Such learning can be exploited more prof-

itably when knowledge and routines are re-employed from project to project and just a limited part of the global offering is tailored to each new client. For example, "Citi caters only to industries in which it has significant business volume and experience – relying as much on standard 'industry templates' as on individual solution tailoring" (Miller et al. 2002). Similarly, Sun Microsystems' i-Force program brings together people from Sun, as well as from its customers and partners, to develop customer-driven 'ready solutions', or 'solutions champions', that serve as templates to rely on when dealing with new customers (Galbraith 2002).

Integrators that are able to offer repeatable solutions enjoy economies of scale and scope simultaneously. If cost reduction associated with the delivery of an increasing number of solutions projects (economies of scale) has been clearly recognized by the specialized literature (e.g., Davies and Brady 2000; Davies et al. 2003), economies of scope have been underestimated and deserve special attention. In fact, solutions providers that manage to create standard modules which can be differently combined and recombined in order to address changing customer needs across several industries, not only harvest and exploit experience from earlier projects, but also acquire increasing flexibility because they can fulfill changing customers' needs or even embark on new projects across different customers, businesses or technologies. Firms that learn how to master the modularization of solution components *standardize modules and customize their combinations*, depending on the specific needs posed by demand. IBM offers an example of how this approach works. IBM's BEUs make sure that R&D investments can be leveraged by a plurality of solutions. To make solutions especially economical, the BEUs work with FEUs to create solutions platforms. BEUs work hard to deepen capabilities economically, "by using common standards and protocols, leveraging research across related products, and leveraging similar solutions across many clients" (Miller et al. 2002).

Organizing for repeatability through the modularization of solutions components implies that firms learn how to "deconstruct, recombine and reuse complex knowledge from one instance to another" (Chesbrough 2004). This task is especially difficult for systems integrators as knowledge recurrently springs from and has to be captured within a variety of actors throughout the solutions network. Moreover, customers can provide a range of important contributions to organizational learning.

The next section will examine in detail what is customers' perspective on solutions and what they can do to feed and enhance the learning processes.

The customer perspective: feeding the learning process

The analysis of the organizational learning process taking place within the solutions business model has focused on the network of industrial relationships emerging and evolving around a central actor (the solutions supplier) and its specialized units (FEUs and BEUs).

Despite incentives to collaborate in order to share long-term value added, customers and suppliers often do not look at the solutions provision in the same way. An interview-based study has recently revealed that, while many solutions suppliers still "view a solution as a customized and integrated combination of goods and services", customers always perceive it as a "set of customer-supplier relational processes" (Tuli et al. 2007). Two of these processes, customer requirements definition and post-deployment customer support, appear particularly significant, as interviewed customers stress that they "frequently are not fully cognizant of their business needs and cannot easily articulate them to a supplier"; furthermore, "post-deployment support includes deploying new products in response to evolving requirements of a customer", and "delivering solutions is better viewed as an ongoing relationship". But how can customers help solutions providers correctly interpret their current and future needs? How can they stimulate the overall learning process and increase the effectiveness of solutions? Tuli et al. (2002) find that three variables can be leveraged by customers in order to sustain the customer-supplier combined organizational learning: customer adaptiveness, political counseling, and operational counseling.

Customer adaptiveness refers to customer willingness to adjust to supplier's specifications. The extent to which the customer opens to the organizational changes a supplier suggests as necessary for redesigning the comprehensive business model directly impact on the effectiveness of integration and customization. If customer does not adapt, "the number of modifications required in supplier products is greater, which makes it more difficult to customize and integrate them" (Tuli et al. 2007); in addition, customer 'closeness' might compel the solution provider to redesign internal processes, as well as alliances and collaborations with (co)suppliers.

Political counseling refers to a customer's willingness and ability to help the supplier fully understand the political relationships among the various stakeholders within the customer organization. The pervasiveness of solutions procurement process is so high within the customer organization that it can affect the sensitivity of internal stakeholders and undermine the balance of power. Recognizing and relieving possible tensions is a fundamental part of the supplier's learning a customer can facilitate.

Finally, operational counseling refers to customer's ability to make the supplier fully aware of operations within the customer firm, i.e., technical constraints, business requirements, organizational processes, environmental issues, policies, etc. The customer plays a major role in speeding up the supplier's learning process and helping it accurately codify the solution requirements. Similarly, the customer can use operational counseling for keeping the supplier up to date on emerging needs, so that it can either amend or fine-tune the solutions.

Embracing the customers' perspective on solutions leads suppliers to conceive their provision as an ongoing relational process which assigns customer a considerable part in iteratively reshaping the solutions network structure and relationships. From this point of view, "customer-centricity has to be translated into process-centricity or customer-process-centricity" (Tuli et al. 2007).

Future trends

Previous sections have provided an overview of the organizational learning process within the solutions business model throughout three life cycle phases. Although progress appears rather uneven across sectors and countries, some firms are already experiencing the third phase of evolution of their business model and are trying to acquire and nurture replication capabilities. These firms are already in the position of considering what the next step is and how their demand-supply networks can be strategically managed and made responsive to forthcoming changes. Thus, a few considerations on further evolution of the phenomenon will close the chapter.

Getting ready for environmental changes: network proactivity

It is likely that firms will continue to make the transition towards the provision of increasingly sophisticated and comprehensive solutions. The drivers behind this trend are becoming even more important: competition in manufacturing and commoditization of products, higher margins generated by services, processes of reorganization and outsourcing. Moreover, the turbulence of environment and the high pace of change justify the increasing adoption of more adaptive project-based models for sustaining organizational learning and pursuing business success (Pinto and Kharbanda 1995; Davies and Hobday 2005).

Predictably, for most advanced systems integrators the next step will be about acquiring agility and flexibility, so that firms will have to anticipate

and respond adequately and rapidly to environmental transformations. In particular, the customers' environment is likely to constitute the main source of threats and opportunities. As suppliers widen the scope of their solutions to undertake larger parts of customers' activities and operations, they may become more exposed to modifications in the clients' external environment, which impacts directly on their systems integration and operational servicing capabilities. Examples of environmental changes can include the new expectations of end-users, a further reconfiguration of the public role in certain economic sectors (Hobday et al. 2005), or the emerging demand for different profiles of sustainability (social, economic, environmental, etc.). At the same time, new challenges might be posed from the supply-side, not only by technological and scientific factors (e.g., emergence of new dominant designs), but also by (co)suppliers' ambitions to dislodge the incumbent systems integrator from its central position within the industry. A broad acceptation of environment is consistent with the network perspective adopted for investigating the learning process within the solutions business models and underlines the importance of paying constant attention to the relationships which tie all together the actors of the solutions network.

Indeed, two intrinsic characteristics of solutions – architectural complexity and functional interdependency – are important in this context. Architectural complexity refers to the fact that each solution is comprised of a variety of components, or modules: physical products and devices, embodied technologies, pieces of software, and service components like consulting, maintenance, or financing. Each of these components also represents a distinct interface with the external world (e.g., commodity, technology, labor, or financial markets) and a potential channel of changes and transformations. Functional interdependency refers to the fact that all solutions components are, to some extent, interdependent: physical products or devices can perform without technologies or software, which in turn cannot be exploited without the support of important services. High interdependency leads to the possibility that an external modification affecting a single component can trigger a domino effect on other components, impelling vast reconfigurations of the global package.

These considerations should lead solutions suppliers to develop additional capabilities for adjusting to changing scientific, technological, market and economic factors. In this respect, we define *network proactivity* as the firm's strategic and organizational ability to anticipate and respond to environmental changes through rapid reconfigurations of the network relationships among its internal units and between those units and either or both demand and supply networks. In order to seize an anticipated market threat, for example, the firm could rapidly recombine solutions compo-

nents and, in parallel, the supply relationships with the providers of those components. Similarly, the same ability would be required if, for instance, that firm had to exploit the arising of a new technology by modifying the way it creates value with customer organizations or restructuring its customer base across new markets. Yielding to the complex and dynamic nature of upstream and downstream relationships, network proactivity is a core capability of the modern networked solutions enterprise.

Toward complete integration

In terms of future evolution of the solutions business model, the concept of network proactivity emphasizes the role of systems integrators not only as hubs of industrial networks, but also as brokers of knowledge and innovation, able to dynamically mediate market demand and industrial offer through a constant reconfiguration of network collaborations and exchanges. Indeed, integrator firms are in a strong position for acquiring new knowledge from the environment through a variety of channels and interfaces, elaborating it together with customers and industrial partners, integrating and leveraging it in order to improve the competitiveness of the overall network. Therefore, the integration of products, systems, services and technologies into complete solutions can be investigated on two levels: first, on a business process level of the systems integrator, it appears as an integration of complex knowledge and skills, iteratively generated, codified and reused through interactions with customers and (co)suppliers; second, on an industrial organization level, it is an integration of different actors' strategies, variably contributing to the fulfillment of the same customer needs.

Conclusions

The analysis of the learning dynamics within integrated solutions has revealed the interplay existing between two levels of organizational learning. First, at the intra-organizational level of the integrator firm, the learning process supports the acquisition and development of a set of four fundamental capabilities: systems integration and operational services, complemented by consulting and financing services. Second, at the inter-organizational level of the relationships between systems integrators and supply and demand actors, far more sophisticated processes of knowledge creation, acquisition and integration happen. At the front end, the focus on

customer's needs and the deep understanding of its pressing business or operational problems set the basis for the ongoing acquisition of new knowledge which is then internalized and combined with internal capabilities within a loop of iterative demand-driven learning. At the back end, additional capabilities and resources are both developed internally and sought externally. Intense and long-term exchanges of information and collaboration with (co)suppliers and competitors generate new knowledge, which is internalized and compared with internal capabilities within a second loop of supply-driven learning. At the intersection between demand and supply-driven learning, a process of knowledge integration occurs, which assigns system integrator the role of mediator of industrial relationships. This firm reduces a series of obstacles to the integration of products, services and technologies (potentially competing technologies, different relative dimensions of (co)supplier firms, contrasting corporate cultures, etc.), and pursues a constant alignment of the integrated offering with evolving customer needs, due to its in-depth knowledge of a client's business and operational problems.

The network perspective adopted in this chapter enables us to identify two challenges that most mature systems integrators face when looking at both customers and (co)suppliers' environment: solutions repeatability and environmental turbulence. As Davies et al. (2006) put it, "repeatability is the measure of a company's progress in providing integrated solutions". The large investment required to acquire and develop new capabilities during the early stages of a journey toward solutions is more likely to yield returns when firms that have already succeeded in the provision of bundles first, and customized solutions then, can enter the third phase of evolution of their business model and put their knowledge to work more productively on a large number of projects. Firms that learn how to customize combinations of standardized modules maximize the exploitation of acquired capabilities, attain higher flexibility in moving from project to project, and enjoy economies of scale and scope simultaneously.

We suggest that the ultimate challenge, however, stems from the environment. Two intrinsic characteristics of solutions – architectural complexity and functional interdependency – will expose suppliers to increasingly rapid change and uncertainty. In future, solutions providers may need to acquire agility and flexibility in order to respond adequately and rapidly to transformations happening in demand and supply. We introduced the concept of *network proactivity* to identify the firm's strategic and organizational ability to anticipate environmental changes and assure ongoing high responsiveness through rapid reconfigurations of the network relationships among internal and external units within both demand and supply networks. As environmental turbulence heightens, the transfer of risk and re-

sponsibility from customers to systems integrators will make solutions even more complex and challenging. New policy and business requirements for sustainability (social, technological, economic, and environmental) are encouraging firms towards a second generation of totally integrated solutions. Their success will depend on how effective they are at establishing enduring network relationships with customers and industrial partners and leveraging those relationships to create innovative and more effective ways of anticipating and responding to forthcoming challenges.

References

Auramo J, Ala-Risku T (2004) Challenges for going downstream. International Journal of Logistics: Research and Applications, 8 (4):333–345

Ballantyne D, Varey RJ (2006) Creating value-in-use through marketing interaction: the exchange logic of relating, communicating and knowing. Marketing Theory, 6 (3):335–48

Bessant J, Tidd J (2007) Innovation and entrepreneurship. John Wiley and Sons, Chichester

Ceci F, Prencipe A (2008) Configuring capabilities for integrated solutions: evidence from the IT sector. Industry and Innovation, 15 (3):227–296

Chesbrough H (2004) A failing grade for the innovation academy. Financial Times, September 24th

Cova B, Salle R (2000) Rituals in managing extrabusiness relationships in international project marketing: a conceptual framework. International Business Review, 9 (6):669–85

Davies A, Brady T (2000) Organisational capabilities and learning in complex product systems: toward repeatable solutions. Research Policy, 29:931–953

Davies A, Brady T, Hobday M (2006) Charting a path toward integrated solutions. MIT Sloan Management Review, 47 (3):39–48

Davies A, Brady T, Tang P (2003) Delivering integrated solution. Brighton: SPRU-CENTRIM, 1–34

Davies A, Brady T, Hobday M (2007) Organizing for solutions: systems seller vs. systems integrator. Industrial Marketing Management, 36:183–193

Davies A (2004) Moving base into high-value integrated solutions: a value stream approach. Industrial and Corporate Change, 13 (5):727–756

Davies A, Hobday M (2005) The business of projects: managing innovation in complex products and systems. Cambridge University Press, Cambridge

Galbraith JR (2002) Organizing to deliver solutions. Organizational Dynamics, 31 (2):194–207

Gummesson E (2006) Implementing the marketing concept: from service and value to lean consumption. In: "Invited commentaries on the service-dominant logic by participants in the Otago Forum". Marketing Theory, 6 (3):289–98

Hobday M, Davies A, Prencipe A (2005) Systems integration: a core capability of the modern corporation. Industrial and Corporate Change, 14(6):1109–1143

Kalwani MU, Narayandas N (1995) Long-term manufacturer–supplier relationships: do they pay off for supplier firms? Journal of Marketing, 59:10–16

Middleton CJ (1967) How to set up a project organization. Harvard Business Review, March-April, 73–82

Miller D, Hope Q, Eisenstat R, Foote N, Galbraith J (2002) The problem of solutions: balancing clients and capabilities. Business Horizons, March-April, 3–12

Nonaka I, Takeuchi H (1995) The knowledge-creating company. Oxford University Press, New York

Pinto JK, Kharbanda OP (1995) Lessons for an accidental profession. Business Horizons, March/April, 41–50

Prencipe A (2003) The business of systems integration. Oxford University Press, Oxford.

Rust R (2004) If everything is service, why is this happening now, and what difference does it make? In "Invited commentaries on evolving to a new dominant logic for marketing". Journal of Marketing, 68:23–24

Rust R (2006) Does the service-dominant logic need to go further? In "Invited commentaries on the service-dominant logic by participants in the Otago forum," Marketing Theory, 6(3):289–98

Sandberg R, Werr A (2003) The three challenges of corporate consulting. MIT Sloan Management Review, Spring, 59–66

Tuli KR, Kohli AK, Bharadwaj SG (2007) Rethinking customer solutions: from product bundles to relational processes. Journal of Marketing 71:1–17

Vargo SL, Lusch RF (2004) Evolving to a new dominant logic for marketing. Journal of Marketing, 68:1–17

Wise R, Baumgartner P (1999) Go downstream: the new profit imperative in manufacturing. Harvard Business Review, 77, September-October, 133–141.

Chapter 5. Towards "Stakeholder University" as dynamic learning network: the Finmeccanica case

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Challenging the new Competitive Landscape: the emerging Networked Learning Organizations

Over the past 10 years, a radical transformation processes is at work, in terms of converging industry boundaries and change rates in global-scale markets and industries, and increasing customer influence (Ghoshal and Bartlett 1997; Cusumano 2001).

The main forces that have determined this transformation process may be identified in (Marquardt 2002; Baets and Linden 2003):

- Globalization and the Global Economy, that have caused the convergence of economic and social practices, of interests, values and tastes, of challenges and opportunities; as a consequence global firms have emerged, that operate at a global level, using global sourcing for human resources, capital, technologies, facilities, row materials.
- Technology, and mainly the Information and Communication Technologies, that have created a so-called "timeless and space less workspace", where people work no longer in fixed groups, at fixed times, and in specific locations: they develop working activities on the basis of exactly what is needed, just in time, and only where it is needed. This creates more convenience and higher productivity of employees, more efficiency, less stress. This type of workspace rests on the concept of

A. Romano and G. Secundo (Eds.), *Dynamic Learning Networks: Models and Cases in Action*, 95
DOI: 10.1007/978-1-4419-0251-1_5,
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coevolution: the idea that by working with direct competitors, customers and suppliers, a company can create new businesses, markets and industries. People and organizations view themselves as part of a sort of virtual business ecosystem, as a player among many, each coevolving with the others. In the global virtual market, organizations have to take use of other players for capacity, innovation, capital.

- Rapidly escalating change and chaos: the current highly dynamic and complex environment produces almost perpetual disequilibrium in the firm. The challenge is then coping with multiple states of coexisting temporary stability and instability. These conditions require flexibility, speed and innovation that allow firms to reduce periods of instability by making rapid and effective changes.
- Emergence of knowledge and learning as major organizational assets. Knowledge has become more important for organizations than financial resources, market position. Knowledge becomes necessary to update products and services, changing systems and structures, finding solutions to problems, increase employees' ability to identify e develop improvements. Organizations not only need knowledge, but also the skills and competencies to dynamically update and put knowledge into practice.

The impact of these forces may be synthesized in the need for organizations to manage discontinuities and unpredictable change in their environments. Organization's transformations concerns mainly:

- their strategic focus, that has moved from efficiency (to reduce costs and time) towards new forms that allow a continuous management of change. These new networked forms are made up of temporary networks of independent firms, suppliers and distributors that share skills, cost and accesses to markets. Each firm maintains a minimal set of permanent core competencies, integrated by independently contracted partners. The general strategy is based on a value-creating logic, focused on a continuous innovation process, aimed at creating new products and technologies to be offered to the final customer. Indeed new products/technologies allow these networks of firms to obtain temporary higher margins, that however rapidly decrease because of the market and competitors pressures. Networks have then to find new opportunities in form of new products and technologies, also by allocating resources to uses that not yield the highest returns: each organization becomes a society's agent of change. As a consequence top managers have to consider themselves not simply as designers of corporate strategies, but as shapers of the context more suitable for speeding up innovation processes trough:

  - Entrepreneurial processes, assuming that individuals can take initiatives, by creating contexts and mechanism that commit and empower people to do so;
  - Integration processes, by creating environments that support collaborative processes;
  - Renewal processes, that capitalize on the human motivation to learn, by creating resources and tools that support people in their learning processes.
- their critical tasks, that are less and less physical assets, and more and more mental assets. The enhanced competition, and the increasing demands from consumers, require that firms continuously reinvent their sources of value creation, so to reach a constant flow of competitive advantage (Eisenhardt and Sull 2001; Eisenhardt and Brown 1998). This new organizational forms are based on the organization's capacity to continuously learn from past experience, to be willing to unlearn and experiment new ways of solving problems. Change equals learning (Dalin and Rolff 1993; Fullan 1991) and survival must be based on a process of constant learning (Morrison 2002), also at an inter-firm level (Bessant and Tsekouras 2001). They can be increasingly assimilated to entities involved in the continuous development of organizational capabilities. In a research conducted on learning departments and organizations, Accenture found (Meister 2002) that companies with high-performance learning processes returned better revenue and profit growth compared to their competitors and industry peers:
  - productivity (measured by sales per employee) was 27% greater;
  - revenue growth was 40% higher;
  - net income growth was 50% higher.

  The main triggers for these outstanding performance were the alignment of learning initiative to business goals, the extension of learning processes outside company's boundaries, a focus on competency development, the integration of learning and knowledge management, and a blended delivery approach.
- Their organizational model, that posits networks, communities of individuals and refusal of a centralised mindset as the core elements of the new frame of reference. Their processes and structures emphasize decentralization and participation trough the all supply chain: flatter hierarchy and team based work characterise these new organizations, so as to respond more quickly to changes in the business environment and customer demands. These new networked learning organization are also called by some scholars as "Web 2.0 enterprises", which use digital

networks to co-operate and compete with other e-business community partners, by exchanging knowledge and information across transnational borders (Harreld 1998; Ticoll et al. 1998; O'Brien 2000).

The opportunities of shared leaning processes mainly come from (Bessant and Tsekouras 2001):

- Potential for challenge and structured reflection from different perspectives, that can bring in also new concepts;
- Shared experimentation, that can reduce perceived and actual cost risks in trying new things;
- Shared experience, that can open new lines of inquiry and exploration;
- An environment for exploring models and concepts outside the normal experience of individual organizations.

We assist to a shift from a traditional organization, conceived as a close and hierarchical system, where value creation is internally-based, learning is mostly adaptive process, leadership is autocratic, jobs and workforce are static, to an organization as a boundaryless and open system, where value creation is network-based, leadership is distributed, focus is on intangible assets, managerial roles constantly evolve (Garvin 1993; Hamel 2008; Morrison 2002).

The overall effect of the above characteristics is the emergence of a new networked organization to be conceived as an "Individualized Corporation" (Ghoshal and Bartlett 1997), where the core elements of the "Strategy-Structure-Systems -3S" approach (focused on the traditional sequence of activities of planning, allocating resources, controlling the efficiency of the execution) is substituted by the "Purpose-Process-People – 3P" approach, based on the premise that the main task of organizing is to create an environment that enables people to take initiatives, cooperate and learn.

The 3S was a deterministic managerial approach, based on the assumption that top managers where the only individuals who owned the information, knowledge and expertise to enable the company to the environmental demands and opportunities: they had then to define priorities and monitor operations trough sophisticated information systems.

The 3P is a dynamic managerial approach, focused on building stronger relationships with those best able to make decisions, by developing their capabilities and by actively supporting their learning processes. This does not mean to abandon the traditional activities of planning, but to focus mainly on the development of key people that are involved in strategic jobs, that have to influence the evolution of the entire firm. Since individual capabilities rarely success in isolation, it becomes necessary to link dispersed expertise, embedding the resulting relationships in a continuous

process of organizational learning. To this end, instead of setting up formal information systems, top managers have to communicate ideas, to sense stable signals and to set-up tools that facilitate the transfer of knowledge trough relationships among their people (Barton 1992). They have to try to influence people's behavior by shaping the context where people work, and by supporting the processes that are developed in the real working environment.

In this way, following the 3P approach, the most successful firms:

- Have established tools, processes and relationships that support horizontal flows of information, to link and leverage individual knowledge and to embed it in a collective process of shared leaning;
- Have built a strong sense of trust both among the employees and between them and the top managers;
- Have invested substantial resources in developing the expertise and capabilities of their employees.

Indeed, the cornerstone of these firms has become the capability of individuals to create value, by reversing the focus from value extraction to value creation.

The central role of Human Capital in Networked Learning Organizations

Networked organizations can be then successful only if they have people able to identify, understand, and support the sharing of information, experience and knowledge coming from all the partners (Goshal and Bartlett 1998). The challenge becomes the creation of three different kinds of people, able to support the development of a Networked Learning Organization (Baets and Van der Linden 2003):

- Generic competencies, such as the so called middle managers, able to articulate a (shared) strategic vision, to communicate and translate that vision throughout the organization;
- Organic competencies, as organization specific competencies that have context-specific roles;
- Changing competencies, that include those competencies that have an increasing importance in the following few years (emerging), competencies whose relevance is fading out (mature) and competencies whose relevance may decrease while their emphasis increases (transitional).

Moreover, it becomes necessary to give employees the behavioral skills needed to communicate, to work as a member of a team, such as negotia-

tion skills, time management, team building, cross-cultural awareness and communication skills. Skills, knowledge, implicit know-how include capacities of (Meister 2002):

- Learning to learn: people must be able to apply existing knowledge to new situations; experiment with new learning from co-workers, customers, suppliers and educational institutions, as well as build this learning into their everyday life;
- Communication and collaboration: skills as knowing how to work in groups, collaborating with team members, networking with customers, suppliers, key members of the value chain;
- Creative thinking and problem solving: capacity to analyze situations, ask questions, seek clarifications, suggest improvements, think creatively;
- Technology literacy: knowing how to operate in a virtual world, by using virtual reality, groupware, networking;
- Global business literacy: know-how related to business literacy skills such as finance, strategic planning, marketing, to run a global business enterprise;
- Leadership development: capacity to face the real-life business challenges of moving an organization in the current global competitive landscape;
- Career self-management: requisite skills, knowledge and competencies for both their current and future jobs.

In order to speed up the development process of these core competencies, the private sector is increasing its commitment to learning and education.

To this end, a growing number of private organizations have begun to shift from the traditional old-time training events to creating a continuous learning culture where employees learn from each other and share innovations and best practices. This paradigm shift in education concerns manly (Meister 2002):

- The goal and the frequency of the educational events: the educational initiative is no more conceived as a one-time event for improving the individuals skills. The desiderate outcome of education is no longer completing a formal course, but rather developing a continuous learning process, to improve the owned core workspace competencies in solving real business issues;
- The methodology used: the traditional training events in a classroom have been integrated with the so-called “action learning”, where learners work on real problems, solve the problems identified and focus on the acquired knowledge. This is done in cooperation with others, in small

groups. In this view, the "raw material" to help learners is no longer a case study, but a business issue the teams usually faces in their day-to-day jobs. This approach is particularly suitable for adults, as it enables each person to reflect on and review the action he/she has taken and the learning points arising. This should then guide future action and improve performance.

- The audience: the traditional audience made up of individual internal employees has evolved towards teams of employees, customers and product suppliers;
- The place of learning: the traditional classroom has been integrated with on-demand learning, where people can learn where and when they like;
- The Faculty: it has changed dramatically from external University Professors and Consultants towards internal senior managers, that have to illustrate the problems the Organization has to face. Moreover, University Professors and Consultants have not only to teach concepts that they use every day, but also to customize these concepts, in order to allow the students to solve specific problems.

The emerging pedagogical approaches for creating Human Capital

The current strategic nature of learning and the transition from considering learning as an event to thinking learning as a strategic process has determined radical innovations in the pedagogical approaches to the development of the human capital, including both learning contents and pedagogical methodologies.

The traditional pedagogical approaches follow what Baets and Van der Linden (2003) call "transfer metaphor", where the student may be a passive actor that has to be filled with the new knowledge or a "shaping approach", where the teacher is a craftsman who has to give shape to the student, considered also here a passive learner.

Only explicit knowledge is here taken in consideration and, for transmitting explicit knowledge, these approaches create hybrid curricula, focused on the integration of different disciplines:

- Human resource management
- Operations management or production management
- Strategic management
- Marketing management
- Financial management

- Information Technology management.

The emerging pedagogical approaches are usually grounded in the hypothesis that each kind of competence is based on two pillars: explicit knowledge (contents that a person has to learn) and experience (skills that he/she needs to have in order to put in practice his/her knowledge).
Explicit knowledge can be expressed in formal and systematic language and can be transmitted trough the traditional pedagogical approaches.

However, the second pillar of a core competence, the experience, is considered highly personal and hard to formalize. As Nonaka, Toyama and Konno highlight (2000) subjective insights, intuitions and hunches fall into experience. Experience is deeply rooted in action, procedures, routines, commitment, ideals, values and emotions. It is difficult to transfer experience, since it is a process that develops during time, mainly trough a "trial and error" process.

To transmit experience, the real business context has to be simulated, through interconnected processes of problem solving, knowledge integration, experimentation and innovation. Learning needs here to be conceived as a dynamic process based on action and interpretation, reflection and inquiry, that can benefit from the presence of different perspectives, the integration of theory and practice, the analysis of cases, and real life projects focused on managerial processes. This pedagogical approach, that enables learner-centered processes, follows more what Baets and Van der Linden (2003) call a "travelling metaphor", where the teacher has the role of the experienced person, who guides the students trough an area that he/she needs to explore, or a "growing theory", an extension of the travelling metaphor that focuses more on student self-initiative. In this last case the subject matter is indeed considered as a set of experiences that the student has to absorb, basing on his/her personality, by dialogues, assessments, self evaluations.

As a consequence, to develop both "knowledge" and "experience", on which a competence is based on, it becomes necessary to integrate the traditional approaches, based on courseware, workshops, individual development plans, personal study, with "experience development" approach, based on a "reflection in action" learning approach, including action learning projects, workgroups, multidisciplinary case studies, practical tools an solutions adopted successfully in the past.

The organizational context for the "experience development" is based on collaborative platforms, learning laboratories, conceived as a "constructed microcosms" (Senge 1990) in which management teams learn how to learn together, enhancing their collective intelligence and capacity to create value (Romano et al. 2001). These processes allow a continual movement between practice and performance, and combine meaningful

business issues with interpersonal dynamics. The laboratory is a way for engaging participants in the sharing in action of knowledge and experiences, as a collaborative forum where different actors mutually develop new understanding, insights, approaches and perspectives. Fig. 5.1 shows the general approach of learning process.

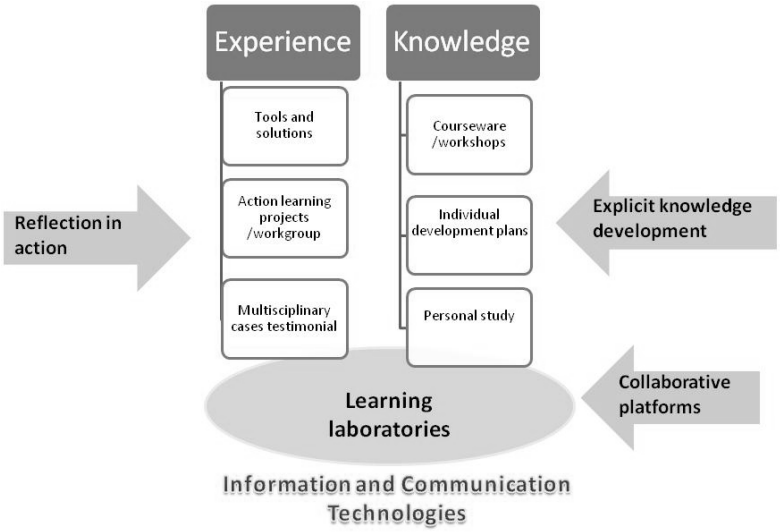


Fig. 5.1 The learning approaches for creating human capital

The learning laboratories may grasp the opportunities coming from the Information and Communication Technologies, that allow them to develop also in different places or in different times. Fig. 5.2 shows an example of the different applications that can be used to develop learning processes in same/different times and places.

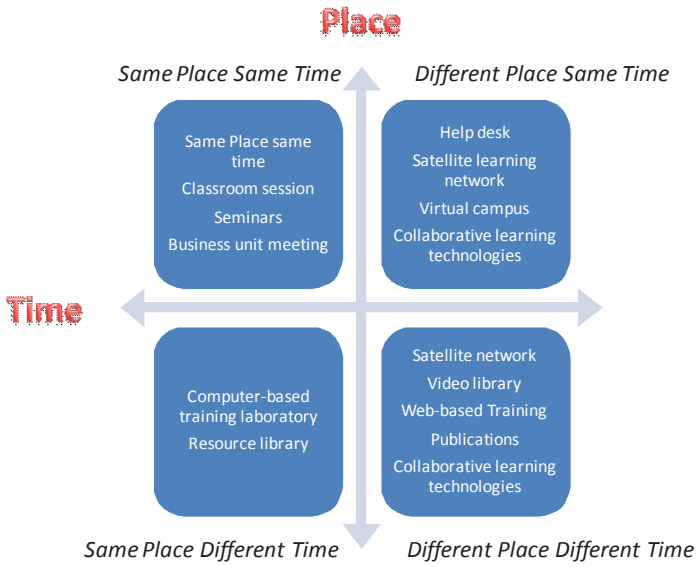


Fig. 5.2. Tools supporting human capital creation process (Source: Elaboration from Baets 2003)

The following sections analyze some education initiatives and the pedagogical approaches they have used in terms of action lines/purpose, processes and business cases.

Towards the Stakeholder Universities for creating suitable Human Capital

The emerging "Individualized Corporations" and their new conception of learning, considered more a process than an event, can explain the phenomenon of the corporate universities (CUs). During the last ten years CUs have emerged as the fastest growing educational entities focused on the development of human capital within private organizations (Paton and Taylor 2002; Meister 1998; Allen 2002). The term "University" has been used both as a metaphor for outlining the importance of its learning initiatives and for branding the organizations' educational programs, courseware, processes. The term "Corporate" has given it because of it also offers learning contents to suppliers and customers, or takes knowledge from them, by utilizing communications and facilitation of social, technological and organizational practices (Meister 1998). Initially, 40% of Fortune 500 firms had a Corporate University. Today, the market has expanded to in-

clude also midsize companies, and a recent survey of 100 CUs deans found that 64% of universities are not in the Fortune's list.

CUs differ substantially from the traditional "training and development" departments, both in their orientation and in their impact on corporations' performances (Wheeler and Clegg 2005). The "training and development" departments usually focus both on the development of single individuals and on the contents to offer, with only incidental impact on business goals and objectives. On the contrary, CUs focus on solving organization-wide issues, or on attracting new businesses. The analysis of outstanding cases of CU and the results of the benchmarking report show that success CUs align learning and development process with the strategic and over-reaching business goals. The main organization-wide issues faced by CUs concern (Wheeler and Clegg 2005):

- Leadership development and building, or employees' skills up-to-dating;
- Customer education, including education in product understanding and use, or in technical assistance, to increase their purchasers of products and services;
- Helping employees understand and interpret the new competitive environment and helping the entire organization move trough the all phases of change, by providing new capabilities and new skills to meet the new challenges;
- Research focus that is usually reserved for research labs in universities, such as exploring ways people can learn faster the things needed to allow their organization to become more effective.

Indeed, CUs have emerged more and more as a strategy to ensure the integration of learning processes within the organization, and to customize them according to the specific needs of the company. As a result their huge business impact concerns mainly product/service quality (improved by 70%), customer service (improved by 70%), operating costs (reduced by 59%), sales efficiency (improved by 49%) and profits (increased by 48%) (Corporate University Xchange, 2007).

Nowadays, CUs are facing the same pressure to outsource as other corporate business units: according to the analysis carried out by the Corporate University Xchange, outsource processes concern mainly technology-enabled course delivery (32%), assessment/competency development (31%), classroom-based course delivery (30%), course design and development (27%), learning technology infrastructure (24%). In a lot of cases cost is less important than establishing a two-way dialogue (59%). CUs want partners that are willing to understand the business (77%), and work with internal subject experts (66%). Moreover, they place medium/high

importance on forming alliances with colleges and universities (51%): 46% of CUs has already partnerships where academic universities offer on-site courses at the organization's facilities, and 45% partners with academic universities to offer customized courses or degree programs. Expert CUs also foster collaborations with academic partners trough educational consortia and share leading-edge practices, methods or technologies (Corporate University Xchange, 2007).

As this analysis highlights, CUs are evolving towards "networked academic configurations", the so-called Stakeholder Universities (SUs) (Bleiklie and Kogan 2007), conceived as hubs of learning networks, with learning processes distributed also outside of the university, with radical changes in (Romano 2008):

- the strong integration of education with research activities and pilot projects. Indeed we are evolving from the paradigm of the "Teacher-expert", to the paradigm of the "Teacher-mentor", that has to facilitate a flexible and learners' centered knowledge diffusion;
- private-public partnerships, because of the necessity to face challenges related both to the new clients of the education processes (adult people), and to their social experience and interaction processes (based not more on the exchange of codified knowledge, but on dynamic learning communities); effective and efficient corporate education projects have been often held also in collaboration with external education entities;
- growing role of the information and communication technologies: lessons are no more based on rigid scheduling, but delivered also in high interaction in a virtual space and time, as well as in an asynchronous way, everywhere 24/7. In this way, SU are evolving towards global assets, where institutions, firms and universities in different locations and countries are converging towards networked configurations.

To this end we assist to a change also in the managerial organization, with the emergence of powerful organizational structures that parallel and in some case replace the traditional chair-faculty systems. The traditional Universities were based indeed on "republic of scholars" ideals, where decision making was based on collegial decisions of independent scholars. Now the "republic of scholars" has been substituted by "stakeholder organizations", where decision making is carried out by councils, boards or trustees that incorporate representations from the world of business/public services and politics, where the traditional organizational emphasis on teaching processes is evolving towards the emphasis on learning outcomes.

SUs have then become skilled not only at creating knowledge for the improvement of individuals teaching and learning, as well as at speeding up the learning processes of a whole organization, but also at modifying

their behaviors to reflect their new learning needs. In short, SUs have become themselves learning organizations (Dill 2004). But how we can develop a SU as a learning organization?

The following section will describe the experiences developed in Finmeccanica case, describing in detail the new pedagogical approaches developed, and the impact that the information and communication technologies had on it.

Creating human capital in Finmeccanica Group: towards the Stakeholder University

The Finmeccanica Group is a world leader in design, development and manufacturing of state-of-the-art technologies in high-tech business sectors: it is a key player in Defence, Aeronautics, Helicopters, Space and Security and also has well-established skills and capabilities in the Transportation and Energy sectors. The Finmeccanica Group has generated a total turnover of e13.4 bn in 2007 and has 300 sites located all over the world with some 61,000 employees[i], with an average growing rate of 49.4%.

Currently Finmeccanica is a multidomestic/multicompany group[ii], aiming at becoming an International Integrated group, trough the valorization of its human and intellectual capital at a worldwide level. To this end, as highlighted in Fig. 5.3, a big challenge becomes the Finmeccanica's capacity to fill the gap of both a critical area (people that have to change their competences/jobs, not requested in the future) and an opportunity area (competencies/jobs that will be necessary in the future, but still missing in Finmeccanica Group). Fig. 5.3 shows the distribution of the employees (in terms of seniority classes) currently working in the Finmeccanica group (yellow line) and the distribution forecasted in next ten years (brown line).

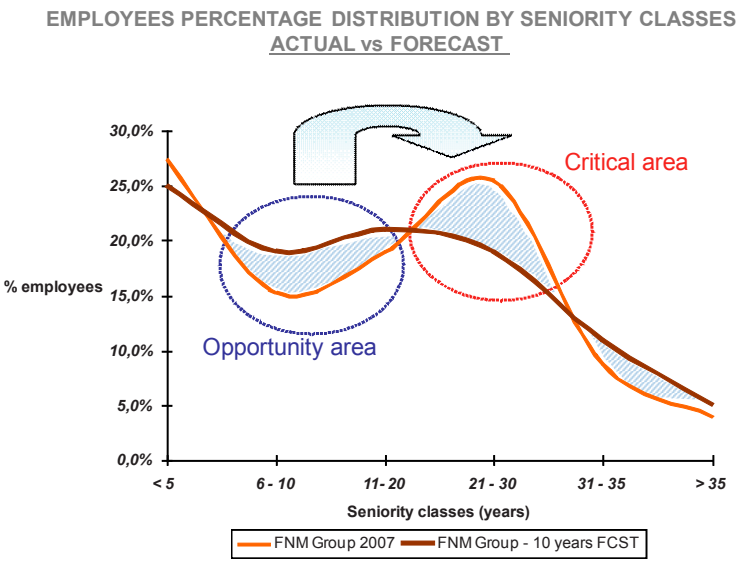


Fig. 5.3. Employes's distribution in Finmeccanica

In order to fill this gap, Finmeccanica has started an evolution process towards a Stakeholder University, by launching a Virtual Corporate University.

The start-up: the Finmeccanica Virtual Corporate University

Since its start-up, Finmeccanica Virtual Corporate University has set up a wide array of education programs[iii], following three main guide lines:

- the international integration of Finmeccanica employees, by developing in them a Common Culture, a Common Behavior and a Common Identity. A wide array of initiatives has been set-up, such as:
    - "FHINK", a master in International Business Engineering, designed to forge a link between academic institutions and industry, with the involvement of prestigious Partner Universities from around the world[iv]. Participants (30 talents selected at a worldwide level) in the Master's Programme gain exclusive access to learning and development opportunities that will be invaluable for their future career in the Finmeccanica Group. The Master's Programme enables high-potential graduates to become world class professionals active in an international Business Engineering environment. The Master's Fac-

ulty includes world class academics as well as industry experts and company executives. Students benefit from interaction between selected Academic Partners and the combined skills and knowledge of the Finmeccanica Group.

- The "Business Culture Project", a system of analysis and intervention on culture and climate, a structured process of "listening" aimed at interpreting the opinions that people have about the environment in which they work, in the various companies at an international level[v]. The objective is to identify the distinctive cultural convergences of each group company to turn them into strengths within the oricess of change and contribute to the development of a distinctive Finmeccanica Identity;
- "Training courses for Finmeccanica Internal Communicators", to encourage the expansion and optimization of the communication flow within the Finmeccanica Group, and to develop a network of Internal Communicators, to facilitate the definition and creation of dedicated projects at a group level;

- the knowledge transfer throughout the group, by developing Integrated Professional Communities. Some initiatives concern:
  - the "Project Management Programme"[vi], that aims at:
    - Establishing a Finmeccanica PM Professional Community and providing support, through Group Knowledge Management systems and the sharing of best practicebetween the Operating Companies.
    - Valuing the professionalism of the Project Management capability within the Finmeccanica Group and, where beneficial in the future, opening the model to Stakeholders.
    - Delivering world-class training paths on key Project Management competencies, in alignment with the highest market standards.
  - "Professional Communities", aiming at encouraging the development of people, by assisting their growth, intragroup mobility and integration trough workshops, specialist seminars, activities of communication and knowledge sharing;
  - "From Technology to Values", aimed at developing leadership skills, cognitive thinking necessary to interpret global situations at a macro-economic and geo-strategic level;
- a continuous change management process, by supporting the development of Open Learning Communities. Main initiatives concern:

- "BEST" (Business Education Strategic Ten), a master for young graduates with potential, who have no more than 3 years working for the company[vii]. The master aims at accelerating their managerial skills and to develop team working skills;
- "Innovation Award", an international initiative destined for all the employees of the Finmeccanica Group to encourage the submission of innovative ideas and projects concerning the companies' business areas[viii]. The objective is promoting innovation and protecting Intellectual Capital.

The educational approach has evolved in terms of:

- Scope of the education process, that has evolved from a "Targeted management of Human Resources" to a systemic management of Intellectual Capital
- Dynamics of the students' learning process, that have evolved from an approach based on "one way" training to an approach based on interactive learning and focusing on problem solving;
- Dynamics of the teaching process, that has evolved from a traditional training based on knowledge towards an integrated development of learners' skills.

The target population of these initiatives may be grouped in:

- executives
- middle managers
- young talents.

Fig. 5.4 shows the general framework of the Finmeccanica Virtual Corporate University.

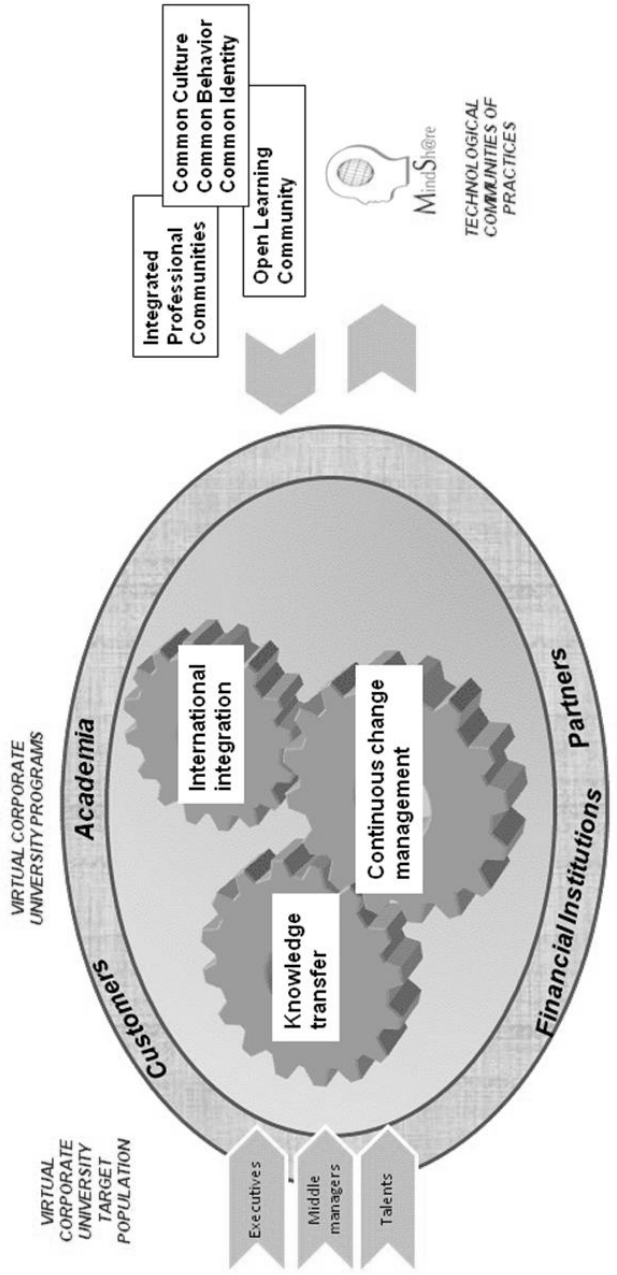


Fig. 5.4. Finmeccanica Virtual Corporate University

As shown in the Fig. 5.4, all the initiatives have been realized in cooperation with a network of external partners:

- Customers of these education initiatives: the Italian Ministry of Defense and other Italian Public Institutions and Organizations;
- Partners: Boeing, Thales, BAE, Eads, Bombardier;
- Academia: MIT, Harvard University, Imperial College, Politecnico di Milano, ISUFI
- Other subcontractors and financial institutions.

Finmeccanica Virtual University has however developed a special virtuous dynamic cooperation with seven technology learning communities, the so-called Mindsh@re communities, able to meet the innovation challenges facing the Finmeccanica Group, even trough a strong Partners' network.

Each of these Mindsh@re communities is required not only to be the engine of an intelligence culture able to give value to a specific innovation, but also to be a cooperative and interconnected animator of a network involving the other Mindsh@re communities, the group companies, the market, clients, complementary suppliers and research centres. As permanent design centres of intelligence and knowledge, Mindsh@re communities constitute the clearest and most tangible expression of the International Integrated group that Finmeccanica is aspiring to become.

The role played by these transversal structures in supporting Finmeccanica Virtual University is twofold: on one end communities are "customers" of the Finmeccanica Virtual University, on the other end they supply Finmeccanica Virtual University with some empirical contexts where the Finmeccanica Virtual University students can develop their "learning in action" processes. In this way, they can be particularly instrumental in allowing the Finmeccanica Group to create a real Stakeholder University (SU), as a dynamic learning network being acknowledged for its leadership in the culture of innovation.

Towards the Finmeccanica Stakeholder University: open issues

The design and implementation of a SU effort require experience, creativity and ability to analyze changes, as well as a deep knowledge and experience on pedagogical, methodological and technological facets of learning. The development of a proper awareness and mindset, and a clear roadmap of operating issues represent thus two key elements for the successful start-up of a SU (Grenzer 2006; Paton 2005). This chapter aims at presenting the main challenges faced by Finmeccanica in setting-up its model of SU.

Finmeccanica has defined a roadmap to designing and implementing a SU initiative, made up of the following activities:

1. Create the Stakeholder University Core Curriculum (Contents and methodologies).
2. Define the Organizational Model.
3. Identify Key Stakeholders.
4. Financing and Timing the Initiative.

For each step, we present in the following the main issues.

Create the Stakeholder University Core Curriculum (Contents and methodologies).

The first point is particularly relevant in the whole process of creating the SU. In particular, the competencies and skills required to people vary sensibly. Competencies and skills central for the change management focus will thus be different to those crucial in the business development, external customer, and internal development focus. A set of competencies and skills which are important irrespectively from the strategic direction is the following:

- Creativity
- Multitasking
- Future orientation
- Strategic thinking
- Global perspective
- Problem analysis and solving
- Coaching
- Ability to adapt
- Mental agility
- Competitive knowledge.

Based on the experience of major companies involved in the creation of curriculum for their SU, the objective is to develop also people in three main broad areas (Meister 1998):

- Corporate Citizenship: the goal is to create a strong attachment to the company, similar to what citizens feel for their city or country. An increasing number of world-class companies are developing learning programs to ensure workforce knowledgeable about the vision, history, culture, tradition and values of the company.
- Contextual Reference: this refers to providing learning opportunities about the company's products, services, and industry dynamics, in the context of what competitors are offering in the marketplace and the best practices of the industry as whole. The goal is for all the employees to

be able to clearly communicate the company's strategic intent to both current and perspective employees.
- Workplace Competencies: this involves training employees in the specific competencies the corporation has deemed critical to its long term success.

Moreover, the following infrastructures and tools can be useful to enhance learning processes of the SU:
- systems of content management, knowledge management and e-learning;
- platforms to support collaborative learning and the creation of learning communities;
- social network analysis tools, to monitor learning interactions inside the stakeholder university.

The use of web 2.0 applications, such as wikis, blogs, and RSS feeds is also increasingly developed for managerial education because of the low investments required to companies and the very promising results in terms of creating and sustaining a learning community inside the organization and outside, with its value network.
The adoption of technology enablers is strictly related to delivery methods used to drive learning, such as traditional classroom experience, web based training, group projects and activities, simulations, coaching, self-study, experiential learning activities, etc. Strictly related to the delivery is a set of issues and preliminary activities to consider:
- Definition of knowledge domains and learning themes/objects to cover
- Establishment of the pedagogical strategy and learning models to apply
- Activation of appropriate learning processes and facilitating laboratories
- Creation of knowledge and learning base.

To face all these challenges, Finmeccanica can use the opportunities coming from its technological and distinctive culture, as well as its common methodologies (long life cycle management and process control) and key competencies (international business engineers and program/project managers). However, there are the problems of sharing its "sensible" knowledge and skills with other stakeholders, and of setting-up the right focus on different target populations.

The solution that Finmeccanica is going to adopt is applying a bottom-up approach, starting from sharing and customize its "Virtual University" education program.

Define the Organizational and the Governance Model

The basic motivations and long-term reasons for creating the Stakeholder University have to be defined at the beginning and strongly supported by top management. Indeed, while middle managers, human resource executives or training directors may see a clear need for the Stakeholder University, they usually do not launch it from the bottom-up. Rather, a senior leader and in many cases the CEO are the real trigger of the initiative. Top management has to communicate the vision defined constantly and consistently to all key stakeholders, which should be made aware of objectives, products and programs.
Considering the different models implemented in practice, as also reported by market analysis and benchmarking reports, the structure of a Stakeholder University can be drawn according to three main models:

1. Centralized, if all activities and people involved in developing and delivering learning, and all those involved in capturing and disseminating knowledge, report to a single person, and there is a tightly defined purpose for the SU, which is established at central/corporate level (ex. the Leadership Institute at General Electric, which is charted directly by its CEO).
2. Decentralised model, if there is no controlling central organization or central direction and independent managers lead different divisions globally dispersed and with very different local needs, also in terms of learning and development.
3. Federal model, if the structure consists of a strong central group that influences, coordinates and ties together disparate training functions worldwide, if there are common needs and a necessity to coordinate duplicated services or curriculum.

Moreover, the governing board of the SU should be composed by senior executives and managers, with the following key roles (Meister 1998):

- identifying and prioritizing current and future needs;
- approving strategic learning plans;
- promoting the university within the company;
- defining the corporate university vision and mission;
- setting the organization's learning philosophy.

Finmeccanica is trying to implement an hybrid model, since it is aware of all the opportunities coming from a SU wide and flexible model, with the involvement of a large number of partners. However, it takes in consideration also the constraints of the total decentralized model, in terms of lack of direct control and low effectiveness of the decision processes.

The solution that it seems possible to them is then to arrange a governance model made up of three governance layers, with centralized procedure and processes:

a) A Scientific Board;
b) A Strategic steering committee;
c) An operating management system

Identify Key Stakeholders

One of the challenges in building a SU is to identify the proper kinds of stakeholders to sustain the success of the initiative. There are some major steps in the process of assessing and selecting the stakeholders staff:

- assess current staff to understand how well it meets company learning needs;
- conduct gap analysis to understand the current strengths and weaknesses
- choose staffing model among program management, mixed and team development models;
- implement staffing and hiring plan according to best practices available.

In terms of people roles needed to launch a Stakeholder University, the different staff roles are relevant, grouped in categories:

Management:

- Account/relationship management
- Learning delivery management
- Project management
- Curriculum management
- Learning technology management
- Vendor management

Instructional:

- Instructor/facilitator
- Classroom-based instructional design development
- e-Learning instructional design development

Performance Improvement

- Performance Consulting
- Measurement and evaluation
- Research & Development

Other
- Back office support
- Knowledge management and Communities of Practices
- Marketing and corporate communications

Finmeccanica can grasp all the opportunities coming from its wide value chain and value network, developed at a global level. However, main issues concern the setting up of suitable criteria for selecting and admitting stakeholders, and the methodologies to follow for setting common views and behaviors with them.

To this end they have identified, as a possible solution, the identification of a small number of founders (no more than 3-4 big players) and a consequent focused selection of the other stakeholders, that have to be appointed step-by-step.

Financing and Timing the Initiative

In detail, the models possible are the following:
- Corporate allocation model, if all expenses are considered part of the general operating overhead of the firm, and costs are allocated to profit centres according to some internal procedures.
- Partial or full cost recovery model, if the university charges individuals and departments for the costs they actually incur for training, and this implies that the learning institution operates more or less as a business, itself.
- Profit centre model, a model based on selling classes and learning services to customers located both internally and externally to the company. The training function of the university does not cost the company nothing.

Finmeccanica is considering solutions of self-funding and offset; because of the high costs forecasted (10 mil € on annual basis) it has decided to start with low-cost project, following an idea of "quick win".

As for the timing issues, for speeding up the SU decision processes, and for defining a right-time schedule, a two-step possible solution has been identified:
- A "quick-win strategy" with a significant and focused pilot project in the short term;
- A "steady consolidation strategy" in the medium/long term.

Conclusions

Leading companies recognize people as key value and greatest strength, and their strategic purpose in terms of human resource management is to prepare next generation leaders and deliver organizational learning companywide. In order to speed up the development process of these core competencies, the private sector is increasing its commitment to learning and education. A growing number of private organizations has begun to shift from the traditional old-time training events to creating a continuous learning culture, where employees learn from each other and share innovations and best practices. To this end, they are creating "networked education configurations", the so-called Stakeholder Universities (SUs), conceived as hubs of learning networks, with learning processes distributed also outside of the company. But how they develop a SU? This paper describes the experiences of the Finmeccanica case, presenting in detail the experiences already developed, and the roadmap designed for implementing its SU initiative, pointing out the issues faced for:

- Creating the Stakeholder University Core Curriculum (Contents and methodologies).
- Defining its Organizational Model.
- Identifying its Key Stakeholders.
- Financing and Timing the Initiative.

References

Allen M (2002) The Corporate University Handbook, AMACOM, New York

Baets WRJ, van der Linden G (2003) Virtual Corporate Universities: A Matrix of Knowledge and Learning for the New Digital Dawn. Kluwer Academic Publishers, Norwell, MA

Bessant J, Tsekouras G, (2001) Developing learning networks. AI & Society 15: 82-98

Bleiklie I, Kogan M, (2007) Organization and Governance of Universities, Higher Education Policy

Corporate University Xchange (2007) Corporate University Xchange 6th-Sixth Annual Benchmarking Study

Cusumano MA, Markides C (2001) Strategic Thinking for the Next Economy. Jossey-Bass, San Francisco

Dalin P, Rolff HG (1993) Changing the school culture, Cassell, London

Dill, DD (1994), 'Rethinking the Planning Process', Planning for Higher Education: pp. 8-13.

Eisenhardt KM, Sull DN (2001) Strategy as simple rules. Harvard Business Review 79: 107-116

Eisenhardt KM, Brown SL, (1998) Time pacing: competing in markets that won't stand still. Harvard Business Review, 76: 59–69

Fullan M (1991) The new meaning ofeducational change, Cassell, London

Garvin DA (1993) Building a Learning Organization, Harvard Business Review, 73:78-91

Ghoshal S, Bartlett CA (1998) The Individualized Corporation: A Fundamentally New Approach to Management. Heinemann, London

Grenzer JW, (2006) Developing and Implementing a Corporate University, Human Resource Development Press

Hamel G, (2008) Title: The Future of Management Journal: Human Resource Management International Digest 16

Harreld JB, (1998) Building faster, smarter organizations. In Tapscott, D, Lowy, A, & Klym, N, Blueprint the digital economy: Creating wealth in the era of e-business., McGraw Hill , New York

Leonard-Barton D (1992) The Factory as a Learning Laboratory, Sloan Management Review, 34: 23-38

Marquardt MJ (2002) Building the learning organization: Mastering the 5 elements for corporate learning. Davies-Black , Palo Alto, CA

Meister JC (1998) Corporate universities: lessons in building a world-class work force, McGraw-Hill Trade

Meister JC (2002) Ten Steps To Creating A Corporate University, Ebsco Publishing

Morrison EW (2002) Newcomers' relationships: The role of social network ties during socialization. Acad. Management J. 45:1149–1160

Nonaka I, Toyama R, Konno N, (2000) SECI, Ba and leadership: a unified model of dynamic knowledge creation, Long Range Planning 33:5–34

O'Brien JA (2000) Introduction to Information Systems: Essentials for the Internetworked E-Business Enterprise, McGraw-Hill Education

Paton (2005). Handbook of Corporate University Development

Paton R, Taylor S (2002) Corporate universities: Between higher education and the workplace, In: Williams, G. (ed.) Enterprise in Universities: Evidence and Evaluation, Open University Press, Buckingham

Romano A, Elia V, Passiante G (2001) Creating business innovation leadership. An ongoing experiment: the e-business management school at Isufi, Edizioni Scientifiche Italiane

Romano A (2008) Costruire l'Università post-fordista, Cacucci Editore, Bari

Senge PM (1990) The Fifth Discipline: The Art and Practice of the Learning. Organization, Doubleday Currency, New York

Ticoll D, Lowy A, Kalakota R (1998), Joined at the Bit. The Emergence of the E-Business Community. In Tapscott, D, Lowy, A, & Klym, N, Blueprint the digital economy: Creating wealth in the era of e-business., McGraw Hill , New York

Wheeler K, Clegg E, (2005) The Corporate University Workbook: Launching the 21st Century Learning Organization, Pleiffer, san Francisco

[i] Italy 43.200 (59%), USA: 12.400 (17%), UK: 10.200 (14%), Francia: 3.600 (5%), RoW 3.600 (5%) – consolidated at the end of October 2008

[ii] Finmeccanica's companies compete in Defence Systems (Oto Melara Wass,_MBDA), Energy/Transportation (Ansaldo Breda, Ansaldo Ener-gia, Ansaldo STS), Defence Electronics(Selex Communications, Selex Sensors and Airborne Systems,Galileo Avionica,Selex Sistemi Integrati, Selex Service Management, Elsag Datamat, Seicos_Orizzonte_Reles), Space (Telespazio_Thales Alenia Space, Aeronautics (Alenia Aeronautica, Alenia Aermacchi,Alenia Aeronavali_ATR_Global Military Aircraft Systems-_Global Aeronautica), Helicopters (AgustaWestland,_BAAC)

[iii] Finmeccanica Virtual Corporate University has been created by Francesco Mantovani (SVP Education and HR Development Human Resources Central Function – Finmeccanica)

[iv] Marta Di Santo, head of services of the Master's Programme, coordinates all the actors involved in for create a new world class professionals, active in an international Business Engineering environment

[v] The "Business Culture Project" is a system of analysis and intervention on culture and climate, delivered by the department of Marina Possi

[vi] the "Project Management Programme", is delivered by the department of Davide Amabile

[vii] BEST is coordinated by Carla Serafini

[viii] "Innovation Award" is coordinated by Silvia Silvetti

Chapter 6. Expanding the Value of Corporate Universities: The Stakeholder Approach

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Introduction

For educational programs at traditional universities, it is clear who the primary stakeholders are: the students. In corporate universities, it is not as obvious. At a minimum, it is the student/employees as well as the organization itself. But a deeper look reveals that in addition to the students, their managers and co-workers benefit from the education, just as we can say that in addition to the organization as a whole, the department or division benefits. Peeling the onion a little deeper, we can expand the stakeholder network to include vendors, suppliers, business partners, customers, and possibly even governments, communities, and society.

Since organizations are comprised of complex networks of stakeholders, both internal and external, the beneficiaries of organizational learning are an equally complex network of stakeholders. Moreover, as we will discuss in this chapter, it is these very networks that enable us to multiply the value of our learning.

In fact the very notion of a corporate university can be seen as being quite comparable to a learning network. A learning network is "a network formally set up for the primary purpose of increasing know edge" (Bessant and Tsekouras 2001). As we will see in the next section, a corporate university is definitely a formal entity with learning and knowledge as its objectives.

A. Romano and G. Secundo (Eds.), *Dynamic Learning Networks: Models and Cases in Action*, 121
DOI: 10.1007/978-1-4419-0251-1_6,
© Springer Science + Business Media, LLC 2009

This chapter will explore the many and varied stakeholders of a corporate university and the ways that each of them, individually and in networks, contribute to organizational success. The chapter begins with a brief history of corporate universities and their evolution from training departments to next generation corporate universities. We will discuss what the various functions of a corporate university are and compare them to traditional universities and their functions. We will then get into the exploration of stakeholders. Who are they? What are their stakes in organizational success? And, most importantly, how does viewing a corporate university as a stakeholder university enable us to discern new ways of generating value.

The history and evolution of corporate universities

The first documented corporate universities date back to World War II. At that time, Northrop Aviation discovered that there were not enough qualified and trained workers to meet the demand for the airplanes that needed to be built to support the war effort in the United States (Eurich 1985). In a classic "make or buy" decision, the company decided that since it could not attract enough qualified workers, it would create Northrop University to "make" a qualified workforce.

In the decades that followed, more corporate universities sprang up to address workforce needs. As the concept gained popularity in the 1980s and especially the 1990s, organizations increasingly adopted the corporate university concept. Unfortunately, as the notion of corporate universities became trendy, many organizations adopted the name without embracing the concept. Thus there were a number of training departments that relabeled themselves as corporate universities without truly transforming anything but their name. This diluted the concept of corporate universities, and if that organization subsequently tried to institute a true corporate university, it was met with skepticism by those who remember the previous, less-effective incarnation

This of course raises the question of what a corporate university really is. Without any generally accepted definition, it was certainly fair game for any training organization to brand itself as a corporate university. In 2002, I offered this definition:

"*A corporate university is an educational entity that is a strategic tool designed to assist its parent organization in achieving its mission by*

conducting activities designed to cultivate individual and organizational learning, knowledge, and wisdom."(Allen 2002, p. 9).

There are some concepts in this definition that warrant further explanation. First, there is the notion that a corporate university is an entity. Some people mistakenly harbor the notion that a corporate university should be a place. While there are some that have sprawling campuses (like Veolia Environnement's Campus Veolia in suburban Paris) or well-equipped buildings (such as the Farmers Insurance's University of Farmers), a campus, building, or place of any sort is not necessary to qualify as a corporate university.

The most important word in the definition is "strategic." To me, this is the word that distinguishes a true corporate university from a traditional training department. While training is important, even vital, to the success of an organization, it is often tactical, operational, and job or task specific. In order to be a true corporate university, the entity is strategic, that is, its tasks, functions, and even very existence are tied directly to the organization's strategy and mission.

That is why the phrase "assist its parent organization in achieving its mission" is a crucial part of the definition. Training organizations exist to help get the day-to-day work done, but are usually not linked directly to organizational mission. Hence training is a part of the functions of a corporate university, but not the sole raison d'être.

The notion of a parent organization is important as well. Corporate universities do not exist as stand-alone entities or to serve themselves. In order to qualify as a corporate university, the entity must be a part of (and provide service to) a parent organization. Thus the Apollo Group's University of Phoenix, a large for-profit university, is both a corporation and a university, but not a corporate university.

The word "corporate" also need not be a part of a corporate university's identity. There are many highly successful corporate universities that reside within non-profit organizations, charities, hospitals, municipalities, and governments. Defense Acquisition University, part of the United States government's Department of Defense, is likely the largest corporate university in the world and it is located within a governmental agency, not a corporation.

The notions of learning, knowledge, and wisdom will be explored in greater detail later in this chapter. To summarize, learning is the process by which knowledge, skills, and attitudes are acquired. Knowledge refers to the information, skills, and abilities that are transmitted and acquired, while wisdom is the ability to creatively apply knowledge, ideally for the

benefit of the organization. All three are within the province of corporate universities.

The last term from the definition that I would like to expand upon is the word "activities." What, specifically, do corporate universities do to cultivate individual and organizational learning, knowledge, and wisdom? While a short answer to that question might be "conduct classes," that answer is rather limiting. While classroom learning (and even learning that takes place virtually in electronic classrooms) might be a big part of what some corporate universities do, it is not the only thing. In fact the list of corporate university activities, functions, and responsibilities is quite long. The next section discusses the many things that corporate universities actually engage in.

Corporate University Functions

As mentioned, training is just one of many functions that a corporate university can undertake. The traditional training functions of needs assessment, designing programs, and delivering them are at the top of the list, the list is quite long and contains many activities that are not normally associated with training departments. These functions appear in the second half of the list and are what I call Next Generation corporate university functions. They fit in more with the word "strategic" from the corporate university definition. The list below is lengthy, but not comprehensive; there are many functions not on this that might be the responsibility of some corporate universities.

Here is the list of functions that corporate universities might be involved in:

Needs assessments
Designing training programs
Delivering training programs
Designing managerial and executive development programs
Delivering managerial and executive development programs
Assessing technology options
Delivering e-learning or blended learning programs
Hiring vendors
Managing vendor relationships
Marketing programs internally
Marketing programs externally
Evaluating programs

Evaluating the corporate university
Managing university partnerships
Executive coaching
Mentoring
Career planning
Strategic hiring
New employee orientation
Succession planning
Culture change
Strategic change
Knowledge management
Wisdom management
Library and electronic collections of information
Research and development (Allen 2007, p. 6)

This list was first published in 2007. Since then, I have added several functions to the list, including fostering an ethical culture, maintaining an organization's heritage center (written and oral histories), and performance management.

I do not mean to suggest that every corporate university should undertake all of these activities. When I work with organizations that are just starting up a corporate university and are in the planning the phase, I present them with this list and provide four options for each item: primary responsibility, outsource, don't do, and involvement. I doubt if any corporate university is involved with each of these functions, but it is important to remember that you do not have to have responsibility for a function to be involved in it. In many cases, efforts around culture change, for example, may reside elsewhere in the organization, but the corporate university may be (and, I would argue, should be) very much involved in this effort. Additionally, a corporate university can be responsible for a function, but outsource it. I have witnessed corporate universities that outsource even the most basic functions of the delivery of training. They do the needs assessment and the training design, but they outsource almost all of the delivery.

Some of the functions, especially the Next Generation functions near the bottom of the list, are not normally associated with a corporate university. Things like strategic hiring, succession planning, career planning, culture change, strategic change, and knowledge management often reside in other parts of the organization. But many savvy organizations are coming to realize that these are fundamentally people processes and processes

involving change and learning, and therefore they are functions that the corporate university should be involved in or responsible for.

I want to expand a bit on the concepts of knowledge and wisdom management. Many organizations thought the best place to place responsibility for knowledge management was the information technology (IT) department. They asked IT to build a database that would enable the recording, retrieval, and communication of data throughout the organization. The problem is that in many cases, these companies confused data with knowledge. Databases are great at storing and communicating data, but knowledge typically exists in the heads of people. Beyond factual information, people's heads are full of other types of knowledge—skills, attitudes, and the general sense of how to get things done. These types of knowledge are generally of much greater value to the organization than the facts and data that can be stored in a database, but they are much harder to capture. The difficulties stem from two sources. First, it is much harder to record and store skills in a database. Secondly, people are often unwilling to share their skill set with others in the organization. They believe that their value to the organization—their personal competitive advantage—derives from what they know how to do. Sharing it, they believe, reduces their value and detracts from their job security. Sadly, in many organizations this is true—these organizations unwillingly have systems in place that reinforce these ideas.

In order to get people to share their knowledge, we need more than a database. We often need a culture change. We need to shift the organizational mindset from one in which knowledge is power to one where there are rewards for sharing knowledge (March 1997). The corporate university can be invaluable in helping people transfer knowledge, especially the skills and know-how types of knowledge that we have discussed. Just as importantly, the corporate university can be a tremendous asset in helping to shift the culture away from one that rewards knowledge hoarding and toward a culture in which knowledge sharing is rewarded, expected, and ultimately the way the company does business every day.

Wisdom management refers to the actual use of knowledge. It is one thing to possess knowledge, it is a far greater challenge to possess the wisdom to know how to apply that knowledge. While most corporate universities are good at transmitting knowledge, often that isn't good enough. Jack Phillips estimates that 60-90% of all knowledge acquired in training programs is never used on the job (Phillips 2007). This represents a tremendous waste of our training efforts and an even greater waste of potential. We need to refocus our efforts on going beyond the transfer of

knowledge to ensuring the use of knowledge. I call this process Wisdom Management and define it as follows:

"Wisdom management is a planned and systematic process by which an organization manages how its employees use and apply their knowledge and skills in ways that benefit the organization." (Allen 2007 p. 391).

This is clearly an important process that in many cases is the missing link that can make or break the success of a program or an entire corporate university. Naturally, ownership of this process should belong to the corporate university and perhaps is the most vital function of a corporate university and its greatest opportunity to add value to the organization.

Finally, it is instructive to notices what is missing from the list of corporate university functions: degree programs and academic research.

There are very few corporate universities that grant degrees. Several years ago, a researcher went in search of corporate universities that grant degrees (Thompson 2000). After a lengthy search of corporate universities throughout the United States and Canada, he was only able to locate fewer than ten that grant degrees. While many corporate universities want to provide degrees for their employees, almost all of these do so in partnership with traditional universities. It is the traditional university that grants the degree, not the corporate university.

While research and development is on the list, this function refers to the idea of fostering creativity, not the traditional corporate function of R&D or the traditional university function of academic research. The way a corporate university can foster creativity is via a function that one corporate university calls "innovation grants." What this corporate university does is maintain and administer a budget for these grants. Employees apply for these grants based on ideas they have for a new invention or process. The employee might need money, facilities, or even a bit of time off from the job to develop a new product or procedure that can benefit the company. The corporate university administers these grants and thereby fosters innovation.

But what corporate universities do not do is academic research in the traditional sense. As a matter of fact, there is little overlap in the functions of corporate universities and traditional universities.

Traditional University Functions

While the scope of disciplines in a traditional university might be quite large, the functions are typically summarized as teaching, research, and service. Different universities might place different priorities on each of the three, but the basic list is fairly consistent across universities.

Teaching in traditional universities typically refers to educating undergraduates and graduate students who are pursuing degrees, a function fundamentally different from the type of teaching that goes on in corporate universities. Some university business schools engage in executive education—non-degree programs in business, management, and leadership offered to corporate clients and individuals. This is closest area of overlap between traditional and corporate universities.

Research, as mentioned above, remains the province of traditional universities. Academic research, while important for society, is typically not a profitable endeavor and therefore is avoided by corporate universities. However traditional universities, both public and private, undertake this endeavor as a core part of their mission.

Service is often a vaguely defined part of universities' missions. However, most universities undertake some form of service to the communities in which they reside. And while many corporations have a robust program of community service, this is typically not a corporate university function. However, many organizations have recently begun using their corporate universities as levers of corporate social responsibility (Renaud-Coulon 2008).

Now that we examined what traditional universities do, it will be instructive to examine whom they do it for, i.e., their stakeholders.

Stakeholders of Traditional Universities

As a professor at a traditional university, when I teach a class, it seems obvious who my stakeholders are: the students who sit before me. If I do my job well and educate them, they benefit from that. However, there are others who may benefit and can claim a stake in the successful execution of this educational process. And moving beyond a single classroom, there are certainly many stakeholders of the varied activities that a traditional university undertakes.

From a financial perspective, we have customers and consumers. The students in the classroom are the consumers of the educational process, but

frequently someone else is footing the bill. The parents might be paying for this education and as such can certainly stake a claim to being stakeholders in the educational process. For many graduate programs, especially MBA programs, the student's employer might pay for all or part of the tuition. This certainly qualifies them as stakeholders. Moreover, in many countries, governments pay for much of higher education and that makes them very involved stakeholders.

Looking beyond the financial aspects, there are the beneficiaries of the education. If we are looking at undergraduates, their future employers will certainly benefit from the education the students receive. For MBA students, their current employers will benefit from the education, regardless of whether they are paying for it.

Expanding the circle even wider, consider medical education. Certainly the medical students themselves benefit, but so do their future patients. Certainly well-educated medical professionals benefit society.

And when we consider universities' service and research functions, we can certainly see that local communities as well as society as a whole are stakeholders of traditional universities. It is generally accepted throughout the world that all of society benefits from a good higher education system. But can we expand the circle of stakeholders of a corporate university as widely? The next section examines this question.

Stakeholders of Corporate Universities

We may again begin our discussion of stakeholders with the student in the classroom. But clearly the list of stakeholders of a corporate university goes beyond the individuals. The mission of every corporate university would involve some sort of organizational benefit. Corporate universities are not created for the benefit of the individual employees; they exist to benefit the organization. Remember, the definition of corporate universities includes the phrase, "assist its parent organization in achieving its mission." Also critical in the definition is the notion of cultivating "individual and organizational learning, knowledge, and wisdom."

If a corporate university does its job well, the employees become more capable or more productive. This certainly benefits the employee, but it also benefits his or her manager and the department or division in which they reside. And clearly the goal of the corporate university is to benefit the organization as a whole. So the stakeholders of a corporate university can be viewed as a series of concentric circles with the employee at the center and radiating outward to the department and ultimate the division.

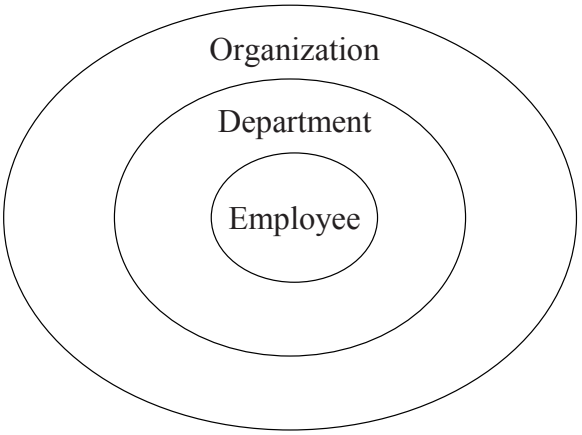


Fig. 6.1. The stakeholders of a corporate university.

But is this the end of the stakeholder list? This might be the case if corporate universities only existed to educate employees. Many corporate universities open their doors to customers, vendors, suppliers, business partners, and even strangers willing to pay. As we consider these groups, we have to expand the circle of stakeholders.

The corporate university adds value to the organization not just by developing employees, but by working with many partners along the value chain. If these partners can benefit from the offerings of the corporate university, that should add value to the organization.

A simple example of this is the University of Farmers at Farmers Insurance. The University offers many courses to its employees. However a large portion of the company's sales come from independent agents. These agents are not employees of Farmers, but independent businesspeople. The University of Farmers has created numerous programs designed to enhance the sales skills of these agents. The agents benefit by having better skills to build their businesses, while Farmers naturally benefits by getting increased sales.

While it seems clear that helping an independent sales force will enhance your bottom line, there are benefits to be derived from working with more subtle relationships in the value chain. Increasingly, corporate universities are making offerings available to suppliers, vendors, and business partners of all types along the value chain. If you can help these partners become more productive or even more familiar with your ways of doing business, there are benefits to be derived all along the value chain.

Customers are another vital stakeholder group that is often the focus of corporate universities. For example, Home Depot has a long-standing

practice of offering home improvement courses to its customers (at no charge). The customers benefit by having more knowledge about how to better do their home improvement projects and the company benefits from being able to sell more products to customers who undertake these projects.

One unusual group that is often given access to corporate university offerings is outsiders—people who have no affiliation with the organization, but are willing to pay to attend programs. Increasingly, corporate universities are willing to open their doors to all who are willing to pay. At first blush, this seems to run counter to the corporate university mission of helping its parent organization to achieve its goals, but in an era of tightening budgets (especially for learning and development), the revenue generated from the outsiders enables the corporate university to have more resources to devote to fulfilling its primary mission. So even though these outsiders are not truly stakeholders in the organizations, they provide revenue that enables the primary stakeholder, the organization, to reap more benefits.

Finally, the last stakeholder group to reap the benefits of an effective corporate university is society as a whole. At a minimum, a good corporate university helps its parent become more profitable. Profitable corporations contribute more to society in taxes, and thus there are more resources available to benefit society at large.

Beyond the contributions of tax dollars, successful corporations give back to their communities in many other ways, ranging from community services to charitable donations and philanthropy. Even more directly, corporations are beginning to look at their corporate universities as a primary lever in their efforts toward social responsibility. Annick Renaud-Coulon (2008) has documented numerous corporate universities on every continent that are their organization's vehicle of corporate social responsibility.

Thus, the initial circles of employee, department, and organization from Fig. 6.1 can be expanded further to include business partners, customers, and society as a whole, see Fig. 6.2.

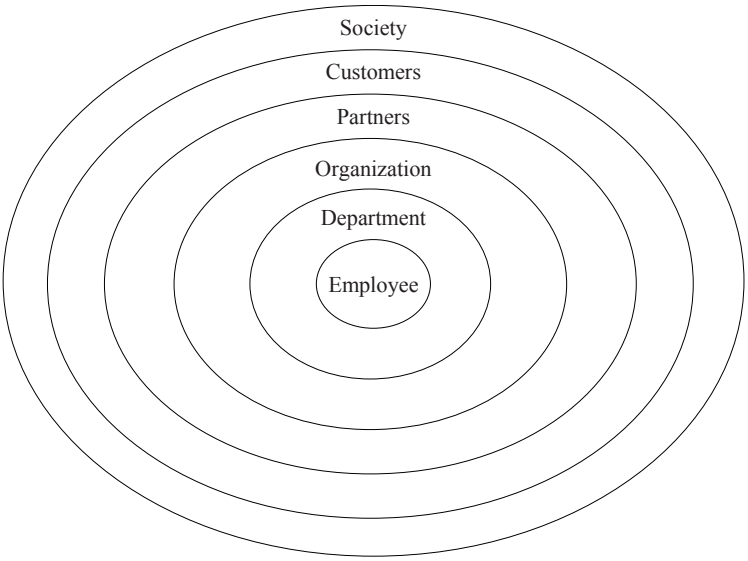


Fig. 6.2. The expanded circles of a true stakeholder university.

Finally, there is one additional group that bears mention: the true stakeholders of an organization—the owners or shareholders. Needless to say, if a corporate university does its job and does it well, the owners will reap the benefits of increased profits. Thus when we look at the circle labeled "organization," we are not just referring to internal organizational benefits, but also to increased wealth in terms of profits, dividends, or increased stock price for shareholders.

The Value of the Stakeholder Approach

By viewing the corporate university as a stakeholder university, we can expand the reach beyond employees to a variety of internal and external stakeholders. Beyond the obvious benefits of increasing the reach of the corporate university, there are numerous other advantages to the network approach.

Firstly, there is the notion of social capital. This is the value derived from relationships. We have long acknowledged that it's not what you know, it's who you know. Organizations are becoming more savvy about hiring people based on "know-who" rather than just "know-how." One of the skills organizations are looking for in managers is the ability to cultivate connections. So by bringing together a variety of people—

employees, suppliers, customers, etc.—in a corporate university setting, we can first add value by building relationships.

Additionally, there is the notion of networked learning. Just as the computing world has moved from the focus being on the node (a single computer or terminal) to the network (the connections between computers), the learning world is recognizing that there is considerable value to be derived from focusing on learning networks rather than individuals. Just as corporate mergers have long attempted to generate synergy of a 1+1=3 nature, networked learning can produce synergies greater than the sums of individual learning. By bringing together learners from the various constituent groups—employees, managers, partners, customers—the effects and benefits of the learning can be multiplied.

Some of this added value comes from having a common language. By having the various members of the value chain in the same learning interventions, it ensures that everyone is learning the same concepts in the same way and speaking the same language.

Moreover, there is value to be derived from having the different stakeholders work together in a learning environment. It can enable them to work together better in productive ways outside the classroom. An ideal type of learning for a network of stakeholders is action learning. In a well-constructed action learning scenario, a team of stakeholders will work together on a real-world project. Action learning combines the joint goals of a collaborative solution to the real project and learning outcomes.

Finally, there is the opportunity to leverage learning. The costs of many learning interventions are fixed. The cost of developing a course and delivering it are pretty much the same regardless of how many people are in it. Inviting stakeholders beyond the initial targeted employee group can leverage the learning at little additional cost. This applies to classroom learning and is even more true for e-learning in which virtually all of the cost is in development and the cost of delivery to additional learners is negligible.

But let me offer some words of caution about both cost-savings and e-learning. The focus of learning efforts should be the learning and the value of it, not doing it on the cheap. Sometimes in our efforts to save money, we end up cheapening the program into ineffectiveness. It is actually better to do nothing than to have ineffective learning programs. If you do nothing, at least people are spending their time working and you're not spending any money. If you do a bad learning program, you have pulled people away from their jobs and wasted money. Even if you have trimmed some costs from an initial plan, if the learning never gets used on the job (as we discussed in the wisdom management section above), the whole

thing is a waste of time and money. It is better to spend an appropriate amount of money wisely than to spend a smaller amount wastefully.

Sadly, much of what is being done in e-learning is still wasteful. Many organizations embark on e-learning programs with an eye toward saving money. While the cost of developing an e-learning program can be substantial, the cost of delivery is minimal. A program can be delivered to thousands of people (from every stakeholder group) at little or no additional cost. Additionally, a company's travel budget can realize tremendous savings from e-learning. In many corporate universities, some of the greatest costs are travel, food, and housing, all of which are eliminated with e-learning. However, too often the learning is eliminated as well.

Many companies are still faced with dismal completion rates for their e-learning elements. Some companies combat this with mandatory completion as a condition of continued employment. This will force people to finish, but it does nothing to make the program any more engaging. And without that engagement, learning does not happen. So in our rush to utilize technology to save costs, meet the requirements of our younger workers (as we have come to believe), or be on the cutting edge, let's just remember to always make sure we always put the learning into our e-learning.

Summary and Conclusion

In the first part of this chapter we examined what a corporate university is and what it does. In the second part, we discussed the various potential stakeholders of a corporate university. In this concluding section, we will tie the concepts together and look at some ways that corporate universities can creatively address the needs of the diverse stakeholders.

At the top of our list of corporate university activities was training. At a minimum, we can develop training programs for our employees and invite other stakeholders to sit in. However, if we want to derive maximum value, these programs will be designed specifically for the diverse groups of stakeholders instead of for a single group with others allowed to attend. This same concept holds true for e-learning programs.

Looking at some of the other items on the list, we can think about applying some of the other corporate university functions to the stakeholder model. For example, evaluation is an important function for most corporate universities. Rather than taking a summative approach to looking backward at programs and evaluating their success, a multi-

stakeholder approach would examine the success of a program across the various stakeholders and stakeholder groups. More importantly, the purpose of the evaluation would go beyond a retrospective look at success, it would include Level 4 measures of results. Those results might go beyond impact on the sponsoring organization and look at results all along the value chain and across the stakeholders.

The very best evaluation is built in to the program design. A program should be designed with specific results in mind. The evaluation should be engineered into the design not only as metrics to evaluate success, but as motivators and mileposts to ensure success. And if success is designed as specific results for a variety of stakeholders, the learning, the evaluation, and the impact can be spread across the entire value chain.

As we examine the role of corporate universities in helping to promote change across an organization in a proactive change leadership way as opposed to a reactive change management way, the success of the change can be facilitated by including external stakeholder groups such as suppliers and customers. If a company is reinventing itself in a substantial way, whether it is through a change in strategies, products, markets, or culture, business partners and customers are likely to be affected. By including them in the change process, the change is more likely to happen, to happen quickly, and, most importantly, to stick.

We can also apply the stakeholder approach to the process I referred to above as wisdom management. This process ensures that learners apply their knowledge in ways beneficial to the organization. If this process can be applied across stakeholders, we can realize benefits across the entire value chain.

This stakeholder approach fits perfectly with the concept of learning networks. While each stakeholder realizes benefits, the value derives from the interactivity between the individual elements. The true value is in the connections.

Our organizations do not exist in a vacuum and neither do our corporate universities. The notion of a closed system is absurd—no organization can exist without interacting with its environment. If we open our systems to an even greater extent and integrate learning and development efforts within an expanded network of stakeholders, then we can fully leverage learning as a driver of value.

References

Allen M (2002) The corporate university handbook. AMACOM, New York.

Allen M (2007) The next generation of corporate universities. Pfeiffer, San Francisco.
Bessant J & Tsekouras G (2001) Developing learning networks. AI & Society 15: 82-98.
Eurich NP (1985) Corporate classrooms. The Carnegie Foundation for the Advancement of Teaching, Princeton, NJ.
March A (1997) A note on knowledge management. Harvard Business School Press, Cambridge, MA.
Phillips J (2007) Next generation evaluation. In Allen M (editor) The next generation of corporate universities. Pfeiffer, San Francisco.
Renaud-Coulon A (2008) Corporate universities: A lever of corporate responsibility. Global CCU Publisher, Paris.
Thompson G (2000) Unfulfilled prophecy: The evolution of corporate colleges. The Journal of Higher Education 71(3): 322-341.

Chapter 7. Corporate Universities 2.0: The Future Networked Learning Organization

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Introduction

"Using corporate learning to drive a company's culture and strategy is a top issue," says Integrated Service Solutions (ISS) CEO Jorgen Lindegaard. ISS is one of the world's largest commercial providers of facility services. Headquartered in Denmark, the company has operations in 50 countries in Europe, Asia, North America, South America and Australia. ISS created its corporate university in 2001 as a way to spread the corporate culture to its far-flung workforce (Corporate 2008). However, the manner in which organizations like ISS accomplish this strategic goal of aligning learning to the needs of the company will need to evolve as the needs of the workforce change over time.

One of the major forces impacting this will be the shift in workforce demographics. Millennials (those born after 1981) are entering the workplace in force at the same time as older workers are deciding in large numbers to remain in the workplace longer. According to the American Association of Retired Persons (AARP), a growing number of older workers simply cannot afford to retire, while others are choosing to stay in the workforce longer for various quality of life reasons: to continue to be engaged in society, to be forced to maintain their creativity, or just to have something to do. In any case, AARP believes working past the traditional retirement age is here to stay (Baby 1999).

This multi-generational workforce will bring with it a differing set of needs, expectations and perceptions for how work is to be accomplished as well as how to best maintain and update one's skills and knowledge.

A. Romano and G. Secundo (Eds.), *Dynamic Learning Networks: Models and Cases in Action*, 137
DOI: 10.1007/978-1-4419-0251-1_7,
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A Worldwide Shift in Demographics

The United States is not the only country facing this shift in the demographics of the workplace. In Europe, an aging population, combined with a significant fall-off in fertility rates, has translated into a one percent decline in the overall population during the 2000–2010 decade. After 2010, the trend is projected to accelerate over the next forty years, with Germany, Italy and Spain all expected to experience population declines ranging from 14 to 25 percent over that timeframe, according to the United Nations Population Division (Delong 2004).

These demographic trends are producing a workforce that is both aging and shrinking. In 2020, it is estimated that Germany will have a workforce that is 20 percent smaller than it was in 2000 and a retired population that is 50 percent larger (Zakaria 2008)

In many Asian countries the demographic situation is similar or more pronounced to that seen in Europe. Fertility rates in Japan, Taiwan, Korea, Hong Kong and China are all well below replacement level. Due to this, estimates indicate that major East Asian countries will consequently face a sizable reduction in their working population as well over the next half-century. For example, in Japan the working age population has already peaked. In 2010 Japan will have three million fewer workers than it did in 2005. And in China the situation looks much the same, by 2030, China will have nearly as many senior citizens aged sixty-five or older as children aged fifteen and younger.

How will corporate learning and development change in light of these shifting workforce demographics? Will companies continue to invest in classroom based learning programs? How will the global economic meltdown of 2008–2009 affect corporate universities?

Corporate Universities: A Target in Tough Economic Times?

The oldest corporate university dates back to 1956 when General Electric opened the first "management school." In fact, General Electric's Jack Welch Management Development Institute has become known as a top leadership academy which has supplied numerous CEOs for companies such as Boeing, Home Depot, and many others.

In the past fifty-plus years, scores of corporate universities have been launched both in the United States and across the world as a vehicle to drive strategically relevant learning across an organization's enterprise to

its employees, customers, suppliers and now even consumers. In fact, by 2010 it is likely that the number of global corporate universities in existence will surpass the number of accredited for-profit and non-profit universities in United States, which are now numbering over 4,300.

But, as corporate universities have grown in numbers, one question looms: are they able to remain relevant to the changing business needs of their parent companies and relevant to the diverse generations in the workforce? Today's business leaders are no longer content with passively funding the learning budgets of their various companies. In fact, a survey conducted by the Corporate Leadership Council asked a sample of 250 Chief Executive Officers which area would suffer the most from the global economic meltdown in 2008. Learning and development was tagged as the number one target for budget cutbacks, at 38% of respondents. Recruiting was second, and information technology was third. In spite of "employees are our greatest asset" lip service, once again learning and development was shown to be the primary target for reduced budgets, and in some cases, across the board cancellations of formal training programs.

But corporate universities and learning departments do not have to be a target in tough economic times. In fact, now is the ideal time to develop an agenda of innovation for your corporate university so as to become as relevant and valuable to the enterprise as possible. This is becoming more important than ever as a perfect storm hovers over many organizations today.

The potential of lost institutional knowledge combined with the entrance of the Millennial generation will create lasting and profound changes in the workplace and to the way enterprise-wide learning is designed and delivered. Leaders of corporate universities must assess their current operations in today's social media landscape where social computing personalization and mobility dominate the business environment.

The Era of Social Computing Arrives

Social computing, or the use of computer technology to facilitate collaboration in groups, has been around for decades. What is different today is the spontaneous emerging of communities who share common interests. Consider the enormous success of such consumer websites:

- Flickr, (www.flickr.com) an online community for sharing photographs.
- Del.icio.us, (www.delicious.com) a social bookmarking service that allows users to tag, save, manage and share web pages from a centralized source.

- Dodgeball, (www.dodgeball.com) dubbed "the mobile friend finder" by *WIRED Magazine*, a site which combines the functions of an instant messenger, a social networking site, and a location based search engine so users can find friends with similar interests at any time through text messaging and RSS feeds.
- MySpace, (www.myspace.com) a social networking site which has become one of the most popular social networking sites in less than 2 years and now has 2 ½ times the traffic of Google Inc.

A new term has been coined to describe those who engage in large amounts of social computing: they have been dubbed the hyperconnected. In a new study conducted by the International Data Corporation (IDC), 2,400 working adults in seventeen countries were asked to describe their usage of social computing. IDC reports that 16% are currently "hyperconnected" and another 36% will be joining them within the next two years.

According to IDC, these hyperconnected workers:

- Have a much higher adoption rate of communications devices in their work and personal lives.
- Come from and are located in all countries although they are concentrated in China, India and Germany.
- Are found in all job functions and occupations but are largely concentrated in the IT and research and development functions.
- Can be any age, although 60% are under 35 years old and only 7% are over 55 years old.
- Have wired homes—63% have wireless internet connections available to them in their homes, versus the average of 40%.
- Push the envelope in adopting new tools such as Amazon's Kindle e-book reader and Apple's iPhone phone and mobile computing unit.
- Finally, and amazingly, they report that they would be more likely to take their laptop before their wallet or mobile phone if they had to leave their house for more than 24 hours.

For these hyperconnected, the boundary between work and personal connectivity is nonexistent. Two-thirds of the 2,400-respondent research sample reported they use text or instant messaging for both work and personal use, and more than one third of the sample reported they use social networking in both arenas as well.

As more working professionals become hyperconnected they will expect, if not demand, access to the same tools in the workplace that they use in their personal lives. Once they enter companies and begin updating their skills for the future, these hyperconnected employees will demand access to the tools which will allow them to learn, share knowledge and

work together on the job as easily and effortlessly as current technology allows them to share photos, find a new restaurant or locate a friend with similar interests in a strange city.

Personal Broadcasting Leads To Personalized Learning

According to market research firm GfK, MP3 players are now the fastest growing electronic consumer goods of all time with over 140 million units forecast to be sold by 2009 (Itfacts 2005). Already widely in use, the MP3 player is increasingly being used for learning. For example, financial services provider Capital One, with more than 20,000 employees worldwide, now hands out iPods as standard equipment to employees enrolled in training courses. According to Capital One Chief Learning Officer Ted Forbes, iPods are now viewed as another productivity tool, similar to laptops and personal digital assistants, allowing employees to work and learn remotely. Forbes points to the startling statistic that more than 90 million Americans drive alone to work everyday. Their eyes are occupied but their ears and minds are mostly idle. Now with learning available as an audio file on an iPods, commuters can advance their careers while driving to work. The ubiquity of iPods will enable the expansion of personal broadcasting (podcasting and vodcasting) to become a reality, allowing broadband content to be ported to iPods and mobile phones.

Sophistication of Mobile Phones Allows For Mobile Learning

According to the Global System For Mobile Communications Association, there are more mobile phones than people in UK. By the end of 2006, for every 100 Britons there were 116.6 mobile connections (Gantz 2008). While there are currently more than 4 billion mobile phones worldwide today, GSM projects that by 2020 there will be at least 20 billion mobile subscriptions in the global marketplace.

Today's high-end mobile phone has the computing power of a mid-1990s personal computer and features a hotbed of innovations such as geo-positioning services, video content availability, and even disposable mobile phones with built in content. As faster networks spring up to support data transfer for mobile phones at speeds that rival the wireless cards in most laptops, these phones will become the ideal device for accessing learning. In China, the BBC is already providing English language training on mobile phones. One can imagine corporate trainers dispensing disposable mobile phones costing less than one dollar, each

pre-loaded with engaging lectures and then supplemented by mobile phone call-ins and further communication and discussions among learners.

The Emergence of Corporate Universities 2.0

In this networked age, corporate learning must adapt or die. This means companies must redefine their corporate university to be accessible at all times and to leverage the latest collaboration tools and technologies to enable learning that is both more personalized as well as more unstructured. Fig. 7.1 shows some of the tools and technologies that can be used to create corporate universities 2.0

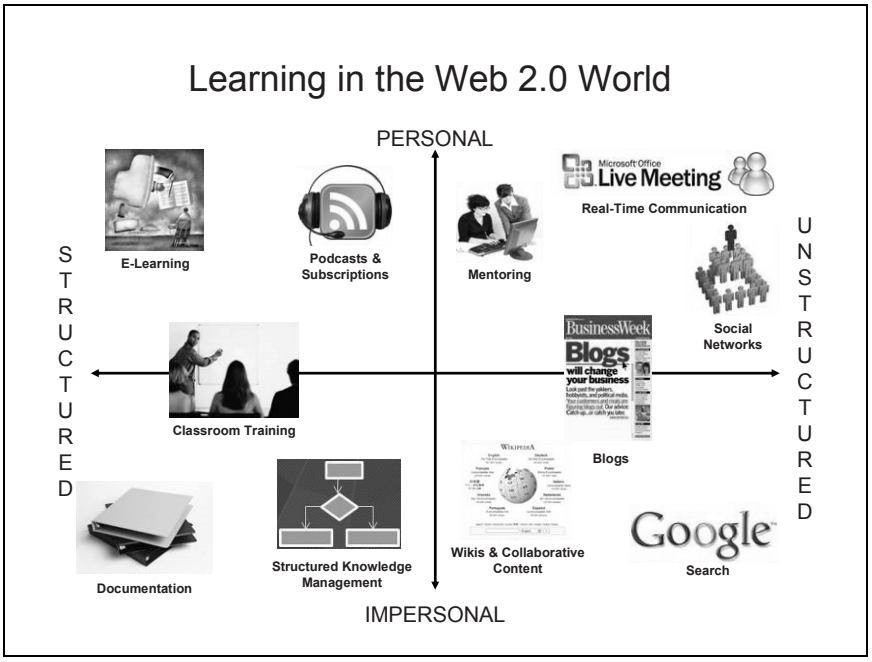


Fig. 7.1. Learning in the Web 2.0 world.

Corporate Universities 2.0, as the name might suggest, use Web 2.0 social media tools such as blogs, wikis, podcasts, online forums and video sharing sites to seamlessly blend learning and working. Tim O'Reilly, CEO of O'Reilly Media, coined the phrase Web 2.0 to refer to the ability to use the web to both publish and create content. This revolution in how the internet is used has changed all functions of a company, putting corporate learning under increased pressure to re-invent itself from a

traditional classroom based event to a network of learning and knowledge sharing.

The challenge is for corporate universities to move to a much more unstructured and open form of learning as depicted in Fig. 7.2.

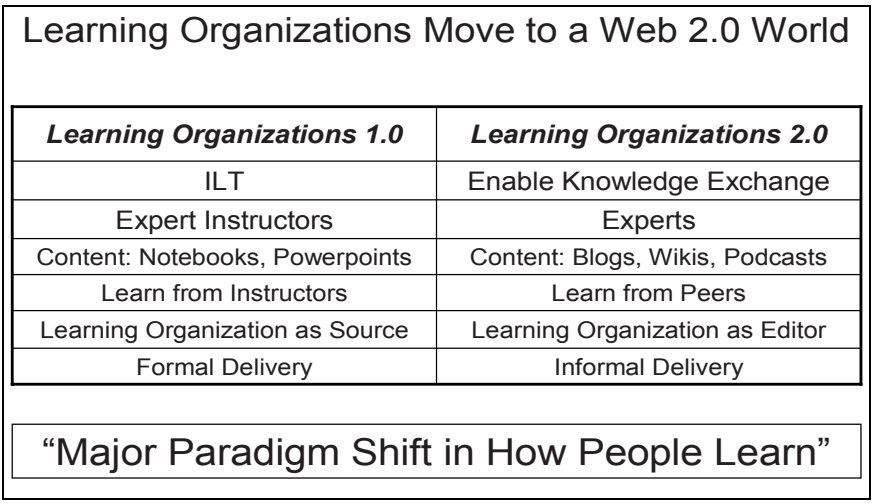


Fig. 7.2. Learning Organizations move to a Web 2.0 World

This new vision of learning depicts a model where networked learning is leveraged from all forms, including blogs, wikis, online communities, mentoring, and tagging of user generated content.

The goal is to move the learning organization from a General Motors production model to a Google search and find model. Learning under this corporate university 2.0 model can be characterized as:

- **Accessible Across Multiple Channels—**allows for learner centric distribution modes including ILT, eLearning, podcasts, wikis, blogs, etc.
- **Flexible To Allow For Content Reuse—**allows for multi purpose learning available online, in the classroom and in flexible audio formats.
- **Dynamic—**allows learners to easily contribute content, share lessons learned and add to their knowledge base.
- **Engaging—**so learners *want to* participate in interactive learning games and simulations that hold their attention and interest.
- **Personalized—**so learning is relevant to an individual's job performance and delivered at the point of need.
- **Intelligent Searching—**so learners will be able to automatically filter searches for "just what is needed" to be successful on the job.

- **Accountable**—so learners as well as faculty can track their contributions and be accountable for managing knowledge capital related to their job and their organization.

The vision for the corporate university 2.0 is one that is highly personalized, role-based, engaging, fun, and in keeping with the needs of a diverse multi-generational workforce. Under this new model, more learning will be achieved through interaction with peers and on the job.

Learning will be more "self-service" and less "formally structured" resulting in learners that will have to, and have the chance to, navigate on their own terms to discover meaning. Increasingly, learning will be viewed as a continuous resource, on demand when and where they need it rather than from a static catalogue of courses and lectures. And finally, learning will more and more be delivered using a two-way format where learners continuously contribute to the total body of knowledge through white papers, discussion forums, chat groups, blogs, wikis and podcasts with call-in features. Learning will no longer be one size fits all, but will be searchable and customized to each learner: their role, workforce needs, and expectations.

Corporate Universities 2.0 In Action

The growth of knowledge work is perhaps the single most important factor driving the future of the workplace. According to the Bureau of Labor Statistics, roughly 34 percent of the entire workforce in 2004 could be classified as workers who "think for a living." In other words, they have high degrees of expertise, education or experience and the primary purpose of their jobs involves the creation, distribution or application of knowledge (Davenport 2005).

So the question becomes, how can learning and development organizations be created that provide workers with the information, knowledge, and access to continuous learning and performance support that they need to succeed while giving them access to the latest tools and technologies?

Corporate Universities 2.0: Blending Learning with Working

McKinsey Research refers to the increase of knowledge workers as "the rise of the tacit workforce." By this, they mean that the number of jobs that include tacit work as an essential component of the job description have been growing at a rate two and half times faster than the number of

transactional jobs and three times faster than employment as a whole in the entire national economy. McKinsey defines tacit work as accessing a complex set of interactions to perform one's job such as problem solving, judgment, listening, collaborating, and communicating with co-workers.

According to research conducted by McKinsey, 48 million of the more than 137 million U.S. workers today can be classified as knowledge workers, making knowledge workers the fastest growing talent pool in most organizations. Put anther way, this is 70% of all U.S. jobs created since 1998—4.5 million, or roughly the combined workforce of the fifty-six largest public companies by market capitalization.

Corporate Universities 2.0: From Cerner Virtual University To Cerner Knowledge Works

This rise of the tacit workforce describes the workforce of Cerner Corporation, a global leader in healthcare information technology solutions. In an industry as complex as healthcare information technology, employee and client training is crucial to a company's success. Cerner committed itself to investing in learning and development with the launch in 1996 of Cerner Virtual University. Robert Campbell, a proven learning and development professional, was brought in to guide and shape this new learning function. The initial focus of Cerner Virtual University was to address the orientation, cultural, and job-specific competency needs of Cerner's growing workforce. During the late 1990s, Cerner was doubling its workforce every 2-3 years. In addition to this growth, Cerner was moving outside the scope of internal associate learning and development to meet client-learning needs related to Cerner's software. Then, starting in 2006, Cerner Virtual University expanded into client education, improving the capabilities of nurses and physicians in core, role-specific competency areas.

Since the inception of Cerner Virtual University, the education function has grown in scope and target audience. Once wholly dedicated to internal employees, Cerner Virtual University has expanded from a focus on providing learning and development to staff members to also designing and delivering learning to Cerner's clients. The new mantra for Cerner has become: "What we learn today we must be able to use tomorrow." This led to re-inventing Cerner Virtual University. Now known as Cerner KnowledgeWorks, this education function has become a hub for the creation and distribution of all the company's intellectual capital. The increased scope of this project has led to the exploration of new tools and

technologies for the delivery and access of learning, including Web 2.0 tools such as wiki's and blogs.

According to Campbell**, "**the focus of KnowledgeWorks now also includes business strategy, knowledge management, and documentation, as well as formal and informal learning. Cerner KnowledgeWorks is responsible for the timely creation, maintenance, and delivery of intellectual capital and the improvement of the performance of Cerner associates, clients, and business partners" (Johnson 2005).

Fig. 7.3 shows how Cerner KnowledgeWorks created a "corporate university 2.0" model leveraging a diverse set of Web 2.0 tools and technologies to blend learning into work.

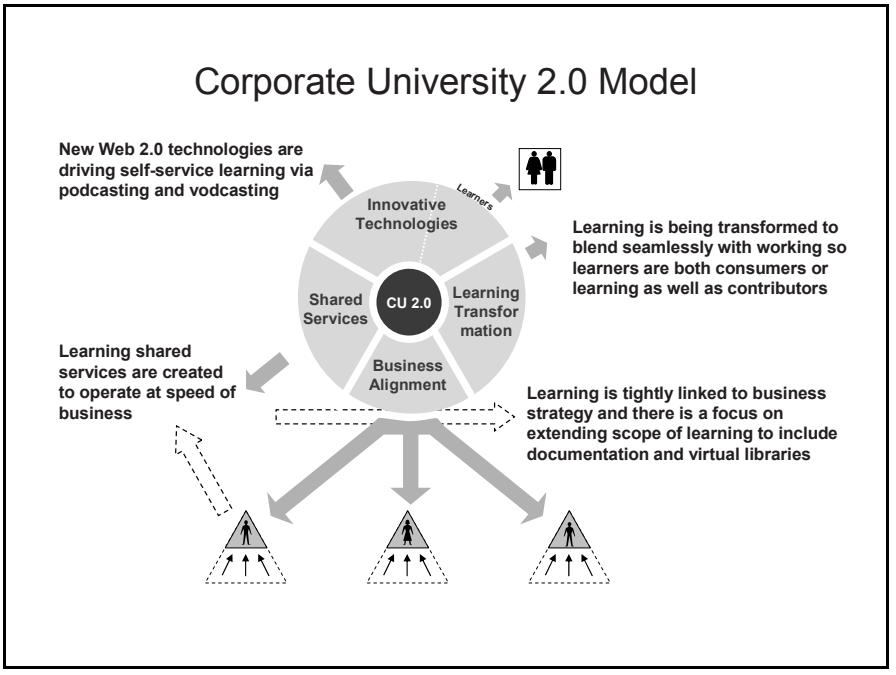


Fig. 7.3. Corporate University 2.0 Model.

In this 2.0 model, learning is captured and then distributed out to the workforce via numerous forms, such as: training modules, face-to-face instructor-led classes, performance support for a help desk, documentation/publications, mentor training, and even a series of 15 minute Podcasts. In essence, Cerner Knowledge Works has become the hub for all learning, performance support, documentation, and content management for the firm.

The overarching goal of Cerner KnowledgeWorks is to move intellectual capital through the organization faster; to get it to the people who need it, when they need it, and in a form that is most easily consumed.

Becoming A Corporate University 2.0

If you want to re-invent your corporate university to leverage the latest tools and technologies, how do you begin your journey?

Here are seven suggestions to move your corporate university to a 2.0 version:

1) Train your entire learning staff on the use of Web 2.0 tools and technologies. Focus on conveying how to move from a linear instructional design methodology to one that embraces learning in the context of working and then use Web 2.0 tools to create and distribute this knowledge. Use your corporate university as a pioneer for testing and vetting these new tools in the workplace.
2) Create new human resources and clear legal guidelines for how to use Web 2.0 tools such as blogs and wiki's and podcasts in the workplace so employees can be responsible users of these new tools and technologies.
3) Audit your current learning programs and assess if they are still relevant to your business strategies, use the latest digital tools that are available in the consumer marketplace to engage the youngest generation of workers, in particular.
4) Create a new vision and set of learning principles that will provide your staff and your governing board with a roadmap for becoming a corporate university 2.0. This includes the need to consider learning as seamlessly intertwined with working and to allow learners to become contributors as well as consumers of knowledge and intellectual content.
5) Brand your new vision for learning & development so it is viewed as a 21st century solution to achieving your organization's business priorities.
6) Create new alliances with the Head of Employee Staffing & Recruiting in your organization so your new vision for learning & development can become integrated into your employer brand. Learning is and can be viewed as an important driver for employee attraction and retention.
7) Seek out business awards that reinforce your investment in learning and development so your organization can become an employer-of-choice.

Implications for Corporate Universities

Heads of corporate universities must put into place a process to drive innovation in learning design, development, delivery that meets the strategic priorities of the business. A survey of 500 CEOs conducted by *The Economist Intelligence Unit* found that fostering innovation in all parts of the organization will be one the keys to growth over next two to three years. In the words of Joel Lemke, CEO of Enovia Corporation, a division of French industrial conglomerate Dassault Systemes, "cutting costs is important in all parts of the organization but if you do a really good job managing innovation this is what gives you an order of magnitude increase over cost savings" (Campbell 2008).

Establishing this innovation mindset is critical to creating a corporate university 2.0. In the book, *Blue Ocean Strategy,* Kim and Mauborgne challenge businesses to ask themselves five questions to create a new innovative mindset for the organization (Kim and Mauborgne 2005). These same questions can be a starting point to think about transforming a traditional corporate university to a corporate university 2.0:

- What can be created and offered that has never been offered before?
- Which factors/services should be raised above industry standards?
- Which factors/services/offerings should be eliminated?
- Which factors/services/offerings should be re-invented?
- How should this new vision be marketed and branded?

These questions can be the starting point for you to analyze how to best transform your current model of a corporate university into a 2.0 version. Keep in mind you are on a journey and that one of the key steps is to create a network of business leaders who can provide advice and counsel in this re-invention process. Close collaboration between those responsible for the development and delivery of learning content and the company's senior management responsible for establishing the business strategy is essential, and is too important to be done in a haphazard or careless manner as is all too often the case.

Instead, what is needed is a formal system of alignment to ensure that business strategy and workforce development are created in tandem. This means the corporate learning function must understand the business dynamics and propose how learning can drive significant improvements in revenue, business productivity and speed to market. True alignment is the job of a network of senior business leaders working in close collaboration with the Chief Learning Officer or Vice President of Learning Transformation to draft a vision for learning which leverages the latest tools and technologies in Web 2.0.

As we look into in the future of 2010 and beyond, we begin to see the

lines blurring between learning, working, communicating and entertaining, with learning becoming increasingly embedded into all the devices we now use in our daily lives as consumers –including the smartphone, the laptop, the MP3 Player and even the car.

References

Baby (1999) Baby Boomers Envision Their Retirement, February 1999, http://boomersint.org/aarp.htm (3 January 2009).

Campbell R (2008) Personal interview Robert Campbell, Vice President and Chief Learning Officer, Cerner

Corporate (2008) Corporate Learning Network Executive Spotlight - ISS CEO Jorgen Lindegaard,, October 23, 2008, http://www.imd.ch/news/IMD-CLN-Spotlight-Jorgen-Lindegaard.cfm (3 January 2009).

Davenport TH (2005) Thinking For A Living, Boston: Harvard Business School Press.

Delong WD (2004) Lost Knowledge: Confronting The Threat of an Aging Workforce, New York: Oxford University Press, 14.

Gantz J (2008) The Diverse and Exploding Digital Universe, Hopkinton: EMC2 Publications, 3.

Itfacts (2005) 140 mln MP3 players sold in 2005, 286 mln by 2010, May 2, 2005, IT Facts, http://www.itfacts.biz/140-mln-mp3-players-sold-in-2005-286-mln-by-2010/6409 (4 January 2009).

Johnson BC, Manyika JM, Yee L (2005) The Next Revolution in Interactions, McKinsey Quarterly, Number 4, 25.

Kim C, Mauborgne R (2005) Blue Strategy Ocean, Boston: Harvard Business School Press.

Zakaria F (2008) The Post American World, New York: W.W. Norton & Company, 196.

Chapter 8. Emergent management learning in dynamic learning networks

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Complexity emergence and a quantum interpretation of business

The idea that many simple, non-linear deterministic systems can behave in an apparently unpredictable and chaotic manner is not new. It was first introduced by the great French mathematician Henri Poincaré. Other early pioneering work in the field of chaotic dynamics is found in the mathematical literature by scientists such as, amongst others, Birkhoff, Levenson and Kolmogorov. More recently, Noble prizes, to Prigogine and Kauffman, have been awarded in this field of research. One of the difficulties for management theory and practice engaging with complexity theory lies in its attachment to causality.

Complexity as an emergent organisational paradigm in the knowledge based economy primarily questions the concept of causality. Despite relativity and quantum mechanics, most physics (and certainly all managerial thinking) is still Newtonian, based on a fixed space-time frame. In the meantime, further developments have taken place in the area of biology (such as the concept of Sheldrake's morphogenetic fields) and mind/body medicine that all seem to point to a federating idea of a quantum interpretation of social phenomena (non-locality, synchronicity and entanglement). Could a-causality form the basis for a quantum ontology of complex systems?

A. Romano and G. Secundo (Eds.), *Dynamic Learning Networks: Models and Cases in Action*, 151
DOI: 10.1007/978-1-4419-0251-1_8,
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The foundational concepts in the complexity realm emerge from such fields as neurobiology, cognitive sciences, physics, and organizational theory. New developments in knowledge management such as connectionist approaches (complex adaptive systems for the visualization of emergence give promising results (Baets 2005). In fact, instead of causality, it appears that the networked economy is ruled by synchronicity (occurring together in time) in line with findings in quantum research. Could it be that economy and management in general, and the more dynamic aspects of it like innovation, in particular, are indeed based on a quantum ontology?

The insight into complexity that developed over the last decade, and its consequences for management, discussed in earlier publications (Baets 2006a, 2006b), provides a platform for this chapter to explore a more ecological view on management aiming for sustainable performance.

What Prigogine and complexity theory in general discussed fundamentally was the existence of any causal relationship. In fact he was surprised that despite the two fundamental revolutions in physics in the last century, relativity theory and quantum mechanics, physics still remained mainly Newtonian. Physics presumes a fixed time and space concept, in which the future is causally related to the past, while complexity theory, and quantum mechanics, show the impossibility of this assumption.

Once holism, constructivism and emergence are accepted as fundaments of a new paradigm in management, a paradox, comparable to the one in science, emerges. Despite the theory of relativity and quantum mechanics, almost the whole scientific management community is still focussed on Newtonian principles, that is to say fixed space and time. If what happened yesterday is known, then what will happen tomorrow is equally well known (as much as what happened the day before yesterday). Management still does very little with the space-time continuum that these revolutions have offered us. In the hard sciences, at least, there are groups of researchers working on this subject. In economy, managerial and social sciences this revolution seems to have been completely side-stepped. Our managerial thinking is still the Marshallian economic thinking of the 19th Century (Arthur 1998).

On to biology: Sheldrake and “implicit order”

The paradigm sketched above refers to the use of quantum concepts, non-locality and synchronicity, as much in physical science as in the science of language and music for instance. This section turns to Sheldrake’s theory,

which is founded on biology. Sheldrake (1995), who is a well-known Cambridge biologist, is now an affiliated Research Fellow at the Noetic Society. Although his theory is controversial (as is often the case with a new paradigm) it has been validated, as his many publications witness, by considerable research.

In a book, which Sheldrake wrote with (Bohm 1982), they broach the subject of "implicit order". Implicit order is something like a ground below time, a tonality, of which each movement is projected in explicit order (what is known). For everything visible, there is something in implicit order which is at the origin of the projection. If an event is repeated a lot, behind that there is a constant built component. A sort of (fixed) link is born. Via this process, the forms of the past can continue to live in the present. This is more or less what Sheldrake calls morphogenetic fields, created by morphogenetic resonance. If something climbs into 'totality' where neither time nor space is fixed, it could be that things of the same nature will attach themselves to one another, or resonate. Because neither time nor space exists in this totality, things which happen at a particular place could therefore also happen elsewhere, or at least have an influence elsewhere.

These ideas are very much in line with the Bogdanovs' singularity (2004) and their observation that a fifth dimension, being a fourth dimension of space, expressed in imaginary time, could exist. Comparing the different "quantum" interpretations in the different sciences seems to converge. The convergence can be understood as an emergent understanding of this quantum world and its consequences.

Although Sheldrake and Bohm's theory, mentioned above, is the scientific topic that Sheldrake vigorously researches these days, his theory of proved morphogenetic fields could also be very useful for us.

Sheldrake's idea of morphogenetic fields complements the later ideas that (Varela 1979) was able to work on (before he died). They engaged with how something like resonance could be responsible as the organising principle in networks. Varela's suggestion has become illustrated by Sheldrake's research. In fact, these characteristics identified from morphogenetic fields are completely in parallel with the complexity paradigm. They could just as well be the characteristics of an economic system, a market, a company.

Sustainability

Sustainability is a concept that is around for roughly 20 years, and though everybody interprets it a bit different, a common understanding seems to emerge. In this contribution we would like to investigate the operationalization of sustainability and in particular sustainable performance.

Publications refer to a growing number of companies that consult in, or are working on the implementation of sustainable development concepts, and we observe that most of them seem to converge to the idea that within the current mainstream managerial thinking, there is no real place for sustainability. The Global Compact Summit in Geneva (July 2007) indeed launched a call for thought leadership in innovating managerial theory, in order to be able to host concepts of responsibility and sustainability.

The concepts proposed and investigated here, accept an ontology, based on recent developments in biology and quantum mechanics, that could be the basis for a managerial concept that integrates naturally sustainability and responsibility. We could enrich this ontology with elements of non-violence, co-creation and values, but we have to pin it down now to a usable concept, translated into diagnostic tools and eventually into a managerial approach: a road book with check lists. In this contribution we are developing a holistic managerial concept and a diagnostic called Cassandra. For the interested reader, a complete methodology, applicable in companies, is available in (Baets and Oldenboom 2009).

Brundtland defined probably as the first in 1987, sustainable development as follows: **development seeking to meet the needs of the present generation without compromising the ability of future generations to meet their own needs.** We do not want to discuss the true nature of sustainable development, since there are many other books doing that. Our aim is ultimately to transform the drive for sustainable development into a concept of sustainable performance. This Brundtland definition introduces at least three dimensions: the economy, the ecology and the society, and it suggests that those are interconnected. It furthermore introduces a time and a space dimension, and it raises the governing issue. By introducing space and time as variables in the equation, Brundtland has introduced a paradox in managerial thinking. Our classical Newtonian view on management cannot cope with a moving and integrated space-time concept, and as long as the society and the economy move slowly, one can make a fixed time-space approximation. But reality is different. The complexity of the world (its non-linear and dynamic character) in connection with the speed of change, does not allow for non-

linear static approximations anymore. Our classical metrics fail and the thermometer becomes the disease itself.

The Brundtland definition introduces the paradox of the short term versus the long term. We need a short term efficiency in order to remain attractive for shareholders, but at the same time we need a longer term sustainability orientation in order to be attractive for the stakeholders. Paradoxes enforce choices; choices and balances between different and sometimes orthogonal interests. Another paradox - that isn't a new one - that Brundtland reinforces is the paradox between reductionism and holism. Classical managerial approaches mainly, if not exclusively, focus on financial performance: the so called bottom-line. By introducing all those extra dimensions: societal, ecological, time, space, governance, no reductionist approach can help a manager to answer the issues raised by the Brundtland definition. We have nevertheless tried for instance by introducing a concept like corporate social responsibility and the request to report on a company's responsibility. We have insisted that companies report on their ecological footprint and ultimately we have even responsibilized companies that they should be ecologically neutral, and if they would not be, they can "pay" for their carbon emission rights. We have ultimately turned responsibility (a value) again into an economic good which we can trade. We have turned responsibility into a tradable emotionless economic good. It doesn't matter if you pollute, as long as you pay for it. And ultimately, there will always be countries that are willing to sell their "non-production" of carbon emission, in order to get some money for their economic development. From a holistic perspective, needless to stress, this is only shifting the problems around and as usual it ends up in the hands of the powerless: nothing new under the sun. A reductionist approach of sustainability responsibility (and even ethics) will only lead to a displacement, not to a solution.

Participating in the Global Compact Summit, one can buy off his carbon emission that he causes by coming by plane to the summit. When are we going to use videoconferencing, and in doing so, open up such a summit to all those that are economically unable to come to such a summit? Why do we still organise higher education at certain localisations, for the happy few that can make it and afford it, and leave millions of people out, that are dying to get education and, via education, development and growth. Technology is available; it becomes an issue of choice. Within a reductionist frame, the solution is found: we have made our misbehaviour an economic good, assuming that everything is an economic good and therefore anything can be commercialised. Is responsibility an economic good?

Sustainability principles

Sustainability, as we understand it today, has developed from the Brundtland understanding. An example of such wider framework are the Core principles of

According to Ben-Eli, sustainability calls for a deep transformation in all aspects of human activity including our worldview, our values, our technology, our governance and more. A growing number of people need little convincing that establishing the concept of sustainability as the organizing principle on our planet, fostering a well-balanced alignment between individuals, society, the economy and the regenerative capacity of the Earth's life-supporting ecosystems, represents a most urgent challenge for our time. It is a challenge unprecedented in scope. It requires a fundamental shift in consciousness as well as in action. It calls for a deep transformation, simultaneously, in all aspects of human activity including our worldview, our values, our technology, current patterns of consumption, production, investment, governance, trade, and more.

The currently prevailing definition of sustainability emphasizes cross-generational equity, clearly an all-important concept for any society that wishes to endure, but one which is operationally insufficient. It often fails to provide unequivocal guidance when specific policy decisions are debated, since actual, specific wishes of future generations are not easy to ascertain. Anchoring an alternative definition directly to the relationship between a population and the carrying capacity of its environment offers a more advantageous approach since it assumes a number of key variables, all potentially measurable. For example: population numbers; a measure of wellbeing; total inventory and rate of consumption of resources; impacts of by-products generated by human activity on the absorption capacity of the environment; impacts of new technologies in opening or hindering new evolutionary possibilities; and the like. Hence, the following definition (Ben-Eli):

Sustainability: A dynamic equilibrium in the processes of interaction between a population and the carrying capacity of an environment such, that the population develops to express its full potential without adversely and irreversibly affecting the carrying capacity of the environment upon which it depends.

This definition points to the dynamic nature of sustainability as a state, a state which has to be calibrated with time, again and again, as changes occur in population numbers or in the resources available for supporting all humans at a desired level of wellbeing. It does not seek to define specifically what such a level is, nor to limit yet unimaginable possibilities

for social evolution. It recognizes, however, boundaries and limits which must be maintained by stone-age tribes and industrial societies alike. As long as the underlying conditions for equilibrium are maintained, the well being of future generations is assured.

The set of sustainability principles which follows is grounded in this definition. The principles are articulated in broad terms but can receive a specific operational meaning in relation to particular sectors of the economy, development issues, business strategies, investment guidelines, or initiatives taken by individuals. They are expressed in relation to five fundamental domains, all representing essential aspect in the interaction of human populations and the environment. These domains include:

The Spiritual Domain: Which identifies the necessary attitudinal orientation and provides the basis for ethical conduct.

The Domain of Life: Which provides the basis for appropriate behavior in the biosphere with respect to other species.

The Social Domain: Which provides the basis for social interactions.

The Economic Domain: Which provides a guiding framework for creating and managing wealth.

The Material Domain: Which constitutes the basis for regulating the flow of materials and energy that underlie existence.

The result is a set of five core principles, each with its own derived policy and operational implications. The set is fundamentally systemic in nature, meaning, that each domain affects all the others and is affected by each in return. Rather than a list, the set should be approached and understood as a coherent whole. Within the framework of these principles, we are able to design a tool that supports management for sustainable performance in companies.

A holistic model

Holism, is yet another term that is loosely defined and interpreted by many people in different ways. Is there a common notion of holism? Is there somebody who one day tried to compile all the theories? Perhaps it is evident that one should have all sorts of critiques here.

We willingly make reference here to one of Ken Wilber's concepts which is, in our way of thinking, and in line with the new geopolitical reality described, very handy and useable. He visualises something which we could call different dimensions of the image of the holistic world. The Fig 8.1 gives a summary of Wilber's concept.

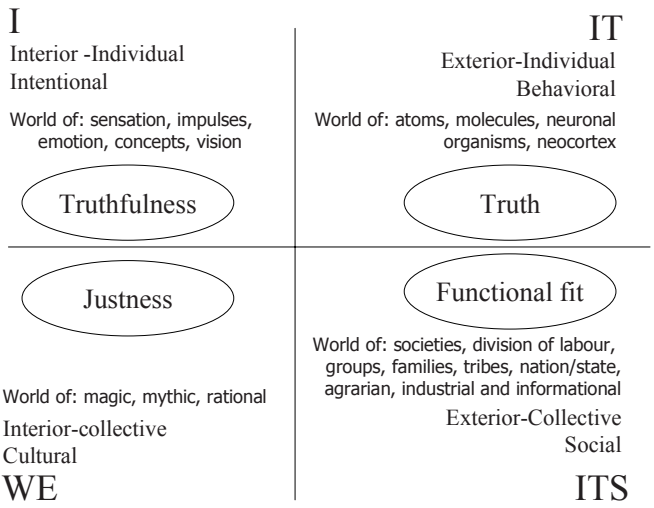


Fig. 8.1. Summary of Wilber's concept

The figure is developed around two dichotomies: external-internal and individual-networked (collective). The quadrants above make reference to the individual level. The quadrants below refer to the collective level. The quadrants on the left have to do with the internalisation of Man (or processes, or things), while the quadrants on the right examine, let us say, the mechanical part (the external). A holistic image is obtained, according to Wilber, if all the quadrants receive sufficient attention. He labels these quadrants the 'I' quadrant, the 'We' quadrant, the 'It' quadrant, the 'Its' quadrants. All the quadrants have to be alive in order to be able to achieve a "life", an observation, a research, a holistic interpretation.

In the right top quadrant, we study the external phenomena, for example how the brain functions and so we naturally reduce it to very specific parts, like atoms, the classical reductionism. Not completely mistaken, there is a reaction to this partial vision in saying that understanding the functioning of a specific atom does not allow us to understand the functioning of the whole (the consciousness of Man).

What we call, at the heart of science, a global approach, is found in the lower right quadrant; however that is nothing more than one of the four dimensions of holism. Here one can think of the systemic approaches (still mainly mechanical), of ecological concepts, sustainable development, etc.

If we really want to understand what the brain produces, we can only find that in the left part of the diagram. The brain causes, in Man, emotions, feelings, concepts, etc. and it is those which we use in daily life.

No matter how detailed our understanding of the right part is, it still says nothing about what Man thinks or feels. To get to the dimensions on the left, the classical approaches are insufficient. Communication is the only means to try to understand how people feel and what emotions they go through.

In the left part there is also a collective dimension: one could label it "culture". That has a relationship with what we accept as a group, the norms and values. So a holistic understanding cannot bypass these internal individual and collective dimensions.

Classical science goes completely in search of the 'truth' (identified top right). More and more we see global approaches (sometimes called systemic) in science: the functional whole. The true notion of Man and his emotions which we call, a little paradoxically, a "flesh and blood" man, does not give us a real understanding of truth and fairness. Here we want particularly to attract attention to the three other quadrants to thus give a more complete understanding than the dominant thinking that our western culture allows. Our attempt is to try a more holistic approach in management research and in the understanding of phenomena.

Whilst summarising this detailed diagram is not only difficult, it does not give full benefit to this diagram either. We would nevertheless like to try. Holism consists of an ensemble of 'I', 'We', 'It' and 'Its'. This is quickly recognised in certain metaphors of holism, such as 'Art meets science and spirituality', the "I", "We" and "It" of Wilber. Another saying is that the hands, head and heart lead to holism. This can also be attributed to Wilber.

Applied to management: A systemic management interpretation

It is clear that the Anglo-Saxon model has brought us a lot of insight into the functioning of markets and companies and therefore it is extremely important to clearly understand this model. But there is more. While the Anglo-Saxon model implicitly wants to wash away diversity as a disturbing factor, it becomes clear in the wave of mergers and acquisitions that currently go around the world, that diversity can be a creative force. Therefore, a creative use of diversity, in our opinion, becomes important in order to tackle the corporate world in the coming years. Openness to

diversity, understanding of different world- and economic views enrich future managers considerably. We gladly take the metaphor of the mosaic. If the world consists of different colours we can either melt the colours together in order to make it one colour, or instead we can create a mosaic, keeping the different colours and have them networked in such a way that it becomes a nice picture. On top of management as a science, we also want to understand management as an art. A holistic management approach clearly aims to enrich the prevailing Anglo-Saxon one.

Therefore in some ways, the proposed vision on management is inherently one of diversity, sustainable development and network structures. Studying the evolution of companies today, we see that increasingly companies are a network of business units that interact in between each other and with the outside world via a series of dynamics links, internally and externally. Not only does the organisation organically, and hence dynamically change, but so do the interactions with outsiders. Indeed, even the larger corporations are in fact a highly dense network of business units, outside providers, individuals, clients, etc., that in reality form a number of often flexible, overlapping and dynamic networks that cut across the company, instead of just a nicely and smoothly organized company.

In a more holistic management approach we cannot exclusively focus on the search for truth, and not only since absolute truth doesn't exist. Furthermore, and despite of the difficulty of measuring the internal quadrants of Wilber's model, it is clear that it are rather those internal quadrants that will make the difference. The values, the culture and personalities of people, be it managers or employees alike, is going to become the intrinsic cause of the emergence of a responsible management approach.

As described in a general way, the Wilber model illustrates the different dimensions of holism in human action, based on a framework of two axes. The two axes oppose on the one hand individual to networked (collective); on the other hand internalized to externalized. Developing a complete, broad and integrated picture of a person (a manager, or a human activity, such as management) one should cover the four quadrants formed in this picture. Hence a valuable and enriching approach, what we believe a systemic view is, indeed covers these four quadrants.

The individual-networked dichotomy illustrates that in all human activity, and though the individual is crucial, there always is a collective dimension. That collective dimension of interacting individuals in companies, markets, societies, is increasingly one that is networked (as opposed to a strict hierarchical organization common during the last decade). The internalized-externalised dichotomy is one that illustrates that

most of our knowledge is experience based and acquired by experimentation, in order to become actionable knowledge via internalization (concepts very popular in knowledge management theory). Nevertheless there is an important aspect of externalized, transferable information that is the visual part of what we read, share, should know, is printed in books, etc.

We can now use this diagram, in order to define a vision for any management development activity (or organisation). In the bottom left quadrant, in between internalized and networked, we have values and culture. Some contemporary value choices are a choice for diversity as a constructive power; the long term perspective with an inherent dimension of social responsibility; humanism and relativism. These are some examples of shared values and cultural dimension, that eventually are going to be the basis of all corporate drive. In this left part of the figure, we are much more concerned with how people (students, future managers, managers, clients) feel culture and shared values, how they co-create and co-construct them and how they are perceived.

Such a vision can only be translated into action by people (students, future managers, managers) who have the personality to take part in this challenge and who have the qualities and the motivation to make a real difference in the world. Not only does this necessitate a strong focus on personal development as a backbone for all managerial approaches. It also highlights the need for companies to organise around life long learning career development for their employees. A key element of management becomes management of human resources, which translates into managerial competencies inventories (assessment centers), personal development programs, coaching, etc. to choose a much more learner-centred approach.

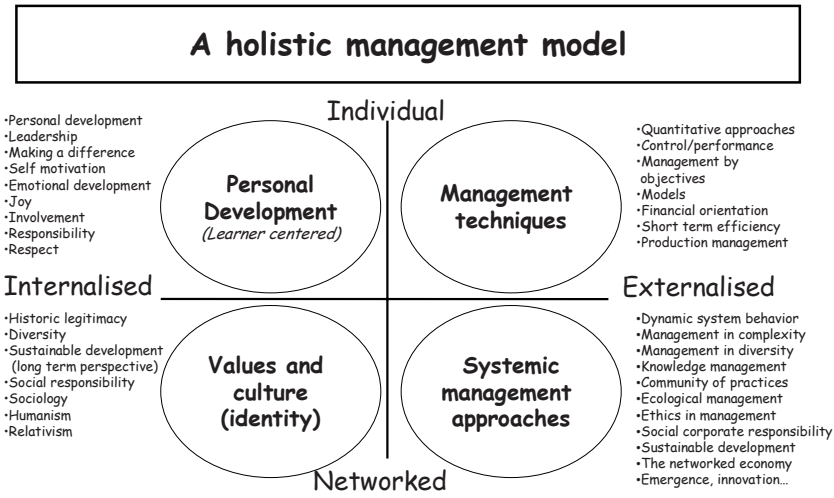


Fig. 8.2. A holistic management model.

These value choices and this focus on personal development, allow us to innovate and improve our management techniques. In the top right quadrant, we find a more Anglo-Saxon management approach (or should we label it a more mechanistic approach?): a minimum condition to be a successful manager. Just to label some characteristics: quantitative approaches; control oriented; performance management; models; financial focus; short term efficiency. Effectiveness and efficiency are and remain important tools for becoming a sustainable company. They are necessary conditions; in no way they are sufficient.

In the bottom right quadrant, this model is complemented by what we call systemic management approaches. We consider systems as interacting elements that create a logic of their own, that surpass the simple addition of the composing elements. In this quadrant we find the more ecological approaches to management, network theories and applications, sustainable development models, complexity theory, concepts around diversity as a constructive force, etc.

A holistic management approach is the "interweaving" of those four quadrants. This is an approach that is based on a clear vision and values, which makes a choice for personal development as the backbone of the corporate activities, in order to allow people to innovate and optimise their management techniques and to enrich it with a more systemic management practice. Only the four quadrants together allow the individual to develop

into a responsible manager who is able to pilot a company for sustainable performance. A manager, just as any other employee, becomes the entrepreneur of his or her own development, within a dense and intensive network of peers. In contrast, the Anglo-Saxon model is focused mainly in the right-upper corner and though companies might pay some attention to culture, or personal development, the focus remains the realisation of financial results.

Leadership: the driving force of sustainability

Once we accept a holistic management approach, we accept that responsible management goes far beyond the mechanistic, control driven view of management. It even goes beyond the desire to create an interrelated economy: that interrelatedness (right lower quadrant in our holistic model) is only a consequence. The basis and drivers are values, purpose and meaning. We need a management style that is value driven, and that gives true meaning and space to each and everyone's need for personal development, improvement, development and ultimately learning. It further considers that personal development of each employee as the driving force, the energy, for the success of the company. In other words, central in management becomes leadership.

After having dealt with the concepts, the ontology and the (new) paradigm that would allow us to integrate responsibility as a key component and driver of our managerial action, we are now able to construct a tool (a metric) and an approach to make all this happen in real life (if we would like to). We describe here a tool, named Cassandra, that is developed in a personal and a corporate version (Baets and Oldenboom 2009). The personal version, giving the user an insight in his or her potential for being a leader for sustainable performance, is validated and the validation will be summarized. In a second stage, this tool can be used by the manager to design and guide his or her personal development path towards a responsible manager, managing for sustainable performance. The corporate version, that can be used as a metric for a company interested in sustainable performance, and that defines sustainable performance, is by now showcased in a number of corporate analysis.

Summary description of the Innovation School and its conceptual model: sustaining learning networks in corporations

The Innovation School targets delivering management education with a focus on innovation and intrapreneurship. Far too long, innovation and intrapreneurship have been considered as separate disciplines that could in some way contribute to the company's result. The development of the knowledge economy, supported by the existence and increasing use of the e-world, seems to bring innovation and new idea generation in a fast lane. Innovation and intrapreneurship are today in many instances the "raison d'être" of most companies, particularly with a view on the long term sustainable development of companies. In order to cater for this new reality, Euromed Management (Marseille) has developed the Innovation School.

The Innovation School is as much a virtual learning platform, as it is an innovative and cutting edge pedagogical approach that brings learning on the workfloor, in the very heart of the managerial action. By doing so, it integrates knowledge management and learning in every day learning-while-doing situations.

The Innovation School is a learning laboratory, that contains theoretical content, cases, course assignments, all kinds of business information sources like Gartner Business Information services, electronic libraries (both for books and articles) and collaborative tools. The content is hypertext linked, based on the semantics of concepts and cases. This allows full freedom of learning at any level of pre-knowledge for each and every individual. Indeed, each learner can learn at its own pace, level and interest. Learning can be completely individualized, taking into account the diversity of each employee, though it is organised in small groups of students (communities). All content is accessible via multiple entries (per concept, via assignments, via a semantic search engine, etc). The learning platform contains personal workspaces, in which the competency management of the individual can be monitored and managed. Collaborative tools are available, as well as Wiki's, chat facilities and phone over IP conferencing tools. Most innovative and unique is the semantic linking between the concepts, and its visual semantic tree representation that goes with it.

But the Innovation School is equally a ground breaking pedagogical innovation. The approach is completely learning-by-doing driven, putting the learner in the center of the action. Each individual can learn what he wants to learn, whenever he wants to learn and following a completely free

and therefore personal learning path. The learner steers his own learning and guidance is given through assignments. All assignments have an individual component, preparing a group discussion. The size and the purpose of those (small) communities can be adapted most flexibly. Faculty becomes tutors, facilitating the learning of individuals. The more the assignments of the students can be brought in the daily work of those students, the more learning and knowledge management flow together. Learning becomes integrated in daily work practice. We have stopped schooling and finally allowed learning. A version of Fig. 8.3, adapted to the Innovation school looks as follows:

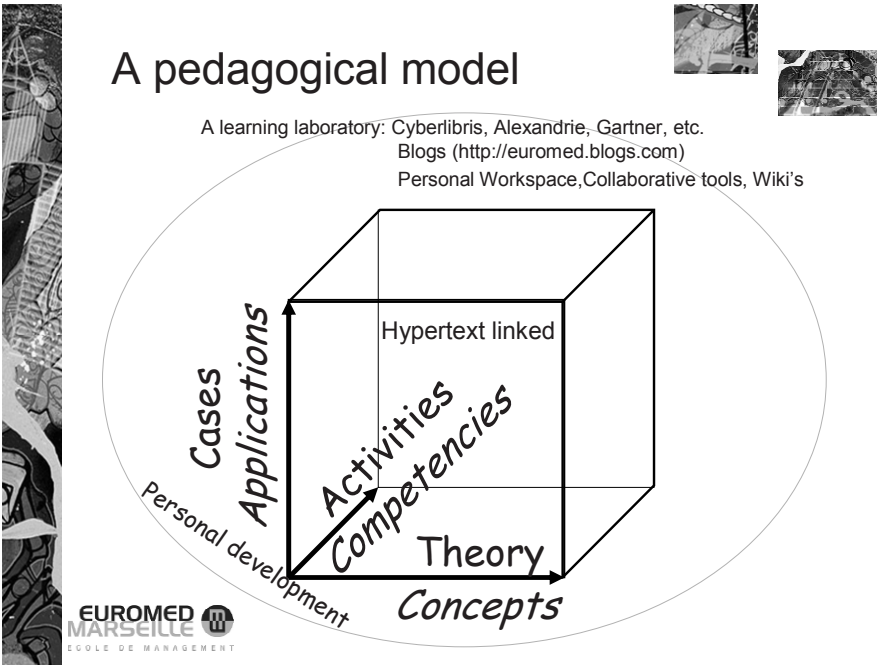


Fig. 8.3. A pedagogical model.

The pedagogical concept of learning-by-doing and learning-while-doing, on which the learning lab is based, is both well researched (see publications mentioned above) and has already proven its value added in real life learning projects. However, we feel that the personal contact and groupwork is certainly equally important: we like to call it a Hybrid Learning Solution. This latter aspect would necessitate an involvement of so called "local partners" that could deliver tutoring (and hence face to face contact) locally.

Some objectives that can be reached with the Innovation School:

- Cater learners with entrepreneurial competencies and knowledge on the basics of management
- Provide a higher "return on education" through a project based learning-by-doing approach (therefore break ground in what we know today in management education)
- Create, manage and support learning networks in companies, either via management learning activities of any kind, and/or via knowledge management
- Create immediate practically relevant output (a company structure)
- Allow for truly international delivery and learning, using a virtual learning laboratory
- Walk ourselves the (internet) talk of innovation and intrapreneurship
- Assist in creating entrepreneurs
- Provide a 'learning space' for innovation
- Create a network of people and ideas, that are able to generate new business proposals and that can contribute to local sustainable development
- Contribute to personal development of people
- Develop and give space for courageous try-outs
- Deliver high-quality

Some conclusions

Learning for sustainable and responsible entrepreneurship - and the entrepreneur to be understood as the manager-leader-creator – is in search for a new paradigm and a new organisation, as argued in this chapter. We clearly need to move away from the teaching paradigm into the learning paradigm, but even that might not be enough to shift the perspective of our economy into a responsible and value creating society. A paradigm shift is needed, that brings us and the "learner" from schooling to wisdom, via the route from causality to emergence. Once we accept a quantum ontology, as described in this chapter, we see how to give the responsibility for learning back to the learner. If we accept that reality is self-organised on the basis of synchronicity, non-locality and entanglement, the responsibility for learning can only be shifted to the learner. An ideal curriculum can never exist; it is up to the learner, using the network of people around, to co-create his or her learning experiences. Schooling becomes a set of activities, probably loosely interconnected, with one aim, that is the personal development of the learner, within his or her search for

"meaning". It should be referred here to the five principles of sustainability, described earlier. Meaning defines purpose and interconnection; in turn, that construct forms the context for the definition of the activities.

A learning agenda becomes a follow up mechanism of a set of predefined management competencies that the learner wants to develop throughout his or her learning experience. Those management competencies could be defined by the learner, but in the case of a corporate setting, those management competencies could be defined in cooperation with HR management. The baseline of the learning agenda becomes the evolution from a certain level of management competencies to another level, a more meaningful one, a more interesting one, a richer one. In order to help realising this, any kind of activity could help.

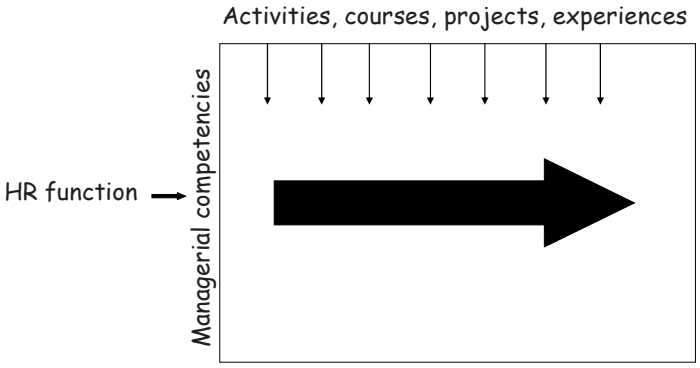


Fig. 8.4. The ideal curriculum.

Projects, courses, any real life activity, or any kind of experience could be of help in order to develop management competencies. The focus is no longer the activities, but the development of competencies. And in order to realise the latter, no formal curriculum is necessary. Workplace learning should typically be such a kind of a setting, just as much as corporate universities all together. However today, most of those corporate universities are mainly copies of what business schools do currently, copying precisely all the weaknesses that for business schools are due to the limitations of structure, accreditations, etc. Companies have a much larger freedom to really organise workplace learning, according to the principles described in this chapter. This network concept of learning, this paradigm shift to one of interconnectedness, entanglement and meaning, is a sound basis to move from a corporate university towards a Stakeholder University. A mere extension of corporate universities to stakeholder

universities, only for the sake of involving more people, has no value added.

For the corporations, the challenge is a paradigm shift and the re-instauration of purposefulness in management. For the business schools, the challenge is probably even larger. Either they arrive to dissipate into companies, where they act as facilitators, knowledge co-creators and pedagogues, or they might disappear in a red ocean of competition between themselves. What business schools do today, can easily be done by corporate universities, and possibly they can do it better.

For those who might think that all this is new, the overall picture and the understanding of its interactions might be. But Ivan Illich, in his book Deschooling Society (popular in the 70s of previous century) came a good end in the same direction.

References

Arthur B (1998) 'The end of certainty in economics', in Aerts D., Broekaert J. and Mathijs E. (eds), Einstein meets Margritte, Kluwer Academic, 1998

Baets W (2005) Knowledge Management and Management Learning: Extending the Horizons of Knowledge-Based Management, Springer

Baets W (2006a) Complexity, Learning and Organisations: A Quantum Interpretation of Business, Routledge

Baets W (2006b) 'Complexity theory: dynamics and non-linearity are the only reason for knowledge management to exist' in Boughzala I., Ermine, J-L., (eds) (2006) Trends in applied knowledge management, Edition Hermes Penton Science

Baets W, Oldenboom E (2009) Rethinking growth: social intrapreneurship for sustainable performance, Macmillan Palgrave

Baets W, Van der Linden G (2000) The Hybrid Business School: Developing knowledge management through management learning, Prentice Hall

Bogdanov I. and Bogdanov G (2004) Avant le Big Bang, Editions Grasset & Fasquelle

Kofman F (2006) Conscious Business: how to build value through values, Sounds True

Sheldrake R (1995) The presence of the past, Park Street Press

Sheldrake R, Bohm (1982) Morphogenetic fields and the implicate order, ReVision, 5:41-48

Stacey R (2000) Strategic Management and Organizational Dynamics, Prentice Hall

Varela F (1979) Principles of Biological Autonomy, Elsevier-North Holland

Wilber K (2000) A Brief History of Everything, Gateway

Glossary

Architectural complexity: Refers to the fact that each solution is comprised of a variety of components, or modules: physical products and devices, embodied technologies, pieces of software, and service components like consulting, maintenance, or financing.

Back-end unit: The integrator firm's unit responsible for the interactions seeking and assuring alignment between customer needs and the set of capabilities available internally or externally within the solutions network.

Bundling: A design or marketing practice aiming to sale sets of products and/or services combined.

Business consulting: One of the services which are likely to be included within an integrated solution package.

Capital goods: Goods employed by firm as means for the production of other goods.

Causality: Denotes a necessary relationship between one event (called cause) and another event (called effect) which is the direct consequence (result) of the first.

Collaborative Social Networks: Dynamic, self-organizing networks of relationships that arise from a desire to collaborate around a shared purpose.

Complexity: Expresses a condition of numerous elements in a system and numerous forms of relationships among the elements. In general usage, complexity tends to be used to characterize something with many parts in intricate arrangement. In science there are at this time a number of approaches to characterizing complexity.

Consumer goods: Goods purchased by the final consumer for usage or consumption.

Corporate Universities 2.0: Corporate Universities which are restructured such as to leverage the latest collaboration tools and technologies to enable learning that is both more personalized as well as being business relevant.

Corporate University (CU): educational institution that typically provides job-specific, indeed company-specific, training for its employees, personnel of the parent corporation, customers, suppliers and, occasionally, consumers. CU is a strategic tool designed to cultivate individual and organizational learning, knowledge, and wisdom.

Co-suppliers: Firms that, being connected to a systems integrator, participate in the design and implementation of a solution by providing specific modules or components (products, services, software or technologies).

Customer-centricity: As opposed to product-centricity, it is an approach to business management that privileges the fulfilment of customer needs independently of the firm's traditional competencies and vocation.

Demand-driven learning: The learning process occurring within an integrator firm at the interface with the customer and aiming to a deep understanding of the customer's strategic and operational problems. This learning process is managed by the supplier's front-end unit and nurtured by its frequent interactions with the customer.

Dynamic learning network: A network formally or informally set up for the primary purpose of enabling any kind of learning to take place over the time for building capabilities, managing change, gain competitive advantage in the perspective of learning organization, eventually supported by the information and communication technologies.

Economies of repetition: Gains in efficiency and effectiveness obtained through the replication over time and across customers of integrated solutions projects.

Environment: In a broader acceptation, the dynamic combination of circumstances evolving around a firm's business and conditioning its activities.

Explorative knowledge: The knowledge acquired by exploring new domains, typically through and within interactions with customers and suppliers.

Extreme events: Rare and large events which are outside the capabilities of routine controls and cannot be predicted using statistical tools of averages.

Five Disciplines (of Organizational Learning): Introduced by Peter Senge a discipline is a skill or practice you can work on for a long time and still continuously improve. Senge's five disciplines are: Personal Mastery, Shared Vision, Mental Models, Team Learning, and Systems Thinking.

Front-end unit: The integrator firm's unit responsible for the interactions with customers. This unit gathers information about customer needs and reports them to the back-end for seeking and assuring alignment.

Functional interdependency: Refers to the fact that all solutions components are, to some extent, interdependent: physical products or devices can perform without technologies or software, which in turn cannot be exploited without the support of important services. High interdependency leads to the possibility that an external modification affecting a single component can trigger a domino effect on other components, impelling vast reconfigurations of the global package.

Gamechanger: A specific approach to managing innovation and research developed by Gary Hamel of Harvard Business School and the Strategos consultants.

Hastily Formed Network: A collaborative social network which forms in the context of a crisis or urgent situation.

High reliability organisations: Organisations which manage major risks, such as nuclear power stations or air-traffic control where the consequences of failure are intolerable.

Holism: Is an approach to research that emphasizes the study of complex systems. This practice is in contrast to a purely analytic tradition (sometimes called reductionism) which purports to understand systems by dividing them into their smallest possible or discernible elements and understanding their elemental properties alone.

Human capital: The stock of knowledge, skills and expertise held by the members of an organization.

Innovation School: Is a combined approach to offer the opportunity to tackle relevant issues and train and educate at the same time. Learners will obtain managerial skills and knowledge as well as the necessary insight in business. It is based on a semantic knowledge base approach, learning by doing, and learning while doing.

Integrated solutions (or customer solutions, or solutions): Offerings of highly integrated bundles of products and services designed and implemented to fulfill a customer outsourcing need. In its most complete form, the provision of integrated solutions involves the transfer of responsibility from the customer to the supplier for the ongoing solution of a business or operational problem.

Knowledge integration: The process of integration and exploitation of the two kinds of knowledge acquired within the demand- and supply-driven learning processes.

Knowledge Workers: Individuals whose work involves the need to access a complex knowledge base to perform their job, such as problem solving, judgment, listening, collaborating, and communicating with co-workers.

Learning: An increase in knowledge that is, an increase in the capacity to produce desired results through reflection and experience.

Mental Models: One of the Five Disciplines, defined as the largely unconscious or preconscious assumptions, beliefs and patterns of thinking that we hold, which determine how we see and act in the world.

Model of industrial organization: Integrated solutions can be analyzed not only as a firm's business model, but also as a model of organizing business relationships within a comprehensive industrial network involving the systems integrator and its customers and suppliers.

Modularization: An approach to solutions design aiming to exploit the advantages of modularity and interchangeability of product and service components.

Network proactivity: The firm's strategic and organizational ability to anticipate and respond to environmental changes through rapid

reconfigurations of the network relationships among its internal units and between those units and either or both demand and supply networks (customers and co-suppliers).

Operational services: Services provided within a comprehensive solution package and aiming to operate, maintain, upgrade and dismiss a complex product or system (generally an infrastructure).

Organizational Learning: A set of concepts, methods and tools popularized in Peter Senge's book *The Fifth Discipline* that enable both individual and collective learning in the context of an organization or social network.

Personal Mastery: One of the Five Disciplines, defined by Peter Senge as "the discipline of continually clarifying and deepening our personal vision, of focusing our energies, of developing patience, and of seeing reality objectively."

Repeatability (or replicability): The property of a specific configuration, or design, of integrated solutions to be replicated over time and across different projects.

Shared Vision: One of the Five Disciplines, recognized when a group shares an image of the future, which is highly motivating and energizing for them.

Social Computing: The use of new computer technologies to facilitate collaboration in groups over the Web.

Solutions business model: The business model adopted by a system integrator for delivering integrated solutions.

Solutions network: The network of firms involved in the supply of integrated solutions. It is comprised of a customer, a systems integrator, and a number of (co)suppliers.

Solutions paradigm: A set of shifts in the approach a firms holds when referring to customers, products and value: from product-centricity to customer-centricity, to output to performances, from transaction to relationship, from value delivery to value co-creation, from competition to co-supply, and from value chain to value network.

Stakeholder University: Educational institution that typically provides job-specific training for the personnel of a networked corporation, including its suppliers, clients and other partners. It is intended as hubs of learning networks, with radical changes in methodologies, processes and organisational structures.

Supply-driven learning: The learning process occurring within an integrator firm at the interface with (co)suppliers and aiming to a deep understanding of the currently available and future technological options which can be leveraged for delivering high-value solutions.

Sustainability: Refers to a dynamic equilibrium in the processes of interaction between a population and the carrying capacity of an environment such, that the population develops to express its full potential without adversely and irreversibly affecting the carrying capacity of the environment upon which it depends.

System: A system is a compound entity (i.e. has parts within) that produces results above and beyond the sum of the contributions of its parts. Systems produce results determined by their structure. Systems mayor may not have been consciously designed.

Systems integration: A typical capability of a systems integrator, i.e. the ability to design and integrate internally and externally developed components – products, services, and software – into unified systems and to coordinate the activities of component, subsystem or product suppliers.

Systems integrator: The typical provider of integrated solutions, able to design and integrate internally and externally developed components – products, services, and software – into unified systems and to coordinate the activities of component, subsystem or product suppliers.

Systems theory: Is an interdisciplinary field of science and the study of the nature of complex systems in nature, society and science. More specifically, it is a framework by which one can analyze and/or describe any group of objects that work in concert to produce some result.

Systems Thinking: Is a discipline for understanding a whole (the system) by examining its parts and how they interrelate to produce system effects.

Team Learning: One of the Five Disciplines recognized when a team is able to have difficult conversations in a way that generates new insights,

perspectives, and learning. Generative and reflective conversations are key practices for team learning to occur.

The Fifth Discipline - A book by Peter M. Senge, 1990, which brought Systems Thinking into the field of Leadership and Management. Recognized by Harvard Business Review in 1997 as one of the five most influential management books of the decade. The fifth discipline is Systems Thinking.

Traditional University: A university, college, or community college which grants associate, bachelor, master, and/or doctoral degrees.

Vendor financing: One of the services which are likely to be included within an integrated solution package.

Vision: An image of the future, which is highly motivation and energizing. See Personal Mastery and Shared Vision. Vision is the key to conservation and expansion of valued accomplishment.

Web 2.0 Tools: The emerging tools in the use of Web technology and Web design that aim to enhance creativity, communication, secure information sharing, collaboration and functionality on the Web. This can apply to tools and technologies such as blogs, wikis, podcasts, online forums and video sharing sites.

Wisdom Management: a planned and systematic process by which an organization manages how its employees use and apply their knowledge and skills in ways that benefit the organization.

Index

A

Adaptive learning; 3

C

causality; 151; 166
change; 23; 25; 27; 36
commercial environment; 28
commercial experts; 33
commercial organisation; 23; 41
competence; 5
complex adaptive system; 30; 152
complex networks; 25
complexity; 24; 152; 162
consumer goods; 76
continuous learning; 3
corporate learning; 138; 142
corporate universities; 104; 121; 122; 124; 138; 148
Corporate Universities 2.0; 15; 137; 142

D

dynamic learning networks; 151
Dynamic Learning Networks; 1

E

effective networks; 49
e-learning; 134
extreme events; 24; 26; 40

F

five disciplines; 46; 52

G

Gaussian distributions; 24
Generative learning; 3

H

Hastily Formed Networks; 47; 48; 52
Hastily Social Networks; 11
high reliability organisations; 26
holism; 155; 157; 159
hyperconnected; 140

I

illycaffè network; 62
increasing returns; 2
information and communication technologies; 106
Innovation School; 164
integrated solutions; 13; 75; 76; 79; 85; 92

K

knowledge; 2; 18; 61; 67; 96; 99; 101; 110; 144
knowledge flow; 66
knowledge management; 126
knowledge network; 68; 69; 70
knowledge workers; 144

L

leadership; 23; 27; 37; 40; 50
leaning organization; 46

learning; 2; 97; 144
learning by doing; 2
learning by interacting; 2
learning by R&D; 3
learning by searching; 3
learning by using; 2
learning departments; 139
learning dynamics; 13; 80; 84; 90
Learning dynamics; 75
learning economies; 2
learning in networks; 4
learning laboratory; 164
learning network; 106; 112; 118; 121; 133; 135
learning organization; 107
Learning organization; 3
life-long learning; 14

M

mechanistic bureaucratic system; 30
mechanistic controls,; 33
Mechanistic models; 24
mechanistic organisational model; 26
multi-generational workforce; 137; 144

N

network; 2; 4; 11; 17; 23; 49; 61
network effects; 66
network organisational model; 26
network proactivity; 76; 88; 89
network society; 68
Network structures; 9
networked enterprise; 13
networked learning; 15
Networked Learning Organization; 99; 137

O

organisational change; 24; 41
organizational learning; 12; 28; 43; 51

P

personal mastery; 51; 52
Peter Senge; 43
positive feedback; 2
power law distributions; 24

S

scientific learning; 3
shared vision; 51
Six Sigma methodology; 24
social computing; 140
Social computing; 139
social networks; 26; 47; 51; 53
solutions business models; 89
solutions network; 13; 83; 85; 86
solutions paradigm; 80
Stakeholder Universities; 14; 104; 106
Stakeholder University; 167
stakeholders; 15
Stakeholders; 116; 129
supplier network; 57
suppliers; 67; 75; 91
sustainability; 154; 155; 156
systems integration; 89; 90
systems integrator; 90
Systems Thinking; 54

T

tacit work; 144
traditional universities; 121; 127

V

value creation; 98
Virtual Corporate University; 108; 110

W

Web 2.0 tools; 147
wisdom management; 126; 134